

Math 121: Commutative Algebra

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Worksheet 1.1: Basics of Rings

Problem 1. Let R be a ring and consider $R[[t]]$ the ring of formal power series with coefficients in R . The goal of this exercise is prove that:

$$(R[[t]])^\times = \{u + tf(t) \mid u \in R^\times, f(t) \in R[[t]]\}.$$

- (a) Elements of $R[[t]]$ can be written as $f = \sum_{k=0}^{\infty} f_k t^k$ for $f_k \in R$. We call f_0 the constant (or lead) term of f . If $f = \sum_{k=0}^{\infty} f_k t^k$ and $g = \sum_{k=0}^{\infty} g_k t^k$ then find a formula for the lead term of fg .
- (b) Using the formula from (a) prove that if f is a unit then $f_0 \in R^\times$.
- (c) Find a formula for the coefficient of t^k appearing in fg .
- (d) Assume that $f_0 \in R^\times$. Prove that f is a unit by explicitly constructing an element $g \in R[[t]]$ such that $fg = 1$. (Hint: the condition that $fg = 1$ places constraints on each coefficient of fg . Use the constraints together with part (c) to recursively solve for the coefficients of g .)

Solution.

- (a) Formula for the constant term of fg is $f_0 g_0$.
- (b) Assume f is a unit. Then there exists some $f' \in R[[t]]$ such that $ff' = 1$. By part (1), the constant term of ff' is $f_0 f'_0$. The constant term of 1 is 1 , so $f_0 f'_0 = 1$. Hence f_0 is a unit, and $f_0 \in R^\times$.
- (c) The formula for the coefficient of t^k is $\sum_{i=0}^k f_i g_{k-i}$.
- (d) Assume $f_0 \in R^\times$. Let its inverse be g_0 . For f to be a unit, every non-leading coefficient of fg must be 0 . That is,

$$fg = f_0 g_0 + \sum_{k=1}^{\infty} \left(\sum_{i=0}^k f_i g_{k-i} \right) t^k$$

where $\sum_{i=0}^k f_i g_{k-i} = 0$ for all k . Then solving for g_k gives us,

$$g_k = -\frac{1}{f_0} \sum_{i=0}^{k-1} f_i g_{k-i} = -g_0 \sum_{i=0}^{k-1} f_i g_{k-i}$$

Problem 2.

- (a) Prove that an element $u \in R$ is a unit if and only if the principal ideal $\langle u \rangle = R$.
- (b) Prove that if ab and b are units, then a is a unit.
- (c) Prove that if u is a unit and $a \in R$, then a and ua generate the same principal ideal.
- (d) In the ring $\mathbb{Z}/\langle 12 \rangle$, determine which elements are units.

Solution.

- (a) ($\implies$) Assume $u \in R$ is a unit.

Then there exists some $u' \in R$ such that $uu' = 1 \in \langle u \rangle$. If 1 is in the ideal generated by u then $\forall r \in R, r \in \langle u \rangle$. Hence $\langle u \rangle = R$.

($\impliedby$) Assume $\langle u \rangle = R$ for some $u \in R$.

Then $1 \in \langle u \rangle$. That is, by definition of an ideal, 1 is a multiple of u . Hence, there exists some $u' \in R$ such that $uu' = 1$. Hence, u is a unit.

- (b) Let $a, b \in R$, and ab, b be units. Then there exist $x, b^{-1} \in R$ such that

$$abx = 1, \quad bb^{-1} = 1$$

Then we have,

$$abx = bb^{-1} \implies \frac{abx}{bb^{-1}} = 1 \implies a \cdot \frac{x}{b^{-1}} = 1$$

Hence, a is a unit and $\frac{x}{b^{-1}}$ is its multiplicative inverse.

- (c) Let $u \in R^\times, a \in R$.

Note that the containment $\langle ua \rangle \subseteq \langle a \rangle$ holds by definition. So we only need to show that $\langle a \rangle \subseteq \langle ua \rangle$. Since u is a unit, there exists some $u^{-1} \in R$ such that $uu^{-1} = 1$. Then, note that $uu^{-1}a = a \in \langle ua \rangle$, i.e. $\langle a \rangle \subseteq \langle ua \rangle$.

Hence a and ua generate the same principal ideal.

- (d) $(\mathbb{Z}/\langle 12 \rangle)^\times = \{1, 5, 7, 11\}$

Problem 3. Given two rings $\mathbf{R}$ and $\mathbf{S}$ we let $\text{Hom}_{\text{Ring}}(\mathbf{R}, \mathbf{S})$ be the set of ring homomorphism from $\mathbf{R}$ to $\mathbf{S}$. When there is little room for confusion we will often drop the subscript Ring and simply write $\text{Hom}(\mathbf{R}, \mathbf{S})$.

- (a) Prove or disprove: If $\mathbf{R}$ and $\mathbf{S}$ are rings, then $\text{Hom}(\mathbf{R}, \mathbf{S})$ is a ring under point-wise multiplication and point-wise addition.
- (b) Give an explicit description of the hom-sets below:
 - (a) $\text{Hom}(\mathbb{Z}, \mathbb{Z})$
 - (b) $\text{Hom}(\mathbb{Z}/\langle 2 \rangle, \mathbb{Z})$
 - (c) $\text{Hom}(\mathbb{Z}, \mathbb{Z}/\langle 2 \rangle)$
 - (d) $\text{Hom}(\mathbb{Z}, \mathbb{C}[x])$
- (c) Based on part (b) state and prove a characterization of $\text{Hom}(\mathbb{Z}, \mathbf{R})$ for any ring $\mathbf{R}$.

Solution.

- (a) Let $f, g \in \text{Hom}(\mathbf{R}, \mathbf{S})$. Note that since ring homomorphisms are unital $f(1_{\mathbf{R}}) = 1_{\mathbf{S}}$ and $g(1_{\mathbf{R}}) = 1_{\mathbf{S}}$. Assume for the sake of contradiction that $\text{Hom}(\mathbf{R}, \mathbf{S})$ is a ring under pointwise addition. Then,

$$1 = (f + g)(1) = f(1_{\mathbf{R}}) + g(1_{\mathbf{R}}) = 1 + 1 = 2$$

A contradiction! Hence $\text{Hom}(\mathbf{R}, \mathbf{S})$ is not a ring under pointwise addition and pointwise multiplication.

- (b)
 - (i) $\text{Hom}(\mathbb{Z}, \mathbb{Z}) = \{\text{id}\}$
 - (ii) $\text{Hom}(\mathbb{Z}/\langle 2 \rangle, \mathbb{Z}) = \emptyset$
 - (iii) $\text{Hom}(\mathbb{Z}, \mathbb{Z}/\langle 2 \rangle) = \{\mathbb{Z} \twoheadrightarrow \mathbb{Z}/\langle 2 \rangle\}$
 - (iv) $\text{Hom}(\mathbb{Z}, \mathbb{C}[x]) = \{\mathbb{Z} \hookrightarrow \mathbb{C}[x]\}$
- (c) Note that since both $\mathbb{Z}$ and $\mathbf{R}$ are rings, any homomorphism $\phi : \mathbb{Z} \rightarrow \mathbf{R}$ is unital, i.e. $\tau(1) = 1_{\mathbf{R}}$. To be a homomorphism we must have $\forall a \in \mathbb{Z}$,

$$\tau(a) = \tau\left(\sum_{i=1}^a 1\right) = \sum_{i=1}^a \tau(1)$$

This completely determines the homomorphism, and thus there is a unique (unital) homomorphism from $\mathbb{Z}$ to any ring $\mathbf{R}$.

Problem 4. Let $\phi : R \rightarrow S$ be a ring homomorphism.

- (a) Prove that if $J \subset S$ is an ideal then $\phi^{-1}(J) \subset R$ is an ideal.
- (b) Assuming ϕ is injective, give a nice description of $\phi^{-1}(J)$ for an ideal $J \subset S$.
- (c) Find a counterexample showing that if $I \subset R$ is an ideal $\phi(I)$ need not be an ideal.
- (d) Prove that if ϕ is surjective then $\phi(I)$ is an ideal of S .
- (e) Let $I \subset R$ be an ideal. Prove that if $\ker(\phi) \subset I$ then $\phi^{-1}(\phi(I)) = I$.
- (f) Is the assumption that $\ker(\phi) \subset I$ needed in part (5)?

Solution.

- (a) Let $a, b \in \phi^{-1}(J)$. Then $\phi(a), \phi(b) \in J$. Since ideals are closed under addition, $\phi(a) + \phi(b) = \phi(a + b) \in J$. Thus $\phi^{-1}(a + b) \in \phi^{-1}(J)$.
Similarly let $r \in R$, and $a \in \phi^{-1}(J)$. Then $\phi(r) \cdot \phi(a) = \phi(r \cdot a) \in J$. Thus, $\phi^{-1}(r \cdot a) \in \phi^{-1}(J)$.

- (b) If ϕ is injective then,

$$\phi^{-1}(J) = \{r \in R \mid \phi(r) \in J\}$$

$$\phi^{-1}(J) = J \cap \phi(R) = J \cap R$$

- (c) Let $R = \mathbb{Z}$, $S = \mathbb{Z}[x]$ and define the homomorphism to be $\phi : R \hookrightarrow S$. Then consider the image of the ideal generated by 3. It does not contain $3x$ and hence is not closed under multiplication.

- (d) Let $I \subset R$ be an ideal and assume that ϕ is surjective. Let $a, b \in I$. We want to show that $\phi(I)$ is a subgroup of the additive group of S and is closed under multiplication by elements of S .

Consider $\phi(a) - \phi(b)$. Note that since ϕ is a homomorphism the following holds,

$$\phi(a) - \phi(b) = \phi(a - b)$$

and since $a - b \in I$, $\phi(a - b) \in \phi(I)$. Moreover since $\phi(0) = 0 \in \phi(I)$, $\phi(I)$ is a subgroup of S under addition.

Now note that since ϕ is surjective, $\forall s \in S \exists r \in R$ such that $s = \phi(r)$. Then since $a \in I$, $r \in R$ and I is an ideal of R we have $ra \in I$. So,

$$s \cdot \phi(a) = \phi(r)\phi(a) = \phi(ra) \in \phi(I)$$

Hence, $\phi(I)$ is closed under multiplication by elements of S and thus is an ideal of S .

(e) Assume $\ker(\phi) \subset I$.

$$I \subset \phi^{-1}(\phi(I))$$

Since $\phi^{-1}(\phi(I))$ is the pre-image of $\phi(I)$, this containment is straightforward.

$$\phi^{-1}(\phi(I)) \subseteq I$$

Let $x \in \phi^{-1}(\phi(I))$. That is $\phi(x) = \phi(a)$ for some $a \in I$.

$$\phi(x - a) = 0 \implies x - a \in \ker(\phi) \subset I$$

Since I is closed under addition $x \in I$. Hence, $\phi^{-1}(\phi(I)) \subseteq I$

(f) Yes. As a counterexample, consider the quotient map $\phi : \mathbb{Z} \rightarrow \mathbb{Z}/\langle 3 \rangle$ and the ideal generated by 2, $I = \langle 2 \rangle$. Note that $0 \in \phi(I) \implies \ker(\phi) = 3\mathbb{Z} \subset \phi^{-1}(\phi(I))$. That is $3 \in \phi^{-1}(\phi(I))$ but $3 \notin I$.

Problem 1. Operations on Ideals.

Throughout this problem let I and J be ideals in a ring R .

- (a) Prove or disprove that $I \cap J$ is an ideal.
- (b) Prove or disprove that $I \cup J$ is an ideal.
- (c) Prove or disprove that the product of ideals, defined below, is an ideal:

$$IJ := \{a_1b_1 + \cdots + a_tb_t \mid a_i \in I, b_i \in J, \text{ for all } t\}.$$

- (d) Prove or disprove that the sum of ideals, defined below, is an ideal:

$$I + J := \{a + b \mid a \in I, b \in J\}.$$

- (e) Prove or disprove that the radical of an ideal, defined below, is an ideal:

$$\sqrt{I} := \{a \in R \mid a^k \in I \text{ for some } k \in \mathbb{Z}_{\geq 1}\}.$$

Solution.

- (a) $I \cap J$ is an ideal.

Let $a, b \in I \cap J$, $r \in R$.

- (i) $I \cap J$ is a subgroup because intersection of subgroups is a subgroup.
- (ii) Note that $a, b \in I, J$. Since I and J are closed under addition and inverses, $a - b \in I, J$. Hence $a - b \in I \cap J$.
- (iii) $r \cdot a \in I \cap J$ since $ra \in I$ and $ra \in J$.

Hence $I \cap J$ is an additive subgroup of R and is closed under multiplication from elements of R .

- (b) $I \cup J$ is not an ideal.

Consider the ideals $I = \langle 2 \rangle$, $J = \langle 3 \rangle$ in the ring $R = \mathbb{Z}$. Note that $5 \notin I \cup J$ even though $2, 3 \in I \cup J$. Hence, $I \cup J$ is not closed under addition and is not an ideal.

- (c) IJ is an ideal.

Let $s \in IJ$, $r \in R$.

- (i) By definition, IJ includes all finite sums of elements in I, J . So it contains 0 and is closed under addition and inverses. Hence it is an additive subgroup.
- (ii) Consider $r \cdot s \in IJ$.

$$r \cdot s = r(a_1 b_1 + \cdots + a_t b_t) = r \cdot a_1 b_1 + \cdots + r \cdot a_t b_t$$

Note that $r \cdot a_i \in I$ for all i and thus $rs \in IJ$ for all r .

Hence IJ is an additive subgroup of R and is closed under multiplication from elements of R .

- (d) $I + J$ is an ideal.

Let $p = a_1 + b_1, q = a_2 + b_2 \in I + J, r \in R$.

- (i) Note that $0 \in I + J$ since $0 \in I, J$.
- (ii) Consider $p - q$.

$$p - q = a_1 + b_1 - (a_2 + b_2) = (a_1 - a_2) + (b_1 - b_2)$$

Note that $(a_1 - a_2) = a$ for some $a \in I$ and $(b_1 - b_2) = b$ for some $b \in I$. Hence, $I + J$ contains zero and is closed under addition and inverses and thus is an additive subgroup.

- (iii) Consider $r \cdot p$.

$$r \cdot p = r \cdot (a_1 + b_1) = ra_1 + rb_1$$

Note that $ra_1 \in I$ and $rb_1 \in J$ by definition of ideals. Hence $rp \in I + J$.

Thus $I + J$ is an additive subgroup of R and is closed under multiplication from elements of R .

- (e) $\sqrt{I}$ is an ideal.

Let $a^n, b^m \in I$, and $a, b, r \in R$.

- (i) Note that $0 \in \sqrt{I}$ since $0^1 \in I$.
- (ii) Consider $(a - b)^{nm}$.

$$(a - b)^{nm} = \sum_{i=0}^{nm} \binom{nm}{i} (a)^i (-b)^{nm-i}$$

- If $i \geq n$, then $a^n \in I \implies \binom{nm}{i} (a)^i (-b)^{nm-i} \in I$.
- If $i < n$, then $nm - i > nm - n = n(m - 1) = nm - i \geq m$. And $b^m \in I \implies \binom{nm}{i} (a)^i (-b)^{nm-i} \in I$.

Since every term in the sum lies in I , the whole sum lies in I . So $a - b \in \sqrt{I}$.

- (iii) Note that if $a^n \in I$ then $(ra)^n \in I$. Hence $ra \in \sqrt{I}$.

Thus $\sqrt{I}$ is an additive subgroup of R and is closed under multiplication from elements of R .

Problem 2. Let $\mathbb{K}$ be a field. The rings $\mathbb{Z}$, $\mathbb{K}[x]$, and $\mathbb{K}[[t]]$ are all principal ideal domains (PIDs) meaning domains where every ideal is generated by a single element. If $I = \langle a \rangle$ and $J = \langle b \rangle$ give, with proof, a careful concrete description of generators for IJ , $I \cap J$, $I + J$, and $\sqrt{I}$ in each of these rings.

Solution.

Let $I = \langle a \rangle$, $J = \langle b \rangle$. Then, we have

(a) $\mathbb{Z}$

- (i) $IJ = \langle ab \rangle$
- (ii) $I \cap J = \langle \text{lcm}(a, b) \rangle$
- (iii) $I + J = \langle \text{gcd}(a, b) \rangle$
- (iv) Let $a = p_1^{e_1} \cdots p_k^{e_k}$, then $\sqrt{I} = \langle p_1 \cdots p_k \rangle$, where p_i is a prime element.

(b) $\mathbb{K}[x]$

- (i) $IJ = \langle ab \rangle$
- (ii) $I \cap J = \langle \text{lcm}(a, b) \rangle$
- (iii) $I + J = \langle \text{gcd}(a, b) \rangle$
- (iv) Let $a = p_1^{e_1} \cdots p_k^{e_k}$, then $\sqrt{I} = \langle p_1 \cdots p_k \rangle$, where p_i is an irreducible element.

(c) $\mathbb{K}[[t]]$ Note that elements of $\mathbb{K}[[t]]$ are of two forms: $a \in \mathbb{K}$ or $a \cdot t^n$, $a \in \mathbb{K}$. So all proper ideals are of the form $\langle 0 \rangle$ or $\langle t^n \rangle$. Let $I = \langle t^n \rangle$, $J = \langle t^m \rangle$. If I is trivial, then $IJ = I \cap J = \sqrt{I} = \langle 0 \rangle$, and $I + J = J$. Assuming neither I nor J are trivial we have,

- (i) $IJ = \langle t^{n+m} \rangle$
- (ii) $I \cap J = \langle t^{\max(n, m)} \rangle$
- (iii) $I + J = \langle t^{\min(n, m)} \rangle$
- (iv) $\sqrt{I} = \langle t \rangle$.

Universal Properties. Throughout this course we will see numerous constructions and operations (e.g., quotients, tensor products, localizations, completions, etc.) each of these will have an explicit construction where we describe the objects precisely as sets. However, there is an alternative perspective, which is just as useful, describing these objects via their *universal properties*, which roughly asks what makes each object special amongst all other such objects. The standard framework for a universal property is, “Object X is unique up to isomorphism among all objects having property P.”

Problem 3. Universal Property of Quotients.

Let R be a ring and $I \subseteq R$ an ideal. Let $\pi: R \rightarrow R/I$ be the quotient map. Suppose $\varphi: R \rightarrow S$ is a ring homomorphism such that $I \subseteq \ker(\varphi)$. Define a function

$$\tilde{\varphi}: R/I \rightarrow S \quad \text{by} \quad \tilde{\varphi}(r + I) = \varphi(r).$$

- (a) Prove that $\tilde{\varphi}$ is well-defined.
- (b) Prove that $\tilde{\varphi}$ is a ring homomorphism.
- (c) Prove that $\tilde{\varphi} \circ \pi = \varphi$.
- (d) Prove that $\tilde{\varphi}$ is the unique ring homomorphism $R/I \rightarrow S$ with the property that

$$\tilde{\varphi} \circ \pi = \varphi.$$

- (e) Explain in words what this says about the quotient ring R/I .

Solution.

- (a) Let $r + I = r' + I \in R/I$. Then, $r - r' \in I$.

$$\tilde{\varphi}(r + I) = \varphi(r) = \varphi(r + (r - r')) = \varphi(r') = \tilde{\varphi}(r' + I)$$

Hence, $\tilde{\varphi}$ is well-defined.

- (b) Let $\bar{a}, \bar{b} \in R/I$, we check if the three conditions hold.

Unital: Since φ is a ring homomorphism, we have

$$\tilde{\varphi}(1_{R/I}) = \varphi(1_R) = 1_S$$

Addition: Again, since φ is a ring homomorphism, we have

$$\tilde{\varphi}(\bar{a}) + \tilde{\varphi}(\bar{b}) = \varphi(a) + \varphi(b) = \varphi(a + b) = \tilde{\varphi}(\overline{a + b})$$

Multiplication: Again, φ is a ring homomorphism, we have

$$\tilde{\varphi}(\overline{a}) \cdot \tilde{\varphi}(\overline{b}) = \varphi(a) \cdot \varphi(b) = \varphi(a \cdot b) = \tilde{\varphi}(\overline{a \cdot b})$$

Hence, $\tilde{\varphi}$ is a ring homomorphism.

(c) Let $a \in R$. Then,

$$\tilde{\varphi} \circ \pi(a) = \tilde{\varphi}(\overline{a}) = \varphi(a)$$

Since a was an arbitrary element in R , this holds for all elements, and hence $\tilde{\varphi} \circ \pi = \varphi$.

(d) Assume there exists some other homomorphism $\sigma : R/I \rightarrow S$ such that $\sigma \circ \pi = \varphi$. Then for any $\overline{a} \in R/I$,

$$\sigma(\overline{a}) = \sigma \circ \pi(a) = \varphi(a) = \tilde{\varphi}(\overline{a})$$

(e) Every homomorphism φ such that $I \subseteq \ker(\varphi)$ can be written as $\tilde{\varphi} \circ \pi$ for some $\tilde{\varphi}$.

Problem 4. In this problem, we wish to consider the evaluation map:

$$\text{ev}_i: \mathbb{Z}[x] \rightarrow \mathbb{C}, \quad f(x) \mapsto f(i)$$

- (a) Prove that ev_i is a ring homomorphism.
- (b) Show that $\langle x^2 + 1 \rangle \subset \ker(\text{ev}_i)$.
- (c) Use Problem 3 to produce a ring homomorphism

$$\widetilde{\text{ev}}_i: \mathbb{Z}[x]/\langle x^2 + 1 \rangle \rightarrow \mathbb{C}.$$

- (d) What does $\widetilde{\text{ev}}_i$ do to the class of x in $\mathbb{Z}[x]/\langle x^2 + 1 \rangle$?
- (e) Prove that $\mathbb{Z}[x]/\langle x^2 + 1 \rangle$ is isomorphic to $\mathbb{Z}[i]$. Is $\langle x^2 + 1 \rangle$ prime and/or maximal?

Solution.

- (a) Let $f, g \in \mathbb{Z}[x]$, we check if the three conditions hold.

Unital: Let f_1 be the constant function 1. Then,

$$\text{ev}_i(f_1(x)) = f_1(i) = 1$$

Addition: Since $\mathbb{Z}[x]$ is a ring under pointwise addition and multiplication, we can write,

$$\text{ev}_i(f(x)) + \text{ev}_i(g(x)) = f(i) + g(i) = (f + g)(i) = \text{ev}_i((f + g)(x))$$

Multiplication: Since $\mathbb{Z}[x]$ is a ring under pointwise addition and multiplication, we can write,

$$\text{ev}_i(f(x)) \cdot \text{ev}_i(g(x)) = f(i) \cdot g(i) = (f \cdot g)(i) = \text{ev}_i((f \cdot g)(x))$$

Hence, ev_i is a ring homomorphism.

- (b) Note that $\text{ev}_i(x^2 + 1) = i^2 + 1 = -1 + 1 = 0$. For any $f(x)(x^2 + 1) \in \langle x^2 + 1 \rangle$ we have $\text{ev}_i(f(x)(x^2 + 1)) = f(i) \cdot 0 = 0 \in \ker(\text{ev}_i)$. Thus the ideal generated by $\langle x^2 + 1 \rangle$ under ev_i maps to 0 and hence is a subset of the kernel.

- (c) From Q3 we know that if $\langle x^2 + 1 \rangle \subseteq \ker(\text{ev}_i)$ there exists a unique homomorphism,

$$\widetilde{\text{ev}}_i: \mathbb{Z}[x]/\langle x^2 + 1 \rangle \rightarrow \mathbb{C}, \quad \widetilde{\text{ev}}_i(\overline{f(x)}) = f(i)$$

- (d) $\widetilde{\text{ev}}_i(\overline{x}) = x(i) = i$.

- (e) Define a map $\varphi : \mathbb{Z}[x] \rightarrow \mathbb{Z}[i]$, $\varphi(f(x)) \rightarrow f(i)$. We will show that $\ker(\varphi) = \langle x^2 + 1 \rangle$.
 $\langle x^2 + 1 \rangle \subseteq \ker(\varphi)$

Note,

$$\varphi(x^2 + 1) = i^2 + 1 = 0 \in \ker(\varphi)$$

$$\ker(\varphi) \subseteq \langle x^2 + 1 \rangle$$

Let $\varphi(f(x)) \in \ker(\varphi)$. If $f(i) = 0$, then $x^2 + 1 \mid f(x)$ in $\mathbb{Z}[x]$ (since $x^2 + 1$ is the minimal polynomial of i over $\mathbb{Q}$ and hence is primitive over $\mathbb{Z}$), and we have,

$$f(x) \in \langle x^2 + 1 \rangle \implies \ker(\varphi) \subseteq \langle x^2 + 1 \rangle$$

Then by the first isomorphism theorem,

$$\mathbb{Z}[x]/\langle x^2 + 1 \rangle \cong \text{im}(\varphi) = \mathbb{Z}[i]$$

Since $\mathbb{Z}[x]$ is an integral domain but not a field, $\langle x^2 + 1 \rangle$ is prime but not maximal.

Problem 5. Universal Property of Polynomial Rings.

The goal of this exercise is to prove the following characterization of polynomial rings. Let $\varphi: R \rightarrow S$ be a ring homomorphism. For any element $s \in S$ there is a unique ring homomorphism $\psi: R[x] \rightarrow S$ such that $\psi(x) = s$ and the following diagram commutes:

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S \\ \downarrow & \nearrow \psi & \\ R[x] & & \end{array} .$$

- (a) Prove there is at most one ring homomorphism $\psi: R[x] \rightarrow S$ with the above properties.
- (b) Construct such a homomorphism explicitly by setting

$$\psi\left(\sum_{i=0}^n a_i x^i\right) = \sum_{i=0}^n \varphi(a_i) s^i.$$

- (c) Conclude that to give a ring homomorphism $R[x] \rightarrow S$ is equivalent to giving:
- a ring homomorphism $R \rightarrow S$, and
 - an element of S .

Solution.

- (a) Suppose there exists another homomorphism $\sigma: R[x] \rightarrow S$ such that $\sigma(x) = \psi(x) = s$. Also, note that since the diagram commutes $\psi|_R = \sigma|_R = \varphi$.

Then for any $\sum_{i=0}^n a_i x^i \in R[x]$ we have,

$$\begin{aligned} \sigma\left(\sum_{i=0}^n a_i x^i\right) &= \sigma(a_1) \cdot \sigma(x) + \cdots + \sigma(a_n) \cdot \sigma(x^n) \\ &= \varphi(a_1) \cdot \psi(x) + \cdots + \varphi(a_n) \cdot \psi(x^n) \\ &= \psi(a_1) \cdot \psi(x) + \cdots + \psi(a_n) \cdot \psi(x^n) \\ &= \psi\left(\sum_{i=0}^n a_i x^i\right) \end{aligned}$$

Hence, $\sigma = \psi$.

(b) Let $\psi : R[x] \rightarrow S$ such that

$$\psi(x) = s, \quad \psi(a) = \varphi(a) \quad \forall a \in R$$

(c) Hence for any ring homomorphism $\sigma : R[x] \rightarrow S$ we have

$$\begin{array}{ccc} R & \xrightarrow{\varphi} & S \\ \downarrow & \nearrow \psi & \\ R[x] & & \end{array}$$

Problem 6. Evaluation maps.

Let R be a ring and let $a \in R$. Define the evaluation map

$$\text{ev}_a: R[x] \rightarrow R, \quad f(x) \mapsto f(a).$$

- (a) Prove that ev_a is a surjective ring homomorphism.
- (b) Prove that $\ker(\text{ev}_a) = \langle x - a \rangle$.
- (c) Use Problem 3 to show that ev_a factors through a ring homomorphism

$$R[x]/\langle x - a \rangle \rightarrow R.$$

- (d) Explain what this means if R is a field or R is an integral domain.
- (e) Give an example of a ring R and prime/maximal ideal $I \subset R[x]$ such that $I \neq \ker(\text{ev}_a)$ for all $a \in R$.

Solution.

- (a) For all $r \in R$ consider the constant functions r represented by $f_r \in R[x]$. Then,

$$\text{ev}_a(f_r(x)) = f_r(a) = r$$

- (b) $\ker(\text{ev}_a) \subseteq \langle x - a \rangle$.

Note that for any $p(x) \in \ker(\text{ev}_a)$ we have

$$p(x) = (x - a)q(x) + r(x)$$

for some $q(x), r(x) \in R[x]$. Evaluating at ev_a we have,

$$\text{ev}_a(p(x)) = (a - a)q(a) + r(a) = 0 \implies r(a) = 0$$

That is

$$p(x) = (x - a)q(x) \implies p(x) \in \langle x - a \rangle$$

$$\langle x - a \rangle \subseteq \ker(\text{ev}_a).$$

Let $p \in \langle x - a \rangle$. Since $\langle x - a \rangle$ irreducible we have $p = f(x)(x - a)$, for some $f(x) \in R[x]$,

$$\text{ev}_a(p) = \text{ev}_a(f(x)(x - a)) = f(a)(a - a) = 0 \in \ker(\text{ev}_a)$$

Hence, $\ker(\text{ev}_a) = \langle x - a \rangle$.

- (c) Since $\ker(\text{ev}_a) \supseteq \langle x - a \rangle$ from problem 3 we know we have an induced ring homomorphism $\widetilde{\text{ev}}_a$ such that

$$\begin{array}{ccc} \mathbf{R}[x] & \xrightarrow{\pi} & \mathbf{R}[x]/\langle x - a \rangle \\ \text{ev}_a \downarrow & \swarrow \widetilde{\text{ev}}_a & \\ \mathbf{R} & & \end{array}$$

- (d) If $\mathbf{R}$ is a domain this means $\langle x - a \rangle$ is prime and also that $\widetilde{\text{ev}}_a = \text{id}$.

If $\mathbf{R}$ is a field $\langle x - a \rangle$ is maximal and $\mathbf{R} \cong \mathbf{R}[x]/\langle x - a \rangle$. Also since all domains are fields we have $\widetilde{\text{ev}}_a = \text{id}$

- (e) $\mathbf{R} = \mathbb{C}, I = \langle 0 \rangle$.

Worksheet 1.3: Gröbner Bases Pt. I

Polynomial Division. Recall the following classical theorem:

Theorem 0.1. *Let $g \in \mathbb{K}[x]$ be a non-zero polynomial. Every polynomial $f \in \mathbb{K}[x]$ can be written uniquely as $f = qg + r$ where $q, r \in \mathbb{K}[x]$, and either $r = 0$ or $\deg(r) < \deg(g)$.*

While this theorem might seem innocuous when first learned as a teenager, it turns out to be very powerful when studying ideals and polynomials in $\mathbb{K}[x]$. The proof of this theorem also provides an algorithm for computing q and r , which when presented in pseudocode looks something like

Algorithm 1 Division Algorithm

Require: g, f

Ensure: q, r

$q \leftarrow 0; r \leftarrow f$

while $r \neq 0$ **and** $\text{LT}(g)$ divides $\text{LT}(r)$ **do**

$q \leftarrow q + \text{LT}(r)/\text{LT}(g)$

$r \leftarrow r - (\text{LT}(r)/\text{LT}(g))g$

end while

The goal of this worksheet is to begin exploring how we might extend this theorem to polynomial rings in more variables. Along the way we will see how these explorations will lead to both computational tools for studying the commutative algebra of polynomial rings over fields, but also gives us useful tools for proving theorems in this setting as well.

Problem 1. Use Theorem 1 to prove the following:

- (a) If $f \in \mathbb{K}[x]$ is a non-zero polynomial then f has at most $\deg(f)$ roots in $\mathbb{K}$.
- (b) If $g \in \mathbb{K}[x]$ is a non-zero polynomial then $f \in \langle g \rangle$ if and only if $r = 0$.

Solution.

- (a) Assume for the sake of contradiction that f has more than $n = \deg(f)$ roots, say m roots: $a_1, \dots, a_m$. Then by Theorem 1, dividing by $(x - a_i)$ for all $1 \leq i \leq m$ we get $f = c(x - a_1) \cdots (x - a_m)$, for some constant c and $r = 0$. But then we have m x terms, i.e., x^m is in f , but since $\deg(f) < m$ this is not possible. Hence a contradiction.

- (b) ($\implies$)

Assume $f \in \langle g \rangle$. Then there exists some $q \in \mathbb{K}[x]$ such that $f = qg$. Then,

$$f = qg + r \implies f - qg = r = 0$$

- ($\impliedby$)

Assume $r = 0$. Then by Theorem 1 we have,

$$f = qg + 0 = qg$$

That is there exists some $q \in \mathbb{K}[x]$ such that $f = qg$ and by definition of an ideal, $f \in \langle g \rangle$.

Problem 2. Implement the algorithm above in Macaulay2, and use your code to compute $\mathbf{q}$ and $\mathbf{r}$ for the following polynomials.

(a) $f = x^3 + x + 1, \quad g = x + 1.$

(d) $f = x^{100} - 1, \quad g = x^2 - 1.$

(b) $f = x^5 - 1, \quad g = x - 1.$

(e) $f = x^{1000} - 1, \quad g = x - 1.$

(c) $f = x^2 + 1, \quad g = x^3 + x.$

(f) $f = \sum_{i=0}^{500} x^i, \quad g = x^2 + x + 1.$

Solution.

(a) $(x^2 - x + 2, -1)$

(d) $\left(\sum_{k=0}^{49} x^{2k}, 0\right)$

(b) $(x^4 + x^3 + x^2 + 1, 0)$

(e) $\left(\sum_{k=0}^{99} x^k, 0\right)$

(c) $(0, x^2 + 1)$

(f) $\left(\sum_{k=0}^{166} x^{3k}, 0\right)$

Definition 0.2. A monomial ordering on $\mathbb{K}[x_1, \dots, x_n]$ is a relation $<$ on $\mathbb{Z}_{\geq 0}^n$ satisfying the following:

- (i) $<$ is a total ordering on $\mathbb{Z}_{\geq 0}^n$,
- (ii) if $\alpha < \beta$ and $\gamma \in \mathbb{Z}_{\geq 0}^n$ then $\alpha + \gamma < \beta + \gamma$, and
- (iii) $<$ is a well-ordering on $\mathbb{Z}_{\geq 0}^n$.

Recall that a *total ordering* means that of any two elements $\alpha, \beta \in \mathbb{Z}_{\geq 0}^n$ exactly one of the following statements is true: $\alpha < \beta$, $\alpha = \beta$, or $\beta < \alpha$. That $<$ is a well-ordering on $\mathbb{Z}_{\geq 0}^n$ means that every non-empty subset of $\mathbb{Z}_{\geq 0}^n$ has a smallest element with respect to $<$.

Once a monomial ordering has been fixed, every non-zero polynomial acquires a distinguished piece of data: its leading monomial, leading coefficient, and leading term. These are the ingredients that drive the multivariable division process just as degree and leading coefficient drive ordinary polynomial division.

Problem 3. Prove that a monomial ordering $<$ on $\mathbb{Z}_{\geq 0}^n$ is equivalent to a relation $<$ on the set of monomials in $\mathbb{K}[x_1, \dots, x_n]$ satisfying analogous conditions to (i), (ii), and (iii). Explain why this justifies the moniker “monomial ordering”.

Solution.

There exists a bijection

$$\mathbb{Z}_{\geq 0}^n \ni \{\alpha = (a_1, \dots, a_n)\} \leftrightarrow \{x^\alpha = x_1^{a_1} \cdots x_n^{a_n}\} \in \mathbb{K}[x_1, \dots, x_n]$$

that satisfies all the conditions because addition of vectors corresponds to addition of exponents during multiplication.

Problem 4. Prove or disprove that the following are monomial orderings:

(a) Lexicographic (Lex) Order: Given $\alpha = (\alpha_1, \dots, \alpha_n)$ and $\beta = (\beta_1, \dots, \beta_n)$ in $\mathbb{Z}_{\geq 0}^n$, we say $\beta <_{\text{lex}} \alpha$ if and only if the left-most non-zero entry of $\alpha - \beta$ is positive.

(b) Graded Lex Order: $\beta <_{\text{grlex}} \alpha$ if and only if

$$|\beta| = \sum_{i=1}^n \beta_i < |\alpha| = \sum_{i=1}^n \alpha_i$$

or $|\beta| = |\alpha|$ and $\beta <_{\text{lex}} \alpha$.

(c) Reverse Lex (RevLex) Order: $\beta <_{\text{revlex}} \alpha$ if and only if the right-most non-zero entry of $\alpha - \beta$ is negative.

(d) Graded RevLex Order: $\beta <_{\text{grevlex}} \alpha$ if and only if

$$|\beta| = \sum_{i=1}^n \beta_i < |\alpha| = \sum_{i=1}^n \alpha_i$$

or $|\beta| = |\alpha|$ and $\beta <_{\text{revlex}} \alpha$.

(e) Weight Order: Fix $w = (w_1, \dots, w_n) \in \mathbb{Z}^n$. Define $\beta <_w \alpha$ if and only if

$$\beta \cdot w = \sum_{i=1}^n \beta_i w_i < \alpha \cdot w = \sum_{i=1}^n \alpha_i w_i.$$

Solution.

(a) It is a monomial ordering. Let $\alpha = (\alpha_1, \dots, \alpha_n)$, $\beta = (\beta_1, \dots, \beta_n) \in \mathbb{Z}_{\geq 0}^n$.

(i) Total Ordering:

If $\alpha = \beta$ we're done. Otherwise, $\alpha - \beta$ has a leftmost non-zero entry. If this entry is positive, $\beta <_{\text{lex}} \alpha$, else $\alpha <_{\text{lex}} \beta$.

(ii) Let $\alpha <_{\text{lex}} \beta$, that is the leftmost non-zero entry of $\alpha - \beta$ is negative. Then for all $\gamma \in \mathbb{Z}_{\geq 0}^n$ we have,

$$(\alpha + \gamma) - (\beta + \gamma) = \alpha - \beta$$

That is the leftmost non-zero entry remains unchanged and hence $(\alpha + \gamma) <_{\text{lex}} (\beta + \gamma)$.

(iii) Let $S \subseteq \mathbb{Z}_{\geq 0}^n$ be non-empty. Since $\mathbb{Z}_{\geq 0}$ is well-ordered we let $m_1 = \min\{\alpha_1 \mid \alpha \in S\}$. Then, among the elements with first coordinate m_1 we let $m_2 = \min\{\alpha_2 \mid \alpha \in S\}$

and so on for all n coordinates. This gives us,

$$\mathbf{m} = (m_1, \dots, m_n) \in S$$

which is the lex-minimal element.

(b) It is a monomial ordering. Let $\alpha = (\alpha_1, \dots, \alpha_n), \beta = (\beta_1, \dots, \beta_n) \in \mathbb{Z}_{\geq 0}^n$.

(i) Total Ordering:

If $\alpha = \beta$ we're done. Else if $|\alpha| = |\beta|$, by part (a) we have an ordering. Otherwise, $|\alpha| \neq |\beta|$ has a leftmost non-zero entry. If $|\alpha| > |\beta|$, $\beta <_{\text{grlex}} \alpha$, else $\alpha <_{\text{grlex}} \beta$.

(ii) Let $\alpha <_{\text{grlex}} \beta$, and $\gamma \in \mathbb{Z}_{\geq 0}^n$.

- If $|\alpha| < |\beta|$, we have

$$|\alpha| + |\gamma| < |\beta| + |\gamma| \implies |\alpha + \gamma| < |\beta + \gamma|$$

- If $|\alpha| = |\beta|$ and $\alpha <_{\text{lex}} \beta$ we have an ordering by part (a).

(iii) Let $S \subseteq \mathbb{Z}_{\geq 0}^n$ be non-empty. Since all entries are non-negative there exists $\mathbf{m} = \min\{|\alpha| \mid \alpha \in S\}$. Then, we take the lex-minimal element in the subset $H \subseteq S$, $H = \{|\alpha| = \mathbf{m} \mid \alpha \in S\}$. This is the grlex-minimal element.

(c) It is a monomial ordering. Let $\alpha = (\alpha_1, \dots, \alpha_n), \beta = (\beta_1, \dots, \beta_n) \in \mathbb{Z}_{\geq 0}^n$.

(i) Total Ordering:

If $\alpha = \beta$ we're done. Otherwise, $\alpha - \beta$ has a rightmost non-zero entry. If this entry is negative, $\beta <_{\text{revlex}} \alpha$, else $\alpha <_{\text{revlex}} \beta$.

(ii) Let $\alpha <_{\text{revlex}} \beta$, that is the leftmost non-zero entry of $\alpha - \beta$ is positive. Then for all $\gamma \in \mathbb{Z}_{\geq 0}^n$ we have,

$$(\alpha + \gamma) - (\beta + \gamma) = \alpha - \beta$$

That is the rightmost non-zero entry remains unchanged and hence $(\alpha + \gamma) <_{\text{revlex}} (\beta + \gamma)$.

(iii) Let $S \subseteq \mathbb{Z}_{\geq 0}^n$ be non-empty. Since $\mathbb{Z}_{\geq 0}$ is well-ordered we let $m_n = \min\{\alpha_n \mid \alpha \in S\}$. Then, among the elements with first coordinate m_n we let $m_{n-1} = \min\{\alpha_{n-1} \mid \alpha \in S\}$ and so on for all n coordinates. This gives us,

$$\mathbf{m} = (m_1, \dots, m_n) \in S$$

which is the revlex-minimal element.

(d) It is a monomial ordering by (b) and (c).

(e) Not a monomial ordering. For instance, if $w = (1, 1), \alpha = (0, 2), \beta = (2, 0)$ we have $\beta \cdot w = \alpha \cdot w$ but $\alpha \neq \beta$.

Problem 5. Leading Terms.

Fix a monomial ordering $<$ on $\mathbb{Z}_{\geq 0}^n$. Let $f = \sum_{\alpha} c_{\alpha} x^{\alpha} \in K[x_1, \dots, x_n]$ be a non-zero polynomial. Define the leading monomial $\text{LM}(f)$, leading coefficient $\text{LC}(f)$, and leading term $\text{LT}(f)$.

- (a) Compute $\text{LM}(f)$, $\text{LC}(f)$, and $\text{LT}(f)$ for

$$f = x^3 + 3x^2y + 5xy^3 - 7z^2 \in K[x, y, z]$$

with respect to Lex, Graded Lex, and Graded RevLex (assume $x > y > z$).

- (b) Prove that if $f, g \in K[x_1, \dots, x_n]$ are non-zero, then

$$\text{LM}(fg) = \text{LM}(f)\text{LM}(g), \quad \text{LT}(fg) = \text{LT}(f)\text{LT}(g).$$

- (c) Explain which property of a monomial ordering is used in part (b).

Solution.

- (a) • Lex:

$$\text{LM}(f) = x^3, \quad \text{LC}(f) = 1, \quad \text{LT}(f) = x^3$$

- Graded Lex:

$$\text{LM}(f) = xy^3, \quad \text{LC}(f) = 5, \quad \text{LT}(f) = 5xy^3$$

- Graded RevLex:

$$\text{LM}(f) = xy^3, \quad \text{LC}(f) = 5, \quad \text{LT}(f) = 5xy^3$$

- (b) Let $f = \text{LT}(f) + \sum_{\alpha < \text{LM}(f)} c_{\alpha} x^{\alpha}$, $g = \text{LT}(g) + \sum_{\beta < \text{LM}(g)} d_{\beta} x^{\beta} \in K[x_1, \dots, x_n]$ be non-zero. Then the product fg is,

$$fg = \sum_{\alpha, \beta} c_{\alpha} d_{\beta} x^{\alpha+\beta}$$

Note that $\text{LT}(f)\text{LT}(g) = c_0 d_0 x^{\alpha_0+\beta_0}$ appears in fg . We know for all α s and β s, $\alpha < \alpha_0$ and $\beta < \beta_0$ so

$$\alpha + \beta < \alpha_0 + \beta_0$$

Hence $x^{\alpha+\beta} < x^{\alpha_0+\beta_0}$, and $x^{\alpha+\beta} = \text{LM}(f)\text{LM}(g) = \text{LM}(fg)$. Moreover $\text{LT}(fg) = \text{LC}(fg)\text{LM}(fg)$, where $\text{LC}(fg) = \text{LC}(f)\text{LC}(g)$ by definition, and we showed $\text{LM}(fg) = \text{LM}(f)\text{LM}(g)$. Hence $\text{LT}(fg) = \text{LT}(f)\text{LT}(g)$.

- (c) (ii) property is used.

Problem 6. Let $G = (g_1, \dots, g_s)$ be an ordered list of non-zero polynomials in $K[x_1, \dots, x_n]$.

- (a) Write pseudocode for a division algorithm which takes as input f and G and returns polynomials $q_1, \dots, q_s, r$ such that

$$f = q_1 g_1 + \dots + q_s g_s + r,$$

where no monomial in r is divisible by any $\text{LM}(g_1), \dots, \text{LM}(g_s)$.

- (b) In the algorithm above, what should happen if the leading monomial of the current polynomial is not divisible by any $\text{LM}(g_i)$?
- (c) Prove that your algorithm terminates. What plays the role of “degree” in the proof?

Solution.

(a) **Require:** $f, g_1, \dots, g_s$
Ensure: $q_1, \dots, q_s, r$
 $q_1, \dots, q_s \leftarrow 0; r \leftarrow 0; p \leftarrow f$
while $p \neq 0$ **do**
 \leftarrow 0
 for $i = 1, \dots, s$ **do**
 if $\text{LT}(g_i)$ divides $\text{LT}(p)$ **then**
 $q_i \leftarrow q_i + \text{LT}(p)/\text{LT}(g_i)$
 $p \leftarrow p - (\text{LT}(p)/\text{LT}(g_i))g_i$
 divides $\leftarrow 1$
 break
 end if
 end for
 if divides $= 0$ **then**
 $r \leftarrow r + \text{LT}(p)$
 $p \leftarrow p - \text{LT}(p)$
 end if
end while

- (b) We add it to the remainder term r , and continue.

- (c) $\text{LM}(f)$ plays the role of degree. At every step, we update either $p \rightarrow p - \text{LT}(p)$ (if no g_i divides $\text{LM}(p)$) or $p \rightarrow \text{LT}(p)/\text{LT}(g_i) \cdot g_i$ (for some g_i dividing $\text{LM}(p)$). So the leading term strictly decreases. Hence, the program must terminate.

Problem 7. Work in $K[x, y]$ with Lex order and $x > y$. Let $f = xy^2 - x$, $g_1 = xy - 1$, and $g_2 = y^2 - 1$.

- (a) Divide f by the ordered list (g_1, g_2) .
- (b) Divide f by the ordered list (g_2, g_1) .
- (c) Show directly that $f \in \langle g_1, g_2 \rangle$.
- (d) What do your computations show about division in several variables? In particular, is the remainder uniquely determined by the ideal $\langle g_1, g_2 \rangle$?

Solution.

(a) $q_1 = y, q_2 = 0, r = -x + y$

(b) $q_1 = 0, q_2 = x, r = 0$

(c) $x \in K[x, y]$ and $f = g_2 \cdot x$, hence $f \in \langle g_2 \rangle \subseteq \langle g_1, g_2 \rangle$.

(d) Remainders are not unique.

Worksheet 2.1: Zariski Topology

Problem 1. Recall that the spectrum and maximal spectrum of a ring $\mathbf{R}$ are the sets:

$$\mathrm{Spec}(\mathbf{R}) := \{ \mathbf{P} \subset \mathbf{R} \mid \mathbf{P} \text{ is a prime ideal} \}$$

and

$$\mathrm{mSpec}(\mathbf{R}) := \{ \mathbf{P} \subset \mathbf{R} \mid \mathbf{P} \text{ is a maximal ideal} \}$$

Given any subset $\mathcal{S} \subset \mathbf{R}$ define

$$\mathbb{V}(\mathcal{S}) := \{ \mathbf{P} \in \mathrm{Spec}(\mathbf{R}) \mid \mathcal{S} \subset \mathbf{P} \}.$$

- (a) Describe $\mathrm{Spec}(\mathbb{K})$ and $\mathrm{mSpec}(\mathbb{K})$ when $\mathbb{K}$ is a field.
- (b) Describe both $\mathrm{Spec}(\mathbb{K}[\mathbf{x}]/\langle \mathbf{x}^3 \rangle)$ and $\mathrm{mSpec}(\mathbb{K}[\mathbf{x}]/\langle \mathbf{x}^3 \rangle)$ when $\mathbb{K}$ is a field.
- (c) For an arbitrary ring $\mathbf{R}$, compute $\mathbb{V}(\{0\})$ and $\mathbb{V}(\{1\})$.
- (d) Explain the natural poset structure on $\mathrm{Spec}(\mathbf{R})$. Describe the poset for $\mathrm{Spec}(\mathbb{Z})$.

Solution.

- (a) Let $\mathbb{K}$ be a field. Then $\mathbb{K}$ only has 1 proper ideal: the ideal generated by 0 . Note that this ideal is prime because $\mathbb{K}$ is an integral domain. Moreover it is also maximal because $\mathbb{K}/\langle 0 \rangle = \mathbb{K}$ is a field. So,

$$\mathrm{Spec}(\mathbb{K}) = \mathrm{mSpec}(\mathbb{K}) = \{ \langle 0 \rangle \}$$

- (b) Note that the ideal generated by 0 is not prime in this ring because $\mathbf{x} \cdot \mathbf{x}^2 \in \langle 0 \rangle$ even though neither $\mathbf{x}$ nor $\mathbf{x}^2$ are in $\langle \mathbf{x} \rangle$. Then we get,

$$\mathrm{Spec}(\mathbb{K}[\mathbf{x}]/\langle \mathbf{x}^3 \rangle) = \mathrm{mSpec}(\mathbb{K}[\mathbf{x}]/\langle \mathbf{x}^3 \rangle) = \{ \langle \mathbf{x} \rangle \}$$

- (c) Let $\mathbf{R}$ be an arbitrary ring, then,

$$\mathbb{V}(\{0\}) = \mathrm{Spec}(\mathbf{R})$$

since every ideal contains 0 and,

$$\mathbb{V}(\{1\}) = \emptyset$$

since 1 is not contained in any proper ideal of $\mathbf{R}$.

- (d) The natural poset structure on $\mathrm{Spec}(\mathbf{R})$ is given by inclusion of ideals, that is, for $\mathbf{P}_1, \mathbf{P}_2 \in \mathrm{Spec}(\mathbf{R})$, we write $\mathbf{P}_1 \leq \mathbf{P}_2$ iff $\mathbf{P}_1 \subseteq \mathbf{P}_2$.

The poset for $\mathrm{Spec}(\mathbb{Z})$ contains prime ideals in $\mathbb{Z}$. The zero ideal is contained in every other ideal and every other prime ideal is maximal.

A *topological space* is a set X together with a collection of subsets τ , which satisfies the following conditions: i) $\emptyset \in \tau$ and $X \in \tau$, ii) τ is closed under arbitrary unions, and iii) τ is closed under finite intersections. We call τ a *topology* on X , the elements of τ the open subsets of X , and a subset $Z \subset X$ is closed if and only if $X \setminus Z$ is open (i.e., $X \setminus Z \in \tau$). If $S \subset X$ is a subset, the *subspace topology* on S is the topology where the open sets of S are all sets of the form $S \cap U \subset S$ where $U \subset X$ is open in X . A map $f : X \rightarrow Y$ of topological spaces is *continuous* if and only if $f^{-1}(U) \subset X$ is open for all open subsets $U \subset Y$, i.e., the preimage of an open set is open.

Problem 2. The Zariski Topology. We will now place a topology on the set $\text{Spec}(\mathbf{R})$, called the Zariski topology after Oscar Zariski.

- (a) Prove that if $\mathcal{S}, \mathcal{T} \subset \mathbf{R}$ are arbitrary sets then $\mathbb{V}(\mathcal{S}) \cap \mathbb{V}(\mathcal{T}) = \mathbb{V}(\mathcal{S} \cup \mathcal{T})$. Explain why your argument extends to an *arbitrary* intersection.
- (b) Prove that if $\mathcal{S}, \mathcal{T} \subset \mathbf{R}$ are arbitrary sets then $\mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T}) = \mathbb{V}(\mathcal{S} \cdot \mathcal{T})$ where

$$\mathcal{S} \cdot \mathcal{T} := \{st \mid s \in \mathcal{S}, t \in \mathcal{T}\}.$$

- (c) Prove that the sets of the form $\mathbb{V}(\mathcal{S})$ are the closed sets of a topology on $\text{Spec}(\mathbf{R})$. We call this topology the Zariski topology on $\text{Spec}(\mathbf{R})$. (Hint: Rephrase the definition of a topology in terms of closed sets instead of open sets.)
- (d) Take a break and Google Oscar Zariski and read about his life.

Solution.

- (a) We prove this for arbitrary intersections. Let $\mathcal{S}_i, i \in \Lambda$ where Λ is some indexing set, be arbitrary subsets of $\mathbf{R}$. Then by definition,

$$\bigcap_{i \in \Lambda} \mathbb{V}(\mathcal{S}_i) = \bigcap_{i \in \Lambda} \{[P] \in \text{Spec}(\mathbf{R}), \mathcal{S}_i \subseteq P\} = \left\{ [P] \in \text{Spec}(\mathbf{R}), \bigcup_{i \in \Lambda} \mathcal{S}_i \subseteq P \right\}$$

And

$$\mathbb{V} \left(\bigcup_{i \in \Lambda} \mathcal{S}_i \right) = \left\{ [P] \in \text{Spec}(\mathbf{R}), \bigcup_{i \in \Lambda} \mathcal{S}_i \subseteq P \right\}$$

That is, $\bigcap_{i \in \Lambda} \mathbb{V}(\mathcal{S}_i) = \mathbb{V} \left(\bigcup_{i \in \Lambda} \mathcal{S}_i \right)$. Hence proved.

(b) Let $\mathcal{S}, \mathcal{T}$ be arbitrary subsets of $\mathbf{R}$. Then by definition,

$$\mathbb{V}(\mathcal{S} \cdot \mathcal{T}) = \{[\mathbf{P}] \in \text{Spec}(\mathbf{R}), \mathcal{S} \cdot \mathcal{T} \subseteq \mathbf{P}\}$$

And

$$\mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T}) = \{[\mathbf{P}] \in \text{Spec}(\mathbf{R}), \mathcal{S} \subseteq \mathbf{P}\} \cup \{[\mathbf{P}] \in \text{Spec}(\mathbf{R}), \mathcal{T} \subseteq \mathbf{P}\}$$

Note that $\mathcal{S} \cdot \mathcal{T} \subseteq \mathcal{S}, \mathcal{T}$. Then the containment $\mathbb{V}(\mathcal{S} \cdot \mathcal{T}) \subseteq \mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T})$ is obvious.

To prove that $\mathbb{V}(\mathcal{S} \cdot \mathcal{T}) \supseteq \mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T})$, it suffices to show that if $\mathcal{S} \cdot \mathcal{T} \subseteq \mathbf{P}$ then either $\mathcal{S} \subseteq \mathbf{P}$ or $\mathcal{T} \subseteq \mathbf{P}$.

Assume $\mathcal{S} \cdot \mathcal{T} \subseteq \mathbf{P}$. Let $s \in \mathcal{S}$ and $t \in \mathcal{T}$. Then, since $\mathbf{P}$ is a prime ideal and $st \in \mathbf{P}$ then either $s \in \mathbf{P}$ or $t \in \mathbf{P}$. If both $s, t \in \mathbf{P}$ we are done. Else wlog, assume that $s \notin \mathbf{P}$. But note that $\forall t' \in \mathcal{T}, s \cdot t' \in \mathbf{P}$. Hence, $t' \in \mathbf{P}, \forall t' \in \mathcal{T}$, i.e. $\mathcal{T} \subseteq \mathbf{P}$.

Hence, $\mathbb{V}(\mathcal{S}) \cup \mathbb{V}(\mathcal{T}) = \mathbb{V}(\mathcal{S} \cdot \mathcal{T})$.

(c) To prove that the sets of the form $\mathbb{V}(\mathcal{S})$ are the closed sets of the Zariski topology on $\text{Spec}(\mathbf{R})$ we must show that (i) $\emptyset, \text{Spec}(\mathbf{R})$ are closed sets, (ii) the finite union of closed sets is closed, and (iii) arbitrary intersection of closed sets is closed.

(i) Note that $\emptyset = \{\mathbb{V}(\{1\})\}$ since 1 is not contained in any proper ideal of $\mathbf{R}$, and $\text{Spec}(\mathbf{R}) = \{\mathbb{V}(\{0\})\}$ since 0 is contained in every ideal and thus in every prime ideal. Both $\mathbb{V}(\{1\})$ and $\mathbb{V}(\{0\})$ are closed sets, hence $\emptyset$ and $\text{Spec}(\mathbf{R})$ are closed.

(ii) From part (b) we know that the finite union of two closed sets is closed, and by induction we can extend this to a finite number of closed sets. Hence, the finite union of closed sets is closed.

(iii) From part (a) we know that any arbitrary intersection of closed sets is closed.

Hence, sets of the form $\mathbb{V}(\mathcal{S})$ are the closed sets of the Zariski topology on $\text{Spec}(\mathbf{R})$.

Problem 3. Describe the closed subsets of $\text{mSpec}(\mathbf{R}) \subset \text{Spec}(\mathbf{R})$ in the subspace topology.

Solution.

Closed subsets of $\text{mSpec}(\mathbf{R})$ look like $\mathbb{V}(\mathcal{S}) \cap \text{mSpec}(\mathbf{R})$, for all $\mathbb{V}(\mathcal{S})$ in the Zariski Topology on $\text{Spec}(\mathbf{R})$. That is, the closed subsets of $\text{mSpec}(\mathbf{R})$ are maximal ideals $\mathbf{M}$ such that $\mathcal{S} \subseteq \mathbf{M}$.

Problem 4. Let $\mathbb{K}$ be an algebraically closed field. This exercise will describe the Zariski topology on $\text{Spec}(\mathbb{K}[x])$.

- (a) Describe the points of $\text{Spec}(\mathbb{K}[x])$. (Hint: There are two types.)
- (b) Prove that if $\mathcal{S} \subset \mathbb{K}[x]$ then $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\{f\})$ where f is the greatest common divisor of all elements of $\mathcal{S}$.
- (c) Show that the subspace topology on $\text{mSpec}(\mathbb{K}[x])$ is the finite complement topology. (Recall the *finite complement topology* on a set X is defined by letting the open sets be the empty set and sets of the form $X \setminus T$ for $T \subset X$ a subset of finite cardinality.)

Solution.

- (a) Points of $\text{Spec}(\mathbb{K}[x]) = \{\langle 0 \rangle\} \cup \{\langle x - a \rangle, a \in \mathbb{K}\}$.
- (b) Note that since every element $s \in \mathcal{S}$ is a multiple of the $\text{gcd}(\mathcal{S})$, the containment $\mathbb{V}(\mathcal{S}) \subseteq \mathbb{V}(\{f\})$ is obvious. To show the opposite containment we use the following algorithm:

Require: A set $\mathcal{S}$

Ensure: $\text{gcd}(\mathcal{S})$

```

Pick  $a_1, a_2 \in \mathcal{S}$ ; set  $d \leftarrow \text{gcd}(a_1, a_2)$ 
while there exists  $a \in \mathcal{S}$  such that  $d \nmid a$  do
    Pick such an  $a \in \mathcal{S}$ 
     $d \leftarrow \text{gcd}(d, a)$ 
end while
return  $d$ 

```

Then, there must exist a_i, a_j such that $\text{gcd}(a_i, a_j) = d$. Then by Bezout's Identity, we have $d \in \langle \mathcal{S} \rangle \implies \langle d \rangle \subseteq \langle \mathfrak{p} \rangle$ for all prime ideals containing $\mathcal{S}$.

Hence, $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\{f\})$.

- (c) **TODO:**

Problem 5. Dense Points. If X is a topological space we say that a set $S \subset X$ is *dense* if $\bigcup \mathcal{S} \neq \emptyset$ for every non-empty open subset $U \subset X$.

(a) Let p be a point in a topological space X . Prove that $\{p\}$ is a dense subset of X if and only if the smallest closed subset which contains $\{p\}$ is X . We call such points *dense points* of X . (Note: no point in $\mathbb{R}^n$ or $\mathbb{C}^n$ with the Euclidean topology is dense in this sense.)

(b) Given a prime ideal $P \subset R$ we will frequently write $[P]$ to denote the corresponding point in $\text{Spec}(R)$. Prove that a point $[P] \in \text{Spec}(R)$ is dense if and only if $\mathbb{V}(P) = \text{Spec}(R)$.

(c) Characterize all of the dense points in the following:

(i) $\text{Spec}(\mathbb{K}[x])$	(v) $\text{Spec}(\mathbb{K}[x, y])$
(ii) $\text{Spec}(\mathbb{Z})$	(vi) $\text{Spec}(\mathbb{Z}/\langle 6 \rangle)$
(iii) $\text{Spec}(\mathbb{Z}[t])$	(vii) $\text{Spec}(\mathbb{Z}/\langle 4 \rangle)$
(iv) $\text{Spec}(\mathbb{K}[x]/\langle x^2 \rangle)$	(viii) $\text{Spec}(\mathbb{Z}/\langle 12 \rangle)$

(d) Prove that if R is a domain then $\text{Spec}(R)$ has a unique dense point.

(e) Repeat part 5(c) replacing Spec with mSpec in the subspace topology. Can $\text{mSpec}(R)$ ever have a dense point?

(f) Consider the following two properties (which together are implied by the Hausdorff condition) for a topological space X : i) for every $p \in X$ the set $\{p\}$ is closed and ii) given two distinct points $p, q \in X$ there exist open subsets $U, V \subset X$ such that $p \in U$ and $q \in V$ and $U \cap V = \emptyset$. Prove that if X is a topological space with a dense point, and X is not a single point, then X fails both parts of this definition.

(a) ($\Rightarrow$)

Assume that $\{\mathbf{p}\}$ is a dense subset of X .

Then $U \cap \{p\} \neq \emptyset$ for all non-empty open subsets $U \subset X$. That is $\{p\} \not\subseteq X/U$ (since all non-empty proper closed subsets of X are complements of non-empty open subsets of X , $\{p\}$ is not contained in any proper closed subset of X). But $\{p\} \subset X$, hence smallest closed subset which contains $\{p\}$ is X .

$$\left(\begin{array}{c} \longleftarrow \\ \parallel \end{array} \right)$$

Assume that the smallest closed subset which contains $\{p\}$ is X .

Then for all proper closed subsets U' of X , $\{p\} \not\subseteq U'$ that is $X/U' = U \cap \{p\} \neq \emptyset$. Note that $\emptyset \neq U$ for any U' since U' is a proper subset of X . Therefore, for every non-empty open subset $U \subset X$, $\{p\} \cap U \neq \emptyset$, and hence $\{p\}$ is dense in X .

Hence proved that $\{p\}$ is a dense subset of X if and only if the smallest closed subset which contains $\{p\}$ is X .

(b) ($\implies$)

Assume $[P] \in \text{Spec}(\mathbf{R})$ is dense. Then the smallest closed set containing $[P]$ is the whole space, i.e. $\text{Spec}(\mathbf{R})$. That is $\mathbb{V}(P) = \text{Spec}(\mathbf{R})$.

($\impliedby$)

Assume $\mathbb{V}(P) = \text{Spec}(\mathbf{R})$. Then $P \subseteq T$ for all prime ideals $T \subset \mathbf{R}$. That is, the smallest closed set containing P is the whole space ($\text{Spec}(\mathbf{R})$). Hence, $[P]$ is dense in $\text{Spec}(\mathbf{R})$. '

Hence proved that a point $[P] \in \text{Spec}(\mathbf{R})$ is dense if and only if $\mathbb{V}(P) = \text{Spec}(\mathbf{R})$.

(c) **TODO:**

(i) S

(ii) 0

(d) Assume that $\mathbf{R}$ is an integral domain.

Existence: Note that $\langle 0 \rangle \subseteq \mathbf{R}$ is a prime ideal, and also $\langle 0 \rangle \subseteq P$ for all prime ideals P in $\mathbf{R}$. Also recall that the smallest closed set containing $[0]$ is $\mathbb{V}([0])$, and

$$\mathbb{V}([0]) = \mathbb{V}(\langle 0 \rangle) = \text{Spec}(\mathbf{R})$$

That is the smallest closed set containing $[0]$ is the whole space, hence $[0]$ is a dense point in $\text{Spec}(\mathbf{R})$.

Uniqueness: Suppose $[P] \in \text{Spec}(\mathbf{R})$ is also a dense point. Then by part(b), we know that $\mathbb{V}([P]) = \text{Spec}(\mathbf{R})$. That is $P \subseteq \langle 0 \rangle$, and $\langle 0 \rangle \subseteq P$ trivially. Hence, $[P] = \langle 0 \rangle$.

Hence if $\mathbf{R}$ is a domain, $\text{Spec}(\mathbf{R})$ has a unique dense point.

(e) **TODO:**

(i)

(f) Assume X is a topological space with a dense point p and X is not a single point.

Then by definition of denseness $p \in U$, for all non-empty open subsets $U \subset X$, i.e. we cannot find two distinct open subsets V, W such that $V \cap W = \emptyset$.

For the sake of contradiction assume $\{p\}$ is a closed set, then $X \setminus \{p\}$ is a non-empty open set (since X is not a single point) not containing p contradicting the definition of denseness. Hence, $\{p\}$ cannot be closed.

Problem 6. Let $\mathcal{S} \subset \mathbb{R}$ be any subset. Prove that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle)$ where $\langle \mathcal{S} \rangle$ denotes the ideal generated by $\mathcal{S}$.

Solution.

Let I be an ideal containing $\mathcal{S}$, then one can see that $\langle \mathcal{S} \rangle \subseteq I$. And trivially $\mathcal{S} \subset \langle \mathcal{S} \rangle \subseteq I$. Hence,

$$\mathcal{S} \subseteq I \iff \langle \mathcal{S} \rangle \subseteq I$$

And it follows that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle)$.

Problem 7. In this exercise we wish to understand the Zariski topology on $\text{Spec}(\mathbb{C}[x, y])$. Fix a point $\mathbf{p} = (a, b) \in \mathbb{C}^2$.

- (a) Use the *evaluation-at-p* map: $\text{ev} : \mathbb{C}[x, y] \rightarrow \mathbb{C}$ to prove that the set of polynomial functions vanishing at $\mathbf{p}$ is a maximal ideal $\mathfrak{m}_{\mathbf{p}} \subset \mathbb{C}[x, y]$.
- (b) Find generators for the ideal $\mathfrak{m}_{\mathbf{p}}$. Is this ideal principal?
- (c) Prove that the following map is injective:

$$\begin{array}{ccc} \mathbb{C}^2 & \xrightarrow{\eta} & \text{Spec}(\mathbb{C}[x, y]) \\ \mathbf{p} & \longmapsto & \mathfrak{m}_{\mathbf{p}} \end{array} .$$

- (d) Show that the map η from part 3c is not surjective by finding at least two distinct points in $\text{Spec}(\mathbb{C}[x, y])$ not in the image of η .
- (e) Describe the closed subsets of $\text{Spec}(\mathbb{C}[x, y])$. (Hint: They come in four types)

Solution.

(a)

A function $f : X \rightarrow Y$ of topological spaces is *continuous* if and only if $f^{-1}(\mathcal{U}) \subset X$ is open for all open subsets $\mathcal{U} \subset Y$, i.e., the preimage of an open is open. A *homeomorphism* $f : X \rightarrow Y$ is a continuous map of topological spaces with a continuous inverse $f^{-1} : Y \rightarrow X$. A useful fact is that a continuous map $f : X \rightarrow Y$ is a homeomorphism if and only if it is bijective and $f(\mathcal{U}) \subset Y$ is open for every open subsets $\mathcal{U} \subset X$.

Problem 8. Ring Maps Induce Continuous Maps. Let $\phi : R \rightarrow S$ be a ring homomorphism. The goal of this exercise is to show ϕ induces a map $\phi^\# : \text{Spec}(S) \rightarrow \text{Spec}(R)$, which is continuous in the Zariski topology.

- (a) Prove that if $[P] \in \text{Spec}(S)$, then $[\phi^{-1}(P)] \in \text{Spec}(R)$.
- (b) Prove that the map below is continuous in the Zariski topology:

$$\begin{array}{ccc} \text{Spec}(S) & \xrightarrow{\phi^\#} & \text{Spec}(R) \\ [P] & \longmapsto & [\phi^{-1}(P)] \end{array}$$

- (c) For the ring map $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ sending each integer to its residue class modulo 2, describe the induced map on spectra explicitly.
- (d) For the ring map $\mathbb{Z} \hookrightarrow \mathbb{Q}$, describe the induced map on spectra explicitly.
- (e) If $\phi : R \hookrightarrow S$ is an injective ring map, give a simple description of the induced map on spectra.

Solution.

- (a) **Claim:** Let $\phi : R \rightarrow S$ be a ring homomorphism, and $P \subseteq S$ be a prime ideal, then $\phi^{-1}(P)$ is a prime ideal.

Proof:

A *category* is a collection of objects together with, for each pair of objects, a set of morphisms between them, equipped with an associative composition law and an identity morphism for each object. A *functor* between categories **C** and **D** is a mapping that assigns to each object of **C** an object of **D** and to each morphism in **C** a morphism in **D**, preserving identity morphisms and composition. A functor is *contravariant* if it reverses the order of composition and *covariant* if it preserves the order of composition.

Problem 9. Spec as a Functor. In this problem we package together much of this worksheet to define the Spec functor. Consider the map:

$$\begin{array}{ccc} \mathbf{comRing} & \longrightarrow & \mathbf{Top} \\ R & \longmapsto & \text{Spec}(R) \end{array}$$

from the category of commutative rings with unit with ring homomorphisms to the category of topological spaces with continuous maps.

- (a) Explain using Question ?? how this mapping defines a contravariant functor from **comRing** to **Top**. To check functoriality, verify

$$(\text{Id}_R)^\# = \text{Id}_{\text{Spec}(R)} \quad \text{and} \quad (\psi \circ \phi)^\# = \phi^\# \circ \psi^\#$$

for ring maps $\phi : R \rightarrow S$ and $\psi : S \rightarrow T$. We call this the Spec functor.

- (b) Do isomorphic rings have Spectra that are homeomorphic topological spaces? Explain.
- (c) If two rings have homeomorphic spectra, are they necessarily isomorphic rings? Explain.

Problem 10. Find a homeomorphism between $\text{Spec}(\mathbb{C}[x])$ and $\text{Spec}(\mathbb{C}[x, y]/\langle x - y \rangle)$. Construct the corresponding ring map explicitly in *Macaulay2* and compute its kernel.

Problem 11. Closed Embeddings I. Let $I \subset R$ be an ideal and $\pi : R \rightarrow R/I$ the natural quotient map. Consider the induced map on spectra

$$\text{Spec}(R/I) \xrightarrow{\pi^\#} \text{Spec}(R)$$

In this exercise we will show this map is a closed embedding. A *closed embedding* is a map of topological spaces whose image is closed and which is a homeomorphism onto its image.

- (a) Using the correspondence between ideals in R and R/I give an explicit description of $\pi^\#$.
- (b) Prove that the image of $\pi^\#$ is equal to $\mathbb{V}(I)$.
- (c) Explain why the ideal correspondence implies that $\pi^\#$ is injective.
- (d) Prove that $\pi^\#$ is a closed embedding onto $\mathbb{V}(I)$, in the subspace topology, by showing that if $J \subset R/I$ is an ideal then

$$\pi^\#(\mathbb{V}(J)) = \mathbb{V}(\pi^{-1}(J)).$$

Worksheet 2.2: Algebraic Sets

Theorem 0.3 (Hilbert’s Nullstellensatz I). *If $\mathbb{K}$ is algebraically closed, then every maximal ideal of $\mathbb{K}[x_1, \dots, x_n]$ is of the form $\langle x_1 - a_1, \dots, x_n - a_n \rangle$ for some point $(a_1, \dots, a_n) \in \mathbb{K}^n$.*

Note this is just the first of a series of theorems that will go by Hilbert’s Nullstellensatz. Recall that a $\mathbb{K}$ -algebra is a ring R together with a specified ring homomorphism $\mathbb{K} \rightarrow R$. A morphism of $\mathbb{K}$ -algebras $\varphi : R \rightarrow S$ is a ring homomorphism that preserves scalars, i.e. $\varphi(c \cdot 1_R) = c \cdot 1_S$ for every $c \in \mathbb{K}$. In particular, $\mathbb{K}[x_1, \dots, x_n]$ is a $\mathbb{K}$ -algebra, every evaluation map $\text{ev}_p : \mathbb{K}[x_1, \dots, x_n] \rightarrow \mathbb{K}$ is a $\mathbb{K}$ -algebra homomorphism, and if $\mathfrak{m}$ is a maximal ideal then $R/\mathfrak{m}$ is naturally a field equipped with a $\mathbb{K}$ -algebra structure. When we say below that a residue field is “isomorphic to $\mathbb{K}$,” we mean *as a $\mathbb{K}$ -algebra*.

Problem 1. Evaluation Maps. For a point $p = (a_1, \dots, a_n) \in \mathbb{K}^n$ the *evaluation-at- p* map is given by

$$\begin{array}{ccc} \mathbb{K}[x_1, \dots, x_n] & \xrightarrow{\text{ev}_p} & \mathbb{K} \\ f & \longmapsto & f(p) \end{array}.$$

- (a) Prove that ev_p is a ring homomorphism and find generators for $\mathfrak{m}_p := \ker(\text{ev}_p)$.
- (b) Prove that there is a bijection, induced by these evaluation maps, between $\mathbb{K}^n$ and the subset of $\text{Spec}(\mathbb{K}[x_1, \dots, x_n])$ consisting of maximal ideals whose residue field is isomorphic to $\mathbb{K}$ as a $\mathbb{K}$ -algebra.
- (c) Find a maximal ideal of $\mathbb{R}[x, y]$ that is not of the form $\mathfrak{m}_p$ for any $p \in \mathbb{R}^2$.
- (d) Restate Hilbert’s Nullstellensatz I (above) in terms of $\mathfrak{m}_p$.

Solution.

- (a) To show that ev_p is a ring homomorphism, we need to show that for all $f_1, f_2 \in \mathbb{K}[x_1, \dots, x_n]$ the three conditions hold,

Additivity:

$$\text{ev}_p(f_1 + f_2) = (f_1 + f_2)(p) = f_1(p) + f_2(p) = \text{ev}_p(f_1) + \text{ev}_p(f_2)$$

Multiplicativity:

$$\text{ev}_p(f_1 \cdot f_2) = (f_1 \cdot f_2)(p) = f_1(p) \cdot f_2(p) = \text{ev}_p(f_1) \cdot \text{ev}_p(f_2)$$

And $\text{ev}_p(1) = 1$ since 1 is a constant function.

(b) **TODO:**

(c) **TODO:**

(d) **TODO:**

Problem 2. Algebraic Sets.

Let $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$ be a subset. Define the corresponding *algebraic set* in $\mathbb{K}^n$ to be:

$$\mathbb{V}(\mathcal{S}) := \{ \mathbf{p} \in \mathbb{K}^n \mid f(\mathbf{p}) = 0 \text{ for all } f \in \mathcal{S} \}.$$

(a) Prove that $\mathbb{V}$ is order-reverse, i.e., if $\mathcal{S} \subset \mathcal{T}$ then $\mathbb{V}(\mathcal{T}) \subset \mathbb{V}(\mathcal{S})$.

(b) Prove that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\langle \mathcal{S} \rangle) = \mathbb{V}(\sqrt{\langle \mathcal{S} \rangle})$.

(c) Describe each of the following algebraic sets, and when possible draw a picture.

(i) $\mathcal{S} := \{x^2 - y\} \subset \mathbb{R}[x, y]$.

(ii) $\mathcal{S} := \{x^2 - xy\} \subset \mathbb{R}[x, y]$.

(iii) $\mathcal{S} := \{x^6 - 3x^5y + 3x^4y^2 - x^3y^3\} \subset \mathbb{R}[x, y]$.

(iv) $\mathcal{S} := \{x^4 - 2x^3y + x^2y^2\} \subset \mathbb{R}[x, y]$.

(v) $\mathcal{S} := \{x^2 + y^2 - 1\} \subset \mathbb{R}[x, y]$.

(vi) $\mathcal{S} := \{n(x^2 + y^2 - z^2) \mid n \in \mathbb{Z}_{\geq 0}\} \subset \mathbb{R}[x, y, z]$.

(vii) $\mathcal{S} := \{x^2 + 1\} \subset \mathbb{R}[x]$.

(viii) $\mathcal{S} := \{x^2 + 1\} \subset \mathbb{C}[x]$.

(ix) $\mathcal{S}$ is a collection of any linear polynomials in $\mathbb{K}[x_1, \dots, x_n]$.

(d) Explain why $\mathrm{SL}_n(\mathbb{K})$ has the structure of an algebraic set.

(e) Explain why the set of $m \times n$ matrices of rank $< t$ is an algebraic set contained in $\mathbb{K}^{m \times n}$.

Solution.

(a) Assume $\mathcal{S} \subset \mathcal{T}$. Let $\mathbf{p} \in \mathbb{V}(\mathcal{T})$.

Then, for all $f \in \mathcal{T}$, $f(\mathbf{p}) = 0$. But since $\mathcal{S} \subset \mathcal{T}$ this implies $g(\mathbf{p}) = 0, \forall g \in \mathcal{S}$. Hence, $\mathbf{p} \in \mathbb{V}(\mathcal{S})$. Since $\mathbf{p}$ was an arbitrary point in $\mathbb{V}(\mathcal{T})$ we conclude that $\mathbb{V}$ is order reversing.

(b) By definition,

$$\mathcal{S} \subseteq \langle \mathcal{S} \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}.$$

And from part (a) we have,

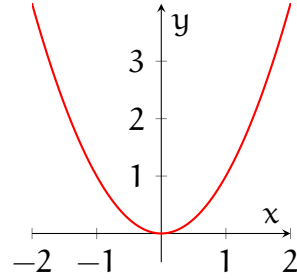
$$\mathbb{V}(\mathcal{S}) \supseteq \mathbb{V}(\langle \mathcal{S} \rangle) \supseteq \mathbb{V}\left(\sqrt{\langle \mathcal{S} \rangle}\right).$$

To show $\mathbb{V}(\langle \mathcal{S} \rangle) \subseteq \mathbb{V}\left(\sqrt{\langle \mathcal{S} \rangle}\right)$ we let $\mathbf{p} \in \mathbb{V}(\langle \mathcal{S} \rangle)$. That is for all $f \in \langle \mathcal{S} \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}$, $f(\mathbf{p}) = 0$, i.e. $\mathbf{p} \in \mathbb{V}\left(\sqrt{\langle \mathcal{S} \rangle}\right) \implies \mathbb{V}(\langle \mathcal{S} \rangle) \subseteq \mathbb{V}\left(\sqrt{\langle \mathcal{S} \rangle}\right)$.

To show that $\mathbb{V}(\mathcal{S}) \subseteq \mathbb{V}(\langle \mathcal{S} \rangle)$ we let $\mathbf{p} \in \mathbb{V}(\mathcal{S})$. That is for all $f \in \mathcal{S}$, $f(\mathbf{p}) = 0$ but since $\langle \mathcal{S} \rangle$ is the set of linear combinations of functions that evaluate to zero at $\mathbf{p}$, they also evaluate to zero at $\mathbf{p}$ and $\mathbf{p} \in \mathbb{V}(\langle \mathcal{S} \rangle)$.

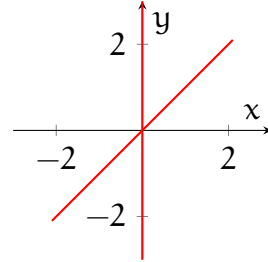
- (c) (i) $\mathcal{S} := \{x^2 - y\} \subset \mathbb{R}[x, y]$.

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 = y\}$$



- (ii) $\mathcal{S} := \{x^2 - xy\} \subset \mathbb{R}[x, y]$.

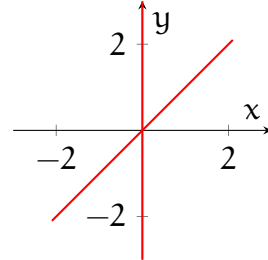
$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0, x = y\}$$



- (iii) $\mathcal{S} := \{x^6 - 3x^5y + 3x^4y^2 - x^3y^3\} \subset \mathbb{R}[x, y]$.

Note that $x^6 - 3x^5y + 3x^4y^2 - x^3y^3 = x^3(x - y)^3$.

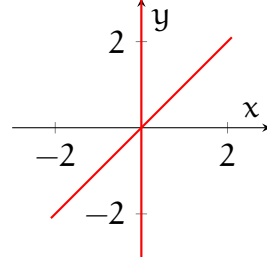
$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0, x = y\}$$



- (iv) $\mathcal{S} := \{x^4 - 2x^3y + x^2y^2\} \subset \mathbb{R}[x, y]$.

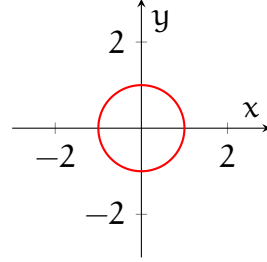
Note that $x^4 - 2x^3y + x^2y^2 = x^2(x - y)^2$.

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x = 0, x = y\}$$



$$(v) \mathcal{S} := \{x^2 + y^2 - 1\} \subset \mathbb{R}[x, y].$$

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}$$



$$(vi) \mathcal{S} := \{n(x^2 + y^2 - z^2) \mid n \in \mathbb{Z}_{\geq 0}\} \subset \mathbb{R}[x, y, z].$$

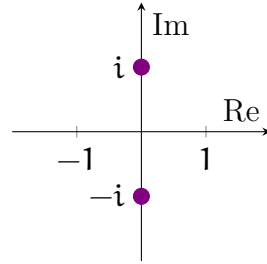
$$\mathbb{V}(\mathcal{S}) = \{(x, y, z) \in \mathbb{R}^3 \mid x^2 + y^2 = z^2\}$$

$$(vii) \mathcal{S} := \{x^2 + 1\} \subset \mathbb{R}[x].$$

$$\mathbb{V}(\mathcal{S}) = \emptyset \text{ since this equation has no solutions over } \mathbb{R}[x]$$

$$(viii) \mathcal{S} := \{x^2 + 1\} \subset \mathbb{C}[x].$$

$$\mathbb{V}(\mathcal{S}) = \{(x, y) \in \mathbb{R}^2 \mid x^2 = y\}$$



$$(ix) \mathcal{S} \text{ is a collection of any linear polynomials in } \mathbb{K}[x_1, \dots, x_n].$$

$$\mathbb{V}(\mathcal{S}) = \ker(A), \text{ where } A \text{ is a } m \times n \text{ matrix where the } i\text{-th row is } (a_{i_1}, \dots, a_{i_n})$$

(d) $SL_n(\mathbb{K})$ is the set of $n \times n$ matrices with determinant 1, and the determinant is a polynomial function on $\mathbb{K}^{n \times n}$. Therefore we can define the polynomial

$$f = \det(A) - 1 \in \mathbb{K}[x_1, \dots, x_{n \times n}]$$

where A is a $n \times n$ matrix, and $SL_n(\mathbb{K}) = \mathbb{V}(\{f\})$.

(e) **TODO:**

Problem 3. Defining Ideal of an Algebraic Set.

Let $V \subset \mathbb{K}^n$ be an algebraic set. The *defining ideal* of V is the ideal in $\mathbb{K}[x_1, \dots, x_n]$ given by

$$\mathbb{I}(V) := \{f \in \mathbb{K}[x_1, \dots, x_n] \mid f(\mathbf{p}) = 0 \text{ for all } \mathbf{p} \in V\}.$$

- (a) Prove that $\mathbb{I}(V)$ is in fact an ideal.
- (b) Prove that $\mathbb{I}(V)$ is radical.
- (c) If $V = \mathbb{V}(\mathcal{S})$ for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, show that $\langle \mathcal{S} \rangle \subset \sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$.
- (d) Let $\mathbb{K} = \mathbb{R}$. Give an example to show that the inclusion $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ can be proper.
- (e) If $V = \mathbb{V}(\mathcal{S})$ for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, show that $\mathbb{V}(\mathcal{S}) = \mathbb{V}(\mathbb{I}(V))$.

Solution.

- (a) Let $g \in \mathbb{K}[x_1, \dots, x_n]$ and $f_1, f_2 \in \mathbb{I}(V)$. That is, $f_1(\mathbf{p}) = f_2(\mathbf{p}) = 0$.

Subgroup Under Addition: Note that,

$$(f_1 + f_2)(\mathbf{p}) = f_1(\mathbf{p}) + f_2(\mathbf{p}) = 0$$

The zero polynomial $0(\mathbf{p}) = 0$ acts as the identity, and

$$f_1(\mathbf{p}) + (-f_1)(\mathbf{p}) = 0(\mathbf{p}) = 0$$

gives us the additive inverse.

Closed Under Multiplication: Notice,

$$(g \cdot f)(\mathbf{p}) = g(\mathbf{p}) \cdot f_1(\mathbf{p}) = g(\mathbf{p}) \cdot 0 = 0$$

Thus $g \cdot f \in \mathbb{I}(V)$ and $\mathbb{I}(V)$ is an ideal.

- (b) We want to show that $\mathbb{I}(V) = \sqrt{\mathbb{I}(V)}$.

Note that the containment $\mathbb{I}(V) \subseteq \sqrt{\mathbb{I}(V)}$ is obvious. To show the opposite containment take $f \in \sqrt{\mathbb{I}(V)}$. Then for some n we have,

$$f^n(\mathbf{p}) = (f(\mathbf{p}))^n = 0$$

But since $\mathbb{K}$ is a domain, this implies that $f(\mathbf{p}) = 0$. Hence $f \in \mathbb{I}(V)$, $\mathbb{I}(V) \supseteq \sqrt{\mathbb{I}(V)}$, and $\mathbb{I}(V) = \sqrt{\mathbb{I}(V)}$.

- (c) Let $V = V(\mathcal{S})$. We know that $\langle \mathcal{S} \rangle \subseteq \sqrt{\langle \mathcal{S} \rangle}$. So showing that $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ suffices. Note that $\mathbb{I}(V) = \mathbb{I}(V(\mathcal{S}))$ which is the set of all functions that evaluate to 0 at $\mathbf{p}$, for all $\mathbf{p} \in V(\mathcal{S})$. And $V(\mathcal{S})$ is the set of all points $\mathbf{p}$ such that all functions $f \in \mathcal{S}$ evaluate to 0 at $\mathbf{p}$. Let $f \in \sqrt{\langle \mathcal{S} \rangle}$. Then by definition for some n , $f^n \in \langle \mathcal{S} \rangle$. We can then write,

$$f^n = \sum_i h_i(\mathbf{p}) g_i(\mathbf{p}), \quad h_i \in \mathbb{K}[x_1, \dots, x_n], g_i \in \mathcal{S}$$

But then,

$$f^n(\mathbf{p}) = \sum_i h_i(\mathbf{p}) g_i(\mathbf{p}) = \sum_i h_i(\mathbf{p}) \cdot 0 = 0$$

Moreover since $\mathbb{K}$ is a domain, $f^n(\mathbf{p}) = 0 \implies f(\mathbf{p}) = 0 \implies f$ is a function that evaluates to 0 at $\mathbf{p}$, hence $f \in \mathbb{I}(V)$.

- (d) Consider the set $\mathcal{S} = \{x^2 + 1\}$. We have,

$$\sqrt{\langle \mathcal{S} \rangle} = \sqrt{\langle x^2 + 1 \rangle}$$

And, because the polynomial has no solutions over $\mathbb{R}$, we have $V = V(\mathcal{S}) = \emptyset$. But $\mathbb{I}(V(\mathcal{S})) = \mathbb{R}[x]$. Also note that since $\langle \mathcal{S} \rangle$ is a prime ideal ($x^2 + 1$ is irreducible), we have $\langle x^2 + 1 \rangle = \sqrt{\langle x^2 + 1 \rangle} \subset \mathbb{R}[x]$. Hence, the containment $\sqrt{\langle \mathcal{S} \rangle} \subset \mathbb{I}(V)$ can be proper.

- (e) Let $V = V(\mathcal{S})$.

$$V(\mathcal{S}) \subseteq V(\mathbb{I}(V(\mathcal{S})))$$

This containment is obvious by construction. $\mathbb{I}(V(\mathcal{S}))$ is the set of all functions that vanish at all points in $V(\mathcal{S})$. And $V(\mathbb{I}(V(\mathcal{S})))$ is the set of all points that all functions in $\mathbb{I}(V(\mathcal{S}))$ vanish at. So, $V(\mathcal{S}) \subseteq V(\mathbb{I}(V))$.

$$V(\mathcal{S}) \supseteq V(\mathbb{I}(V(\mathcal{S})))$$

Note that by definition $\mathcal{S} \subseteq \mathbb{I}(V(\mathcal{S}))$. Then by 2(a) $V(\mathbb{I}(V(\mathcal{S}))) \subseteq V(\mathcal{S})$.

Hence, $V(\mathcal{S}) = V(\mathbb{I}(V(\mathcal{S})))$.

So far you have built two order-reversing operations:

$$\mathcal{S} \longmapsto \mathbb{V}(\mathcal{S}) \quad \text{and} \quad V \longmapsto \mathbb{I}(V).$$

You have shown that passing from equations to a zero set only remembers the radical of the ideal they generate, and Question 4d shows that over a non-algebraically closed field this can lose information. The next theorem says that over an algebraically closed field this is the only ambiguity.

Theorem 0.4 (Hilbert's Nullstellensatz II). *If $\mathbb{K}$ is algebraically closed and $V = \mathbb{V}(\mathcal{S}) \subset \mathbb{K}^n$ is an algebraic set, for some subset $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$, then $\mathbb{I}(V) = \sqrt{\langle \mathcal{S} \rangle}$.*

At this point the basic affine dictionary is taking shape; Question (4) asks you to make the correspondence between algebraic sets and radical ideals completely explicit. After that we turn to coordinate rings, which package an algebraic set V into algebra and will let us recover from $\mathbb{K}[V]$ its points, its topology, and ultimately its morphisms.

Problem 4. Let $\mathbb{K}$ be algebraically closed. Use Hilbert's Nullstellensatz II (above) to give an explicit inclusion-reversing bijection between algebraic sets in $\mathbb{K}^n$ and radical ideals.

Solution.

Let $\mathcal{S} \subset \mathbb{K}[x_1, \dots, x_n]$.

$$\mathbb{V}(\mathcal{S}) \xrightarrow{\mathbb{I}} \sqrt{\langle \mathcal{S} \rangle}$$

$$\mathbb{V}(\mathcal{S}) \xleftarrow{\mathbb{V}} \sqrt{\langle \mathcal{S} \rangle}$$

TODO: Diagram.

Problem 5. Coordinate Rings.

Let $V \subset \mathbb{K}^n$ be an algebraic set and $\mathcal{R} = \text{Hom}_{\text{Set}}(V, \mathbb{K})$ be the ring of all (set) functions from V to $\mathbb{K}$. In this question we study the restriction map from $\mathbb{K}[x_1, \dots, x_n]$ to $\mathcal{R}$ given by:

$$\begin{array}{ccc} \mathbb{K}[x_1, \dots, x_n] & \xrightarrow{\rho} & \mathcal{R} \\ f & \longmapsto & f|_V \end{array}.$$

Here we are viewing $f \in \mathbb{K}[x_1, \dots, x_n]$ as defining a function $\mathbb{K}^n \rightarrow \mathbb{K}$ given by $\mathbf{p} \mapsto f(\mathbf{p})$. We call the image of ρ the *coordinate ring* of V and denote it $\mathbb{K}[V]$.

- (a) Prove that ρ is in fact a ring homomorphism.
- (b) Verify that each of the following functions lies in the coordinate ring of the given algebraic set.
 - (i) Let $V = \mathbb{V}(x^2 + y^2 - 1) \subset \mathbb{R}^2$ and consider the function $V \rightarrow \mathbb{R}$ given by $(x, y) \mapsto e^{x^2+y^2}$.
 - (ii) Let $V = \mathbb{V}(xy - 1) \subset \mathbb{K}^2$ and consider the function $V \rightarrow \mathbb{K}$ given by $(x, y) \mapsto \frac{1}{x}$.
- (c) Identify the kernel of ρ and use it to give an explicit presentation for $\mathbb{K}[V]$ as a $\mathbb{K}$ -algebra.
- (d) The i -th coordinate of $\mathbb{K}^n$ is the linear functional $e_i^\vee : \mathbb{K}^n \rightarrow \mathbb{K}$ sending $(a_1, \dots, a_n) \mapsto a_i$. Explain why $e_i^\vee \in \mathbb{K}[\mathbb{K}^n]$, and under the presentation given in part (c), which polynomial corresponds to $e_i^\vee$.
- (e) Explain why $\mathbb{K}[V]$ is called the coordinate ring of V . Hint: Think about the previous part.
- (f) Describe the coordinate ring for the following algebraic sets:
 - (i) $\mathbb{V}(y - x^2) \subset \mathbb{K}^2$.
 - (ii) $\mathbb{V}(y^2, xy, x^{17}) \subset \mathbb{K}^2$.
 - (iii) $\mathbb{V}(x^6 - 3x^5y + 3x^4y^2 - x^3y^3) \subset \mathbb{K}^2$.
 - (iv) The set of 3×3 matrices over $\mathbb{F}_5$ of determinant 1 and trace 0. (Hint/Warning: There will likely be 9 more polynomials in the defining ideal than you expect, coming from the fact that finite fields have a special automorphism. This highlights another way we might get a counterexample for Question 4d.)

Solution.

- (a) Let $f_1, f_2 \in \mathbb{K}[x_1, \dots, x_n]$. Note that addition and multiplication is defined pointwise in $\mathbb{R}$. To see if ρ is a ring homomorphism, we check if the three properties hold:

Addition:

$$\rho(f_1) + \rho(f_2)(p) = f_1|_V(p) + f_2|_V(p) = (f_1 + f_2)|_V(p) = \rho(f_1 + f_2)(p)$$

Multiplication:

$$\rho(f_1) \cdot \rho(f_2)(p) = f_1|_V(p) \cdot f_2|_V(p) = (f_1 \cdot f_2)|_V(p) = \rho(f_1 \cdot f_2)(p)$$

Unital:

$$\rho(1)(p) = 1|_V(p) = 1$$

- (b) (i) Let $f : (x, y) \mapsto e^{x^2+y^2}$. Note that on V , $x^2 + y^2 = 1$ then, $f|_V = e^{x^2+y^2} = e^1$. Which behaves like the constant polynomial function $g(x, y) = e$ and hence lies in the ring of coordinates of V .
- (ii) Let $f : (x, y) \mapsto \frac{1}{x}$. Note that on V , $\frac{1}{x} = y$, and $x \neq 0$ because $\mathbb{K}$ is a domain. Then, $f|_V = \frac{1}{x} = y$. Which behaves like the polynomial function $g(x, y) = y$ and hence lies in the ring of coordinates of V .
- (c) The kernel of ρ is $\mathbb{I}(V)$.

$$\mathbb{K}[V] \cong \mathbb{K}[x_1, \dots, x_n] / \mathbb{I}(V)$$

- (d) The coordinate ring of $\mathbb{K}^n$, $\mathbb{K}[\mathbb{K}^n]$ is the ring of all set functions $\mathbb{K}^n \rightarrow \mathbb{K}$.

$$e_i^\vee(x_1, \dots, x_n) = x_i$$

which corresponds to the polynomial function $g(x_1, \dots, x_n) = x_i$. Hence, $e_i^\vee \in \mathbb{K}[\mathbb{K}^n]$. Under the presentation given in part (c) we have,

$$\mathbb{K}[\mathbb{K}^n] \cong \mathbb{K}[x_1, \dots, x_n] / \mathbb{I}(\mathbb{K}^n) = \mathbb{K}[x_1, \dots, x_n] / \{0\} = \mathbb{K}[x_1, \dots, x_n]$$

So, $e_i^\vee$ corresponds to the coset $[x_i] = x_i$.

- (e) $\mathbb{K}[V]$ is the coordinate ring of V because every polynomial is a combination of coordinate functions restricted to V . And $\mathbb{K}[V]$ is generated by these functions on V .

$$\mathbb{K}[V] = \mathbb{K}[e_1^\vee|_V, \dots, e_n^\vee|_V]$$

- (f) (i) Let $V = \mathbb{V}(y - x^2)$.

$$\mathbb{K}[V] \cong \mathbb{K}[x, y] / \mathbb{I}(y - x^2) = \mathbb{K}[x, x^2] = \mathbb{K}[x]$$

(ii) Let $V = \mathbb{V}(y^2, xy, x^{17})$.

Note that $\mathbb{I}(V) = \sqrt{\langle y^2, xy, x^{17} \rangle}$.

Since $\mathbb{K}$ is a field, if $(a, b) \in V$ then

$$a^{17} = 0 \implies a = 0$$

and

$$b^2 = 0 \implies b = 0$$

Then, $V = \{(0, 0)\}$. And $\mathbb{I}(V) = \mathbb{I}(\{(0, 0)\}) = \langle x, y \rangle$.

$$\mathbb{K}[V] \cong \mathbb{K}[x, y] / \mathbb{I}(V(y^2, xy, x^{17})) = \mathbb{K}[x, y] / \langle x, y \rangle = \mathbb{K}$$

(iii) Let $V = \mathbb{V}(x^6 - 3x^5y + 3x^4y^2 + x^3y^3) = \mathbb{V}(x^3(x - y)^3)$.

Then if $(a, b) \in V$, either $a^3 = 0 \implies a = 0$ or $(x - y)^3 = 0 \implies x = y$.

Then $V = \{(0, a), (a, a) \mid a \in \mathbb{K}\} = \mathbb{V}(x) \cup \mathbb{V}(x - y)$.

$$\mathbb{I}(V) = \mathbb{I}(\mathbb{V}(x) \cup \mathbb{V}(x - y)) = \mathbb{I}(\mathbb{V}(x)) \cap \mathbb{I}(\mathbb{V}(x - y)) = \langle x \rangle \cap \langle x - y \rangle = \langle x(x - y) \rangle$$

And

$$\mathbb{K}[V] \cong \mathbb{K}[x, y] / \langle x(x - y) \rangle$$

(iv) Let $A = (x_{ij}) \in \mathrm{SL}_3(\mathbb{F}_5)$, and $V = \mathbb{V}(\det(A) - 1, \mathrm{tr}(A))$. Note that the trace formula is $x_{11} + x_{22} + x_{33}$, and we use fermat's little theorem to enforce staying in $\mathbb{F}_5$. Then the coordinate ring is,

$$\mathbb{K}[V] = \mathbb{F}_5[x_{11}, \dots, x_{33}] / \langle \det(A) - 1, x_{11} + x_{22} + x_{33}, x_{ij}^5 - x_{ij} \rangle$$

You should now think of $\mathbb{K}[V]$ as the algebraic avatar of V : its elements are precisely the polynomial functions on V , and the coordinate functions on $\mathbb{K}^n$ descend to generators of $\mathbb{K}[V]$. The next step is to recognize points of V purely algebraically, either as $\mathbb{K}$ -algebra maps $\mathbb{K}[V] \rightarrow \mathbb{K}$ or as maximal ideals whose residue fields are isomorphic to $\mathbb{K}$ as $\mathbb{K}$ -algebras.

Problem 6. Let R be a finitely generated $\mathbb{K}$ -algebra. Fix a presentation $R \cong \mathbb{K}[x_1, \dots, x_n]/I$. Describe a natural bijection between the following three sets:

- (i) points in $V(I) \subset \mathbb{K}^n$,
- (ii) maximal ideals in R with residue field isomorphic to $\mathbb{K}$ as a $\mathbb{K}$ -algebra, and
- (iii) elements of $\text{Hom}_{\mathbb{K}\text{-alg}}(R, \mathbb{K})$, the set of morphisms from R to $\mathbb{K}$ in the category of $\mathbb{K}$ -algebras.

Solution.

Let $\mathbb{K}$ be algebraically closed.

Claim: Let $\pi: \mathbb{K}[x_1, \dots, x_n] \rightarrow \mathbb{K}[x_1, \dots, x_n]/I = R$, and let $J \subset R$ be maximal. We claim that $\pi^{-1}(J)$ is maximal in $\mathbb{K}[x_1, \dots, x_n]$.

Proof. We know that $\pi^{-1}(J)$ is an ideal. Let M be a maximal ideal containing $\pi^{-1}(J)$. Since π is surjective, we know that $\pi(M)$ is an ideal (worksheet 1.1, Q4.4), and $J \subseteq \pi(M)$. Then, since J is maximal, either $\pi(M) = J$ or $\pi(M)$ is the whole ring. $\pi(M) \neq R$ since ring homomorphisms are unital and $1 \notin M$. Hence, $M = \pi^{-1}(J)$ or $\pi^{-1}(J)$ is maximal. $\square$

(i) $\iff$ (ii)

Recall that $V(I)$ contains points $\mathbf{p} = (a_1, \dots, a_n)$ such that $f(\mathbf{p}) = 0$ for all $f \in I$. Also,

$$\text{ev}_{\mathbf{p}}: \mathbb{K}^n \rightarrow \mathbb{K}$$

By Hilbert's Nullstellensatz I, we know that maximal ideals of $\mathbb{K}[x_1, \dots, x_n]$ are of the form $\langle x_1 - a_1, \dots, x_n - a_n \rangle$, for some point $(a_1, \dots, a_n)$.

Problem 7. Zariski Topology on Algebraic Sets. Let $V \subseteq \mathbb{K}^n$ be an algebraic set. Define the *Zariski topology* on V by declaring the closed sets to be the subsets of V of the form $V(I)$ for some ideal $I \subset \mathbb{K}[x_1, \dots, x_n]$. In other words, the closed sets are precisely the *algebraic subsets* of V .

- (a) Verify that this definition indeed determines a topology on V .

- (b) For $f \in \mathbb{K}[V]$, define

$$D(f) = \{p \in V \mid f(p) \neq 0\}.$$

Show that $\{D(f)\}$ form a basis for the Zariski topology on V .

- (c) For each point $p \in V$, let

$$\mathfrak{m}_p = \{f \in \mathbb{K}[V] \mid f(p) = 0\}$$

be the corresponding maximal ideal. Show that the map

$$\begin{array}{ccc} V & \longrightarrow & \text{Spec}(\mathbb{K}[V]) \\ p & \longmapsto & \mathfrak{m}_p \end{array}$$

is a homeomorphism onto its image, which consists of those maximal ideals whose residue field is isomorphic to $\mathbb{K}$ as a $\mathbb{K}$ -algebra.

- (d) If $\mathbb{K}$ is algebraically closed how can the preceding part be rephrased in terms of $\text{mSpec}(\mathbb{K}[V])$?
- (e) Is the Zariski topology on V Hausdorff in general? If not, when is it?

Solution.

- (a) $V = \mathbb{V}(I)$ for some $I \subset \mathbb{K}[x_1, \dots, x_n]$. The closed sets of V are of the form $\mathbb{V}(J)$, where $I \subset J \subset \mathbb{K}[x_1, \dots, x_n]$. We need to check the following:

- (i) $\emptyset$ and V are closed.

Note $\mathbb{V}(I) = V$, and $\mathbb{V}(\mathbb{K}[x_1, \dots, x_n]) = \emptyset$ is closed.

- (ii) Finite union of closed sets is closed.

Let $\mathbb{V}(J), \mathbb{V}(K) \subseteq V$ be subsets. Then, $\mathbb{V}(I) \cup \mathbb{V}(J)$ is the set of points in $\mathbb{K}^n$ such that for all $p \in \mathbb{V}(I) \cup \mathbb{V}(J)$ either $f(p) = 0$ or $g(p) = 0$.

- (iii) Arbitrary intersection of closed sets is closed.

By now you have recovered from $\mathbb{K}[V]$ not just the points of V , but also its Zariski topology. The next goal is to understand maps in the same way: a morphism of algebraic sets should be encoded contravariantly by a $\mathbb{K}$ -algebra homomorphism of coordinate rings.

Problem 8. Morphisms of Algebraic Sets. A *polynomial map* from $\mathbb{K}^n$ to $\mathbb{K}^m$ is a function of the form:

$$\Phi : \mathbb{K}^n \longrightarrow \mathbb{K}^m \quad \mathbf{p} \mapsto (f_1(\mathbf{p}), \dots, f_m(\mathbf{p})).$$

where $f_1, \dots, f_m \in \mathbb{K}[x_1, \dots, x_n]$. A *morphism of algebraic sets* $V \rightarrow W$ is defined to be the restriction of such a polynomial map Φ with $\Phi(V) \subset W$. Fix a polynomial map Φ as above.

- (a) Show that there is an induced $\mathbb{K}$ -algebra homomorphism

$$\phi : \mathbb{K}[\mathbb{K}^m] \longrightarrow \mathbb{K}[\mathbb{K}^n] \quad \text{given by} \quad g \mapsto g \circ \Phi.$$

Describe ϕ explicitly in terms of the coordinate function generators of $\mathbb{K}[\mathbb{K}^m]$.

- (b) Let $V \subset \mathbb{K}^n$ and $W \subset \mathbb{K}^m$ be algebraic sets. Show that $\Phi(V) \subset W$ if and only if $\phi(\mathbb{I}(W)) \subset \mathbb{I}(V)$.
- (c) Deduce that Φ restricts to a morphism of algebraic sets $\Phi|_V : V \longrightarrow W$ if and only if the composition

$$\mathbb{K}[\mathbb{K}^m] \xrightarrow{\phi} \mathbb{K}[\mathbb{K}^n] \xrightarrow{\rho} \mathbb{K}[V]$$

factors through $\mathbb{K}[W]$. Here ρ is the restriction map.

- (d) For each of the given maps of coordinate rings, describe the corresponding map of algebraic sets. When possible draw pictures. Compare to the corresponding maps on spectra.
- (i) $\mathcal{C}[x, y] \rightarrow \mathcal{C}[t]$ given by $x \mapsto t$ and $y \mapsto t$.
- (ii) $\mathcal{C}[t] \rightarrow \mathcal{C}[x, y]$ given by $t \mapsto x$.
- (iii) $\mathcal{C}[t] \rightarrow \mathcal{C}[x, y, t]/\langle xy - t \rangle$ given by $t \mapsto t$.

Problem 9. Prove that a morphism of algebraic sets is continuous with respect to the Zariski topology.

Problem 10. Show that the image of $\mathbb{K} \rightarrow \mathbb{K}^3$ sending $t \rightarrow (t, t^2, t^3)$ is an algebraic set defined by two polynomials. Verify your answer by constructing the corresponding ring map in *Macaulay2*.

Problem 11. Prove or disprove: The image of every polynomial map $\mathbb{K}^n \rightarrow \mathbb{K}^m$ is an algebraic set. (Warning: this might be field dependent. As a hint, if $\mathbb{K}$ is finite, which subsets of $\mathbb{K}^n$ are algebraic sets?)

You now have the main algebra-geometry dictionary in hand. Over any field $\mathbb{K}$, every algebraic set $V \subset \mathbb{K}^n$ has a coordinate ring $\mathbb{K}[V]$, points of V are the same as $\mathbb{K}$ -algebra maps $\mathbb{K}[V] \rightarrow \mathbb{K}$, and morphisms $V \rightarrow W$ correspond contravariantly to $\mathbb{K}$ -algebra homomorphisms $\mathbb{K}[W] \rightarrow \mathbb{K}[V]$. Thus the assignment

$$V \longmapsto \mathbb{K}[V]$$

always gives a contravariant bridge from geometry to algebra. When $\mathbb{K}$ is algebraically closed, Hilbert's Nullstellensatz says that every finitely generated reduced $\mathbb{K}$ -algebra comes from an algebraic set, so this bridge becomes an anti-equivalence of categories.

Problem 12. Putting It All Together. Assume that $\mathbb{K}$ is algebraically closed. Use the previous parts together with Hilbert's Nullstellensatz II to show that every finitely generated reduced $\mathbb{K}$ -algebra is isomorphic to $\mathbb{K}[V]$ for some algebraic set V . Conclude that the assignment $V \mapsto \mathbb{K}[V]$ defines a (anti-) equivalence from the category of algebraic sets over $\mathbb{K}$ to the category of finitely generated reduced $\mathbb{K}$ -algebras.

Worksheet 2.3: Valuation Rings

Fix a field $\mathbb{K}$. In this worksheet we will explore a special, but important type of rings called valuation rings.

Definition 0.5. A valuation ring of a field $\mathbb{K}$ is a subring $\mathbf{R} \subset \mathbb{K}$ with the property that for all $x \in \mathbb{K}^\times$ either $x \in \mathbf{R}$ or $x^{-1} \in \mathbf{R}$.

Note that, since a valuation ring $\mathbf{R}$ of a field $\mathbb{K}$ is a subring of $\mathbb{K}$ it is necessarily a domain. We first study valuation rings directly, then show how certain functions that broadly measure some notion of divisibility, called valuations, produce them. Lastly, we compare discrete and non-discrete examples.

Problem 1. Examples of Valuation Rings.

Recall $\mathbb{C}(x, y)$ is the field of formal rational functions in x and y , which concretely is the set of elements f/g where $f, g \in \mathbb{C}[x, y]$ with $g \neq 0$. For each of the following subsets $\mathcal{S} \subset \mathbb{C}(x, y)$ determine whether or not they are valuation rings by showing whether for every $h \in \mathbb{C}(x, y)^\times$ either $h \in \mathcal{S}$ or $h^{-1} \in \mathcal{S}$.

- (a) Hint: Use that $\mathbb{C}[x, y]$ is a UFD to understand when g is divisible by y .

$$\mathcal{S}_1 := \left\{ \frac{f}{g} \mid f, g \in \mathbb{C}[x, y] \text{ and } g \notin \langle y \rangle \right\}$$

- (b) Hint: Consider the element $\frac{x}{y}$.

$$\mathcal{S}_2 := \left\{ \frac{f}{g} \mid f, g \in \mathbb{C}[x, y] \text{ and } g(0, 0) \neq 0 \right\}$$

- (c) Define the *order* of a non-zero polynomial $f \in \mathbb{C}[x, y]$ to be $\mathbf{ord}(f)$, the lowest degree of a monomial in f and $\mathbf{ord}(0) = \infty$.

$$\mathcal{S}_3 := \left\{ \frac{f}{g} \mid f, g \in \mathbb{C}[x, y], g \neq 0, \text{ and } \mathbf{ord}(f) \geq \mathbf{ord}(g) \right\}$$

Solution. **TODO:**

(a) yes

(b) no

(c) yes

Problem 2. Ideals in Valuation Rings.

Let R be a valuation ring of $\mathbb{K}$.

- (a) Show that for any two non-zero elements $x, y \in R$, either $x|y$ or $y|x$.
- (b) Show that every finitely generated ideal of R is principal.
- (c) Prove that the ideals of R form a totally ordered set. That is, for any two ideals I, J in R , either $I \subseteq J$ or $J \subseteq I$. (Hint: Given $x \in I$ with $x \notin J$ consider whether an element $y \in J$ can divide x .)
- (d) What does the previous parts say about the poset structure on $\text{Spec}(R)$?

Problem 3. Local Rings & Valuation Rings.

An arbitrary ring R is said to be a *local ring* if it has exactly one maximal ideal. We often write a local ring R as a pair $(R, \mathfrak{m})$ where $\mathfrak{m} \subset R$ is the unique maximal ideal.

- (a) Describe $\text{mSpec}(R)$ if R is a local ring.
- (b) Describe the closed points of $\text{Spec}(R)$ if R is a local ring.
- (c) Prove that a valuation ring is a local ring.

Solution.

- (a) $\text{mSpec}(R) = \{\mathfrak{m}\}$
- (b) the only closed point is the maximal ideal.
- (c)

Problem 4. p-Adic Integers.

Fix a prime p . The *p-adic numbers* $\mathbb{Q}_p$ are the completion of $\mathbb{Q}$ with respect to the *p-adic absolute value* $|\cdot|_p$, defined as follows: for a nonzero rational number x , write $x = p^k \frac{a}{b}$ where $a, b \in \mathbb{Z}$ are not divisible by p . Then

$$|x|_p = p^{-k}, \quad |0|_p = 0.$$

Note $|\cdot|_p$ extends uniquely to $\mathbb{Q}_p$. The *p-adic integers* are defined by

$$\mathbb{Z}_p = \{x \in \mathbb{Q}_p \mid |x|_p \leq 1\}.$$

Similar to decimal expansion in $\mathbb{R}$, it is a fact that every element $x \in \mathbb{Q}_p$ can be written

uniquely as a formal series

$$x = \sum_{i=k}^{\infty} a_i p^i \quad \text{where} \quad a_i \in \{0, 1, \dots, p-1\}, \quad a_k \neq 0,$$

and that

$$\mathbb{Z}_p = \left\{ \sum_{i=0}^{\infty} a_i p^i \mid a_i \in \{0, 1, \dots, p-1\} \right\}.$$

- (a) Prove that the p -adic absolute value is well-defined. (Hint: Use the fact that $\mathbb{Z}$ is a UFD to show that every element $x \in \mathbb{Q}$ can be written uniquely (up to sign and reordering) as $p_1^{a_1} \dots p_t^{a_t}$ where the p_i are distinct primes and $a_i \in \mathbb{Z}$.)
- (b) Prove the p -adic absolute value satisfies the *non-Archimedean (ultrametric) inequality*:

$$|x + y|_p \leq \max\{|x|_p, |y|_p\}.$$

- (c) Prove the p -adic absolute value is multiplicative: $|xy|_p = |x|_p |y|_p$.
- (d) Use parts 2b and 3c to show that $\mathbb{Z}_p$ is closed under addition and multiplication.
- (e) Conclude that $\mathbb{Z}_p$ is a ring (called the ring of p -adic integers).
- (f) Prove that $\mathbb{Z}_p$ is a valuation ring.

The previous exercises suggest why valuation rings are called *valuation* rings: they arise from functions that measure divisibility, generalizing the p -adic absolute value. Such functions are called valuations. An *ordered abelian group* is an abelian group $(\Gamma, +)$ together with a total order $<$ on Γ that is compatible with addition, in the sense that if $\alpha < \beta$ then $\alpha + \gamma < \beta + \gamma$ for all $\alpha, \beta, \gamma \in \Gamma$. We will frequently extend the total ordering on an ordered abelian group Γ to $\Gamma \cup \{\infty\}$ by making the convention that $\gamma < \infty$ for all $\gamma \in \Gamma$. Similarly, we will extend addition to $\Gamma \cup \{\infty\}$ by declaring $\infty + \gamma = \gamma + \infty = \infty$ for all $\gamma \in \Gamma$.

Definition 0.6. Let $(\Gamma, +, <)$ be an ordered abelian group. A Γ -valued valuation, or just a valuation, on $\mathbb{K}$ is a function: $v : \mathbb{K} \rightarrow \Gamma \cup \{\infty\}$, which satisfies the following axioms for all $x, y \in \mathbb{K}$:

1. $v(xy) = v(x) + v(y)$
2. $v(x + y) \geq \min\{v(x), v(y)\}$, and
3. $v(x) = \infty$ if and only if $x = 0$.

Note that this means ν defines a group homomorphism $\nu : \mathbb{K}^\times \rightarrow \Gamma$. Indeed, it is standard to define a valuation as a group homomorphism $\nu : \mathbb{K}^\times \rightarrow \Gamma$ satisfying axiom (ii), and then extend it to all of $\mathbb{K}$ by setting $\nu(0) = \infty$. This convention is implicit throughout the remainder of the worksheet.

The *value group* of a valuation $\nu : \mathbb{K}^\times \rightarrow \Gamma$ is the image of ν . A valuation is a *discrete valuation* if its value group is isomorphic to $\mathbb{Z}$. If $\mathcal{F} \subset \mathbb{K}$ is a subfield, we say that ν is a $\mathcal{F}$ -valuation if $\nu(\lambda) = 0$ for all $\lambda \in \mathcal{F}^\times$.

Problem 5. Examples of Valuations.

Prove that the following are valuations.

- (a) The p -adic valuation:

$$\nu_p : \mathbb{Q}^\times \rightarrow \mathbb{Z}, \quad \nu_p\left(p^k \frac{a}{b}\right) = k$$

where $a, b \in \mathbb{Z}$ are not divisible by p .

- (b) The order valuation on $\mathbb{C}(x)$:

$$\nu_0 : \mathbb{C}(x)^\times \rightarrow \mathbb{Z}, \quad \nu_0(f/g) = \text{ord}_{x=0}(f) - \text{ord}_{x=0}(g).$$

- (c) More generally, if $a \in \mathbb{C}$, define

$$\nu_a : \mathbb{C}(x)^\times \rightarrow \mathbb{Z}$$

by order of vanishing at $x = a$. Show this is again a valuation.

- (d) On $\mathbb{C}(x, y)$, define

$$\nu(f/g) = \text{ord}(f) - \text{ord}(g),$$

where $\text{ord}(h)$ is the degree of the lowest-degree nonzero term of $h \in \mathbb{C}[x, y]$. Show that this is a valuation with value group $\mathbb{Z}$.

Problem 6. Prove that every ordered abelian group is torsion-free. (Hint: Reduce to the case when $x > 0$.)

Problem 7. Basic Properties of Valuations.

Let $\nu : \mathbb{K}^\times \rightarrow \Gamma$ be an arbitrary valuation. Fix $f, g \in \mathbb{K}^\times$.

- (a) Show that $\nu(f^{-1}) = -\nu(f)$.

- (b) Show that if $v(g) < v(f)$, then $v(f+g) = v(g)$. (Hint: Use that $(f+g)+(-f) = g$.)
- (c) Show that $v(f) = v(-f)$. (Hint: Use part 2b.)

Problem 8. Consider the field $\mathbb{C}(x, y)$. Throughout this question π refers to the number $\pi = 3.1415926\dots$

- (a) Show that there is a unique $\mathbb{C}$ -valuation $v_\pi : \mathbb{C}(x, y)^\times \rightarrow (\mathbb{R}, +)$ such that

$$v_\pi(x^a y^b) = a + \pi b.$$

What is the value group of v_π ? (Hint: Use Question 3.)

- (b) Consider the group $\Gamma = \mathbb{Z} \oplus \mathbb{Z}$ ordered by the lexicographical ordering. Show that there exists a unique $\mathbb{C}$ -valuation $v : \mathbb{C}(x, y)^\times \rightarrow \Gamma$ such that $v(x^a y^b) = (a, b)$.
- (c) Are either of these valuations constructed in parts 1a or 2b discrete?

Problem 9. Valuations Determine Valuation Rings.

Let v be an arbitrary valuation on $\mathbb{K}$.

- (a) Prove that the set $R \subset \mathbb{K}$ of elements $f \in \mathbb{K}$ with $v(f) \geq 0$ is a valuation ring of $\mathbb{K}$. We call this the *valuation ring* of v . (Hint: Remember our convention about $v(0)$.)
- (b) Show that the following is an ideal of R :

$$\mathfrak{m} := \{f \in \mathbb{K} \mid 0 < v(f)\}.$$

- (c) Prove that if $f \in R$ and $f \notin \mathfrak{m}$ then f is a unit.
- (d) Combine the previous parts to deduce $\mathfrak{m}$ is the unique maximal ideal of R .
- (e) Fix an element $\gamma \in \Gamma$. Prove the sets below are ideals of R :

$$\{f \in R \mid v(f) > \gamma\} \quad \text{and} \quad \{f \in R \mid v(f) \geq \gamma\}.$$

Problem 10. Discrete Valuation Rings.

Let $v : \mathbb{K}^\times \rightarrow \mathbb{Z}$ be a discrete valuation on $\mathbb{K}$. Let $(R, \mathfrak{m})$ be the corresponding *discrete*

valuation ring.

- (a) Let $I \subset R$ be any non-zero ideal, explain why we can choose an element $f \in I$ such that $v(f)$ is minimal among elements in I .
- (b) Prove that every ideal in R is principal. In particular, a discrete valuation ring is a PID.
- (c) By part 2b there exists an element $t \in R$ such that $\langle t \rangle = \mathfrak{m}$. Show that every ideal of R is either 0 or of the form $\langle t^n \rangle$ for some $n \geq 0$.

Problem 11. Consider the valuation v_π on $\mathbb{C}(x, y)$ constructed in Question 1a. Let $(R, \mathfrak{m})$ be the corresponding valuation ring for v_π .

- (a) For a real number $x \in \mathbb{R}_{\geq 0}$ let $\{x\} := x - \lfloor x \rfloor$ denote its fractional part. Fix a positive integer N and consider the sequence consisting of the fractional parts of the first $N + 1$ multiples of π : $\{0\}, \{\pi\}, \{2\pi\}, \dots, \{N\pi\} \in [0, 1)$. Divide the interval $[0, 1)$ into N evenly spaced subintervals:

$$[0, 1) = \left[0, \frac{1}{N}\right) \cup \left[\frac{1}{N}, \frac{2}{N}\right) \cup \dots \cup \left[\frac{N-1}{N}, 1\right).$$

Prove there exist integers $0 \leq k, j \leq N$ such that $\{k\pi\}$ and $\{j\pi\}$ lie in the same subinterval. Deduce that for any positive integer N there exist integers $0 \leq k, j \leq N$ such that $|\{k\pi\} - \{j\pi\}| \leq \frac{1}{N}$. (Hint: Think about pigeons.)

- (b) Using that π is irrational show that $\{j\pi\} \neq \{k\pi\}$ whenever $j \neq k$.
- (c) Prove for every real number $\epsilon > 0$, there exist integers $a, b \in \mathbb{Z}$ such that $0 < a + b\pi < \epsilon$.
- (d) Prove that $\mathfrak{m}$ is not finitely generated. (Hint: Use part 3c.)
- (e) Prove that $\mathfrak{m}$ is the only non-zero radical ideal of R . (Hint: If $I \subset R$ is a non-zero radical ideal, let $f \in I$ be non-zero. For an arbitrary element $h \in \mathfrak{m}$ show that $h^n \in \langle f \rangle$ for $n \gg 0$.)
- (f) Describe $\text{Spec}(V)$.

Problem 12. Consider the valuation v on $\mathbb{C}(x, y)$ constructed in Question 2b. Let $(R, \mathfrak{m})$ be the corresponding valuation ring for v .

- (a) Show that the maximal ideal of R is

$$\mathfrak{m} = \{f \in \mathbb{C}(x, y) \mid v(f) \geq (0, 1)\} = \langle y \rangle.$$

- (b) Show that the ideal $\langle x \rangle \subset R$ is not prime. (Hint: $x = xy^{-1} \cdot y$.)
(c) Show that the set below is a prime, non-maximal ideal of R :

$$P := \{f \in R \mid v(f) = (a, b) \text{ where } a \geq 1\} \cup \{0\}.$$

- (d) Use the valuation to construct an infinite strictly ascending chain of ideals in R :

$$\langle xy^{-1} \rangle \subset \langle xy^{-2} \rangle \subset \langle xy^{-3} \rangle \subset \cdots.$$

Explain why all inclusions are strict.

Problem 13. Valuation Rings Determine Valuations.

Let R be a valuation ring of $\mathbb{K}$ with $R \neq \mathbb{K}$. This problem shows how to recover a valuation from the divisibility structure of R .

- (a) Let Γ' be the set of nonzero principal ideals of R :

$$\Gamma' = \{\langle a \rangle \subset R \mid a \in R, a \neq 0\}.$$

Define a multiplication on Γ' by ideal multiplication: $\langle a \rangle \cdot \langle b \rangle = \langle ab \rangle$.

- (a) Show that this operation is well-defined and makes Γ' into an abelian monoid (associative, commutative, binary operation with an identity element).
(b) Show that Γ' is totally ordered by inclusion: for distinct elements $\langle a \rangle, \langle b \rangle \in \Gamma'$, either $\langle a \rangle \subseteq \langle b \rangle$ or $\langle b \rangle \subseteq \langle a \rangle$.
(b) Define an equivalence relation $\sim$ on $\mathbb{K}^\times$ by $x \sim y$ if and only if $x = uy$ for some unit $u \in R^\times$. Let $\Gamma = \mathbb{K}^\times / \sim$ denote the set of equivalence classes. Given $x \in \mathbb{K}^\times$ write $[x]$ its equivalence class in Γ .
(i) Show that $\sim$ is a well-defined equivalence relation.
(ii) Show that multiplication in $\mathbb{K}^\times$ induces a well-defined operation on Γ , making it into an abelian group.
(iii) Define $[x] \leq [y]$ if and only if $yx^{-1} \in R$. Show this is a well-defined total order on Γ .
(iv) Deduce that Γ is an ordered abelian group.

- (v) Show that the map $\Gamma' \rightarrow \Gamma$ given by $\langle \mathfrak{a} \rangle \mapsto [\mathfrak{a}]$ is an order-preserving embedding whose image consists exactly of the elements $\geq [1]$.
- (c) Define $\mathfrak{v} : \mathbb{K}^\times \rightarrow \Gamma$ by $\mathfrak{v}(\mathfrak{x}) = [\mathfrak{x}]$. Prove that $\mathfrak{v}$ is a valuation on $\mathbb{K}$ with value group Γ and

$$\mathbf{R} = \{ \mathfrak{x} \in \mathbb{K} \mid \mathfrak{v}(\mathfrak{x}) \geq [1] \}.$$

Moreover, $\mathfrak{v}(\mathfrak{x}) \in \Gamma'$ if and only if $\mathfrak{x} \in \mathbf{R}$.

Worksheet 3.1: Gröbner Bases Pt. II

Fix a field $\mathbb{K}$. Unless otherwise specified the ring throughout this worksheet will be $\mathbb{K}[x_1, \dots, x_n]$. In the previous worksheet we introduced monomial orderings and leading terms. In this worksheet we return to the multivariable division algorithm, and then use it to study monomial ideals, initial ideals, and Gröbner bases. The main goal is to push the story all the way to a major finiteness theorem: every ideal of $\mathbb{K}[x_1, \dots, x_n]$ is finitely generated.

Recall a *monomial ordering* on $\mathbb{K}[x_1, \dots, x_n]$ is a total well-ordering $<$ on $\mathbb{Z}_{\geq 0}^n$ such that if $\alpha < \beta$ then $\alpha + \gamma < \beta + \gamma$ for all $\gamma \in \mathbb{Z}_{\geq 0}^n$. Using the bijection between monomials in $\mathbb{K}[x_1, \dots, x_n]$ and elements of $\mathbb{Z}_{\geq 0}^n$ we think of a monomial ordering as giving us a way to compare monomials by saying $x^\alpha < x^\beta$ if and only if $\alpha < \beta$. In Worksheet 1.3 we saw several important monomial orders: lexicographic (lex) order, graded lexicographic (grlex) order, and graded reverse lexicographic (grevlex) order. Please review that worksheet for their definitions.

Returning to our goal of having a division algorithm in $\mathbb{K}[x_1, \dots, x_n]$ we are faced with two challenges compared to the familiar setting of $\mathbb{K}[x]$:

- (i) In several variables the notion of “leading term” is ambiguous. In one variable, the leading term is simply the term of highest degree. In several variables this no longer suffices: for instance, what should the leading term of $x^2 + xy + y^2$ be? All three terms have total degree 2.
- (ii) We want a division algorithm that can answer questions about ideals, i.e., membership, intersections, sums, products, radicals. In $\mathbb{K}[x]$ every ideal is principal, so dividing by a single generator is enough. For $n > 1$, however, $\mathbb{K}[x_1, \dots, x_n]$ is no longer a PID, so ideals generally require multiple generators. We therefore need a way to divide a polynomial by an entire *list* of polynomials.

As seen on Worksheet 1.3, once a monomial ordering has been fixed, we give a meaningful definition of leading term, leading monomial, and leading coefficient; solving issue (i). Note while we normally write $\text{LT}(f)$, $\text{LM}(f)$ and $\text{LC}(f)$ for these quantities; they do depend on the chosen order $<$, and so would more properly be called $\text{LT}_{<}(f)$, $\text{LM}_{<}(f)$ and $\text{LC}_{<}(f)$ to highlight this fact.

To resolve issue (ii), we generalize long division as follows. In single-variable polynomial long division, we repeatedly ask: “Does the leading term of the divisor divide the current leading term of the dividend?” If yes, we subtract off the appropriate multiple; if not, we are done. The multivariable algorithm works the same way, except that instead of checking a single divisor we scan through an *ordered* list of divisors $G = (g_1, \dots, g_s)$ and use the first one whose leading term divides the current leading term. More precisely, the algorithm maintains a “working polynomial” p (initialized to f), quotients $q_1, \dots, q_s$ (initialized to 0), and a remainder r (initialized to 0), then repeats:

- **Try to Divide.** Walk through the list $g_1, g_2, \dots, g_s$ in order. If you find some g_i whose leading term $LT(g_i)$ divides $LT(p)$, subtract the appropriate multiple: update $q_i \leftarrow q_i + LT(p)/LT(g_i)$ and $p \leftarrow p - (LT(p)/LT(g_i))g_i$. Then return to the start of the list and try again with the new p .
- **Move to Remainder.** If *no* g_i in the list has a leading term dividing $LT(p)$, then the current leading term of p cannot be reduced further. Move it to the remainder: $r \leftarrow r + LT(p)$ and $p \leftarrow p - LT(p)$.

We continue until the working polynomial p is equal to 0. Proving this algorithm terminates as expected, which you will do later in this worksheet, amounts essentially to proving the following theorem

Theorem 0.7. *Fix a monomial ordering $<$ on $\mathbb{K}[x_1, \dots, x_n]$ and let $G = (g_1, \dots, g_s)$ be an ordered list of nonzero polynomials. If $f \in \mathbb{K}[x_1, \dots, x_n]$ then there exist $q_1, \dots, q_s, r \in \mathbb{K}[x_1, \dots, x_n]$ such that*

$$f = q_1 g_1 + \dots + q_s g_s + r,$$

where no monomial appearing in r is divisible by any of $LM(g_1), \dots, LM(g_s)$.

Written in pseudocode the algorithm for multivariable polynomial division is shown below.

Algorithm 2 Division by an ordered list $G = (g_1, \dots, g_s)$

Require: $f, g_1, \dots, g_s$

Ensure: $q_1, \dots, q_s, r$

$q_1, \dots, q_s \leftarrow 0; r \leftarrow 0; p \leftarrow f$

while $p \neq 0$ **do**

\leftarrow 0

for $i = 1, \dots, s$ **do**

if $LT(g_i)$ divides $LT(p)$ **then**

$q_i \leftarrow q_i + LT(p)/LT(g_i)$

$p \leftarrow p - (LT(p)/LT(g_i))g_i$

\leftarrow 1

break

end if

end for

if divides = 0 **then**

$r \leftarrow r + LT(p)$

$p \leftarrow p - LT(p)$

end if

end while

Problem 1. Review of Monomial Orders.

Work in $\mathbb{K}[x, y, z]$ and assume $x > y > z$.

- (a) Order the monomials x^2y , xy^2 , xz^2 , y^3 , yz^4 from largest to smallest with respect to lex, graded lex, and graded reverse lex.
- (b) Compute $\text{LM}(f)$, $\text{LC}(f)$, and $\text{LT}(f)$ for $f = 4x^2y - 3xy^2 + y^4 + 2z^5$ with respect to lex and grevlex.
- (c) Prove that 1 is the smallest monomial for every monomial ordering.
- (d) Deduce that every strictly decreasing sequence of monomials terminates.
- (e) Explain why part (d) should make you optimistic that a division algorithm might terminate.

Solution.

- (a)
 - Lex: x^2y , xy^2 , xz^2 , y^3 , yz^4
 - Graded Lex: yz^4 , x^2y , xy^2 , xz^2 , y^3
 - Graded RevLex: yz^4 , x^2y , xy^2 , y^3 , xz^2

- (b) Lex:

$$\text{LM}(f) = x^2y, \quad \text{LC}(f) = 4, \quad \text{LT}(f) = 4x^2y$$

Grevlex:

$$\text{LM}(f) = z^5, \quad \text{LC}(f) = 2, \quad \text{LT}(f) = 2z^5$$

- (c) The monomial 1 corresponds to $(0, \dots, 0)$. For the sake of contradiction assume there exist some $\alpha = (\alpha_1, \dots, \alpha_n)$ such that $\alpha \neq 1$ and $\alpha < 1$. Then, by property (ii) of monomial orders we have

$$\alpha + \alpha < 0 + \alpha$$

Applying (ii) again,

$$3\alpha < 2\alpha$$

And so on,

$$\dots < 4\alpha < 3\alpha < 2\alpha < \dots$$

But this violates the well-ordering property (iii) of monomial orderings, a contradiction! Hence, 1 must be the smallest monomial in every monomial ordering.

- (d) Since we have a minimal element, every strictly decreasing sequence of monomials terminates.

- (e) In a division algorithm, at each step we reduce the polynomial f by replacing its leading term with a smaller combination of terms coming from the divisors. This produces a strictly decreasing sequence of leading monomials and by part (d) every such sequence must terminate. Hence, eventually the algorithm must terminate.

Problem 2. Examples of Multivariable Division.

For this question, we will work in $\mathbb{K}[x, y]$ considered with the lex order where $x > y$.

- (a) Divide $f = x^2y + xy + y - x - 2$ by the ordered list $G = (xy - 1, y - 1)$. Record the successive values of p , the quotients, and the remainder after each update.
- (b) Divide $f = xy^2 - x$ by the ordered list $G = (xy - 1, y^2 - 1)$. Record the successive values of p , the quotients, and the remainder after each update.
- (c) Repeat (b), but with the ordered list $(y^2 - 1, xy - 1)$, i.e. where the order of G is swapped.
- (d) Show directly that $xy^2 - x \in \langle xy - 1, y^2 - 1 \rangle$.
- (e) What do your computations show about division in several variables? In particular, what is always true when the remainder is 0, and what can go wrong when the remainder is non-zero?

Problem 3. Termination of Division Algorithm

Let $G = (g_1, \dots, g_s)$ be an ordered list of non-zero polynomials.

- (a) Show that throughout the algorithm one always has $f = q_1g_1 + \dots + q_sg_s + p + r$.
- (b) Show that after each pass through the `while` loop, the leading monomial of p strictly decreases with respect to the given monomial order.
- (c) Use Question 1(d) to prove that the algorithm terminates.
- (d) Prove that on termination the output satisfies $f = q_1g_1 + \dots + q_sg_s + r$, and no monomial appearing in r is divisible by any of $\text{LM}(g_1), \dots, \text{LM}(g_s)$.
- (e) Explain why a zero remainder certifies ideal membership: if dividing f by G produces remainder 0, then $f \in \langle g_1, \dots, g_s \rangle$.

Question 2 reveals an important new feature of polynomial division in several variables. Unlike the one-variable case, the remainder obtained by dividing a polynomial f by polynomials $g_1, \dots, g_t$ depends on two choices: i) the order of our list of divisors and ii) the monomial order we have fixed. Because of this, division does *not* immediately solve the ideal membership problem in several variables. In other words, even if dividing f by the list $(g_1, \dots, g_t)$ produces a nonzero remainder, it may still happen that $f \in \langle g_1, \dots, g_t \rangle$. So, in several variables, a nonzero remainder does *not* by itself show that $f \notin \langle g_1, \dots, g_t \rangle$. We will eventually develop a way to fix this problem. Before we do, however, it is useful to study a special class of ideals

for which ideal membership *is* controlled purely by divisibility of monomials.

Definition 0.8. An ideal $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ is a monomial ideal if it can be generated by monomials.

We shall see shortly that a monomial ideal is entirely determined by the set of monomials belonging to the ideal. Since monomials in $\mathbb{K}[x_1, \dots, x_n]$ are in bijection with $\mathbb{Z}_{\geq 0}^n$ this means we can – and often do think of monomial ideals as corresponding to subsets of $\mathbb{Z}_{\geq 0}^n$. With this viewpoint, the following classical result guarantees that monomial ideals are well-behaved.

Theorem 0.9 (Dickson’s Lemma). Every subset of $\mathbb{Z}_{\geq 0}^n$ has only finitely many minimal elements with respect to the coordinatewise order $\leq$.

Motivated by this theorem we say a *minimal monomial generating set* for a monomial ideal I is a finite set of monomials $\{m_1, \dots, m_t\}$ such that $I = \langle m_1, \dots, m_t \rangle$ and no m_i divides m_j for $i \neq j$. Equivalently, no generator is redundant: removing any m_i from the set yields a strictly smaller ideal.

Problem 4. Properties of Monomial Ideals.

- (a) Decide which of the following ideals are monomial ideals:

$$\langle x^2, xy^3, z \rangle, \quad \langle x + y, y^2 \rangle, \quad \langle x^2 - y, y^3 \rangle, \quad \langle x^2, xy, y^4 + x \rangle.$$

- (b) Let $m_1, \dots, m_t$ and m be monomials. Prove that $m \in \langle m_1, \dots, m_t \rangle$ if and only if $m_i \mid m$ for some i .
- (c) Deduce that a polynomial $f \in \mathbb{K}[x_1, \dots, x_n]$ lies in I if and only if every monomial appearing in f is divisible by one of $m_1, \dots, m_t$.
- (d) Find a minimal monomial generating set for each of the following ideals:

$$J = \langle x^3, x^2y, xy^2, y^4, x^4y, x^2y^3 \rangle \quad \text{and} \quad L = \langle x^2z, xyz, y^2z, z^3, xz^4 \rangle.$$

Draw a picture of the subset of $\mathbb{Z}_{\geq 0}^2$ corresponding to the monomials in J .

Problem 5. Understanding Dickson’s Lemma.

- (a) Let $A \subseteq \mathbb{Z}_{\geq 0}^n$ and set $I_A := \langle x^\alpha \mid \alpha \in A \rangle$. If $A_{\min}$ denotes the set of minimal elements of A , show that every monomial in I_A is divisible by some x^α with $\alpha \in A_{\min}$.
- (b) Explain why the statement “every subset of $\mathbb{Z}_{\geq 0}^n$ has finitely many minimal

elements with respect to $\leq$ ” is equivalent to the statement “every monomial ideal in $\mathbb{K}[x_1, \dots, x_n]$ is finitely generated.”

- (c) Restate Dickson’s Lemma entirely in terms of monomial ideals in $\mathbb{K}[x_1, \dots, x_n]$.
- (d) Prove your monomial ideal version of Dickson’s Lemma in the case when $n = 1$.

Problem 6. Proving Dickson’s Lemma. We now prove the monomial ideal version of Dickson’s Lemma by induction on the number of variables. Your work in Question 4d serves as the base case. Assume Dickson’s Lemma holds in $n - 1$ variables. Let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be a monomial ideal. For $k \geq 0$ let

$$J_k = \left\{ f \in \mathbb{K}[x_1, \dots, x_{n-1}] \mid fx_n^k \in I \right\} \cup \{0\}.$$

- (a) Prove that J_k is a monomial ideal in $\mathbb{K}[x_1, \dots, x_{n-1}]$ for all $k \geq 0$.
- (b) Show that the J_k form an ascending chain.

$$J_0 \subseteq J_1 \subseteq J_2 \subseteq \dots.$$

(Hint: What happens when you multiply something in J_k by x_n ?)

- (c) Let $L = \bigcup_{d \geq 0} J_d$. Show that L is a monomial ideal in $\mathbb{K}[x_1, \dots, x_{n-1}]$.
- (d) Prove that there exists $N \in \mathbb{Z}_{\geq 0}$ such that $L = J_k$ for all $k \geq N$. (Hint: Use the inductive hypothesis to show that L is finitely generated.)
- (e) For each $0 \leq k \leq N$, choose monomial generators $m_{k,1}, \dots, m_{k,r_k}$ for J_k . Via the inclusion $\mathbb{K}[x_1, \dots, x_{n-1}] \subset \mathbb{K}[x_1, \dots, x_n]$ view these as elements in $\mathbb{K}[x_1, \dots, x_n]$. Prove that I is generated by the finite set

$$\{m_{k,i}x_n^k \mid 0 \leq k \leq N, 1 \leq i \leq r_k\}.$$

- (f) Conclude Dickson’s Lemma for all n .

Problem 7. Minimal monomial generators.

Let I be a monomial ideal, and let $M(I)$ be the set of monomials in I which are minimal under divisibility.

- (a) Use Dickson’s Lemma to show that $M(I)$ is finite.

- (b) Prove that $I = \langle M(I) \rangle$.
- (c) Prove that every monomial generating set of I contains $M(I)$. Conclude that $M(I)$ is the unique minimal monomial generating set of I .

Monomial ideals are easier to study because membership is determined entirely by divisibility of monomials, and Dickson's Lemma gives strong finiteness properties. For a general ideal, we would like to recover similar structures by extracting the leading term of each polynomial. This leads to the following.

Definition 0.10. Let $<$ be a monomial order on $\mathbb{K}[x_1, \dots, x_n]$. The initial ideal of a nonzero ideal $I \subset \mathbb{K}[x_1, \dots, x_n]$ with respect to $<$ is:

$$\text{in}_{<}(I) := \langle LT(f) \mid f \in I \rangle.$$

A finite subset $\mathcal{G} = \{g_1, \dots, g_s\} \subset I$ is a Gröbner basis for I with respect to $<$ if $\text{in}_{<}(I) = \langle LT(g_1), \dots, LT(g_s) \rangle$.

It is standard to set $\text{in}_{<}(\langle 0 \rangle) = \langle 0 \rangle$. Note a priori there is no reason for Gröbner bases to exist in general or be interesting. In the next few exercises you will show that Dickson's Lemma not only implies that Gröbner bases exist, but also they will give an answer to the ideal membership problem.

Problem 8. Initial Ideals of Monomial Ideals.

Let $I \subset \mathbb{K}[x_1, \dots, x_n]$ be a monomial ideal. Prove that $\text{in}_{<}(I) = I$ for every monomial order $<$.

Problem 9.

For this problem consider $\mathbb{K}[x, y]$ with the lex order where $x > y$. Let $I = \langle xy - 1, y^2 - 1 \rangle$.

- (a) Find explicit generators for the ideal generated by the leading terms of $xy - 1$ and $y^2 - 1$.
- (b) Prove that $\{xy - 1, y^2 - 1\}$ is not a Gröbner basis for I by finding an element in I not contained in the ideal computed in 1a. (Hint: Consider the element $x - y$.)
- (c) Use *Macaulay2* to find a Gröbner basis for I .

Problem 10. Existence of Gröbner Bases.

Let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be an ideal and $<$ be a monomial order.

- (a) Show that $\text{in}_<(I)$ is a monomial ideal.
- (b) By Dickson's Lemma, $\text{in}_<(I) = \langle m_1, \dots, m_t \rangle$ for a minimal monomial generating set $m_1, \dots, m_t$. Show that there exist elements $g_1, \dots, g_t \in I$ such that $\text{LT}(g_i)$ divides m_i for all i .
- (c) Show that the minimality of $m_1, \dots, m_t$ implies $\text{LT}(g_i) = c_i m_i$ for some scalar $c_i \in \mathbb{K}^\times$.
- (d) Conclude that $\mathcal{G} = \{g_1, \dots, g_t\}$ is a Gröbner basis of I .

Problem 11. Ideal Membership & Gröbner Bases.

Let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be an ideal and $<$ be a monomial order. Fix a Gröbner basis $G = (g_1, \dots, g_t)$ for I , i.e. elements $g_i \in I$ such that $m_i := \text{LT}(g_i)$ generate $\text{in}_<(I)$.

- (a) Given $f \in \mathbb{K}[x_1, \dots, x_n]$ consider the division of f by $G = (g_1, \dots, g_t)$ obtaining

$$f = q_1 g_1 + \dots + q_t g_t + r,$$

as in Theorem 0.7. Show that $f \in I$ if and only if $r \in I$.

- (b) Assume $f \in I$ and $r \neq 0$. Show that $\text{LT}(r) \in \text{in}_<(I)$, and conclude $\text{LT}(r)$ is divisible by one of $m_1, \dots, m_t$. Why does this contradict the defining property of the remainder? Conclude that every $f \in I$ has remainder 0 on division by G .
- (c) Membership Test. Prove that if G is a Gröbner basis for I , then for any polynomial $f \in \mathbb{K}[x_1, \dots, x_n]$

$$f \in I \iff \text{the remainder on dividing } f \text{ by } G \text{ is } 0.$$

- (d) Prove that $I = \langle g_1, \dots, g_t \rangle$. (Hint: The inclusion $\langle g_1, \dots, g_t \rangle \subset I$ should be clear. For the other, let $f \in I$ then apply the division algorithm and the membership test.)

Notice that Questions 10 and 11 together show; given any ideal $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ and any monomial order $<$, a Gröbner basis for I exists (Question 10), and every Gröbner basis generates I (Question 11(d)). Since a Gröbner basis is finite by construction, every ideal of $\mathbb{K}[x_1, \dots, x_n]$ is finitely generated. This naturally leads us to our first encounter with the

following important finiteness condition.

Definition 0.11. A ring R is Noetherian if and only if every ascending chain of ideals:

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$

stabilizes in the sense that there exists $N \in \mathbb{Z}_{\geq 1}$ such that $I_n = I_{n+1}$ for all $n \geq N$.

It is common to say a ring is Noetherian if it has the ACC property where ACC is short for ascending chain condition. You will show below a ring R being Noetherian is equivalent to every ideal of R being finitely generated. Thus, you have proven that $\mathbb{K}[x_1, \dots, x_n]$ is Noetherian. Since a field $\mathbb{K}$ is Noetherian, you have actually proven a special, but important case of the following theorem.

Theorem 0.12 (Hilbert's Basis Theorem). *Let R be a ring. If R is Noetherian, then $R[x]$ is Noetherian.*

The remainder of this worksheet is devoted to proving the equivalence between the ACC and finite generation of ideals, and proving Hilbert's basis theorem in general.

Problem 12. Properties of Noetherian Rings.

For this question let R be any commutative ring, and consider an ascending chain of ideals in R :

$$I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$$

- (a) Show that $J = \bigcup_{k \geq 1} I_k$ is an ideal of R .
- (b) If J is finitely generated, say $J = \langle a_1, \dots, a_t \rangle$, explain why there exists an integer $N \in \mathbb{Z}_{\geq 1}$ such that $a_1, \dots, a_t \in I_N$. Conclude that the chain of ideals stabilizes.
- (c) Explain why you just showed that if every ideal of R is finitely generated then R is Noetherian.
- (d) Let I be an ideal of R and suppose every ascending chain of ideals in R stabilizes. Pick any $a_1 \in I$. If $\langle a_1 \rangle \neq I$, pick $a_2 \in I \setminus \langle a_1 \rangle$, obtaining $\langle a_1 \rangle \subsetneq \langle a_1, a_2 \rangle$. Continue in this way. Use the ascending chain condition to show this process must terminate, and conclude that I is finitely generated.
- (e) Show that if R is Noetherian and $I \subseteq R$ is an ideal, then R/I is Noetherian.

Problem 13. Hilbert's Basis Theorem.

Let R be a Noetherian ring, and let $I \subseteq R[x]$ be an ideal. For $k \in \mathbb{Z}_{\geq 0}$, define

$$J_k := \left\{ a \in R \mid \begin{array}{l} \text{there exists } f \in I \\ \text{LT}(f) = ax^k \end{array} \right\} \cup \{0\}.$$

Equivalently, $a \in J_k$ if and only if there exists $f \in I$ with $\text{LT}(f) = ax^k$ (or $a = 0$). Thus J_k is the set of all leading coefficients of polynomials of degree k in I , together with 0 . Note that we are viewing $R[x]$ as a polynomial ring in the single variable x over R . We define the leading term of a polynomial $f = a_k x^k + a_{k-1} x^{k-1} + \cdots + a_1 x + a_0$ (with $a_k \neq 0$) to be $\text{LT}(f) = a_k x^k$.

- (a) Show that J_k is an ideal of R for all $k \geq 0$.
- (b) Show that the J_k form an ascending chain $J_0 \subseteq J_1 \subseteq J_2 \subseteq \cdots$.
- (c) Let $L = \bigcup_{d \geq 0} J_d$. Show that L is an ideal in R .
- (d) Prove that there exists $N \in \mathbb{Z}_{\geq 0}$ such that $L = J_k$ for all $k \geq N$. (Hint: Use that R is Noetherian.)

For each $0 \leq k \leq N$, choose generators $a_{k,1}, \dots, a_{k,r_k}$ for J_k and polynomials $f_{k,i} \in I$ such that $\text{LT}(f_{k,i}) = a_{k,i} x^k$. We will now show that I is equal to

$$M := \langle f_{k,i} \mid 0 \leq k \leq N, 1 \leq i \leq r_k \rangle.$$

The inclusion $M \subset I$ is clear. We prove the reverse inclusion $I \subseteq M$ by induction on degree. For the base case suppose $g \in I$ and $\deg(g) = 0$. Since $g \in I$ and $\text{LT}(g) = cx^0 = c$, we have $c \in J_0$. Using that $J_0 = \langle a_{0,1}, \dots, a_{0,r_0} \rangle$ we may write c as $c = b_1 a_{0,1} + \cdots + b_{r_0} a_{0,r_0}$ with $b_i \in R$. For each i we have $\text{LT}(f_{0,i}) = a_{0,i} x^0 = a_{0,i}$ so $f_{0,i} = a_{0,i}$. Hence $a_{0,i}$ is in M for all i implying $g \in M$ proving the base case.

Now suppose every element of I with degree less than d belongs to M . Let $g \in I$ with $\deg(g) = d$ and $\text{LT}(g) = cx^d$. Our goal is to find $h \in M$ with $\text{LT}(h) = cx^d$, so that $g - h \in I$ has degree less than d and the inductive hypothesis applies. Set $n = \min\{d, N\}$. Since the ascending chain $J_0 \subseteq J_1 \subseteq \cdots$ stabilizes at J_N , we have $J_d = J_n$, so $c \in J_n = \langle a_{n,1}, \dots, a_{n,r_n} \rangle$. Write

$$c = b_1 a_{n,1} + \cdots + b_{r_n} a_{n,r_n}$$

for some $b_i \in R$, and define

$$h = \sum_{i=1}^{r_n} b_i x^{d-n} f_{n,i}.$$

(Note: if $d \leq N$ then $n = \min\{d, N\} = d$ and the $x^{d-n} = 1$.) Each $f_{n,i} \in M$, so $h \in M$.

- (a) Show that $\text{LT}(h) = cx^d$. From this deduce that $g - h \in I$ and either $\deg(g - h) < d$ or $g - h = 0$.

- (b) Using the inductive hypothesis conclude that $g - h \in M$, and show this implies $g \in M$. Conclude that $I \subset M$. Explain why this concludes the proof of Hilbert's Basis Theorem.
- (c) Prove via induction on n that if R is Noetherian then $R[x_1, \dots, x_n]$ is Noetherian.
- (d) Compare this proof to the proof of Dickson's Lemma we gave above.

This worksheet has introduced two important concepts that will recur throughout the course: the usefulness of Gröbner bases and the importance of finiteness conditions such as Noetherianity. A significant open thread raised by this worksheet is that while we know theoretically that Gröbner bases exist, we do not yet know how to actually compute them. We will return to this question later.

It is fitting that Gröbner bases and Noetherianity are so linked. Hilbert's original proof of his Basis Theorem was non-constructive, even over $\mathbb{K}[x_1, \dots, x_n]$, which – as the story goes – came as a shock to many mathematicians of the time. Paul Gordan, a prominent algebraist, reportedly remarked, perhaps seriously, perhaps in jest, *“This is not mathematics. This is theology.”* (Gordan later gave another proof of the theorem, one which in a sense also proved Dickson's Lemma some 10 years before Dickson.) On the other hand, Gröbner bases have come to serve as the foundational tool of computational commutative algebra, allowing us to compute and make explicit a great many things that Hilbert's methods could not.

Worksheet 3.2: The Nullstellensatz

Throughout this course, “ring” means *commutative* ring with unity. In this worksheet $\mathbb{K}$ will denote a field. In the previous worksheet you built the order-reversing maps

$$I \longmapsto \mathbb{V}(I) \quad \text{and} \quad V \longmapsto \mathbb{I}(V)$$

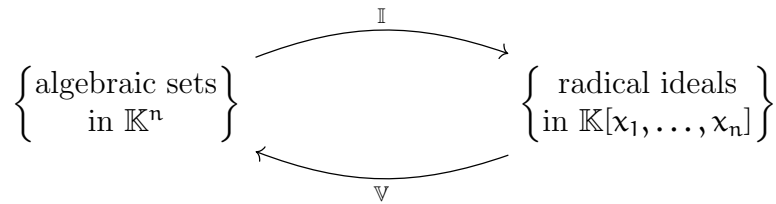
connecting ideals in $\mathbb{K}[x_1, \dots, x_n]$ with algebraic sets in $\mathbb{K}^n$. Our goal now is to prove that over an algebraically closed field these constructions *almost* recover each other exactly. The only restriction being that since $\mathbb{I}(V)$ is always a radical ideal, these functions are only bijections when restricted to the set of radical ideals in $\mathbb{K}[x_1, \dots, x_n]$. More precisely, our goal is to prove the first two versions of Hilbert’s Nullstellensatz under the condition that $\mathbb{K}$ is uncountable (i.e. $\mathbb{C}$). Although our work will also show the second version follows from the first for any algebraically closed field.

Theorem 0.13 (Hilbert’s Nullstellensatz I). *If $\mathbb{K}$ is algebraically closed, then every maximal ideal of $\mathbb{K}[x_1, \dots, x_n]$ is of the form $\langle x_1 - a_1, \dots, x_n - a_n \rangle$ for some point $(a_1, \dots, a_n) \in \mathbb{K}^n$.*

Theorem 0.14 (Hilbert’s Nullstellensatz II). *If $\mathbb{K}$ is algebraically closed and $V = \mathbb{V}(S) \subset \mathbb{K}^n$ is an algebraic set, for some subset $S \subset \mathbb{K}[x_1, \dots, x_n]$, then $\mathbb{I}(V) = \sqrt{\langle S \rangle}$.*

For a point $\mathbf{p} = (a_1, \dots, a_n) \in \mathbb{K}^n$, we will often write $\mathfrak{m}_{\mathbf{p}}$ for the ideal $\langle x_1 - a_1, \dots, x_n - a_n \rangle \subset \mathbb{K}[x_1, \dots, x_n]$, which by the previous worksheet is the kernel of the evaluation-at- $\mathbf{p}$ map $\text{ev}_{\mathbf{p}} : \mathbb{K}[x_1, \dots, x_n] \rightarrow \mathbb{K}$. An alternative statement of Hilbert’s Nullstellensatz I is to say there is a bijection between $\mathbb{K}^n$ and maximal ideals in $\mathbb{K}[x_1, \dots, x_n]$ given by $\mathbf{p} \mapsto \mathfrak{m}_{\mathbf{p}}$. We say that this statement can fail when $\mathbb{K}$ is not algebraically closed, e.g., $\langle x^2 + 1 \rangle \subset \mathbb{R}[x]$.

Since points are themselves algebraic sets, the second version of Hilbert’s Nullstellensatz can be thought of as a generalization of the first. In particular Nullstellensatz II can be thought of as saying that $\mathbb{V}$ and $\mathbb{I}$ are mutual inverses giving bijections



Somewhat amazingly, the entire difficulty of this theorem comes down to only one of the four inclusions needed. In particular, last week we showed that $\mathbb{V}(\mathbb{I}(V)) = V$ for any algebraic subset $V \subset \mathbb{K}^n$, regardless of the field. Similarly, we also proved that the inclusion $\sqrt{I} \subseteq \mathbb{I}(\mathbb{V}(I))$ holds over an arbitrary field. Thus, the hard part of the proof of Hilbert’s Nullstellensatz II is to show that

$$\mathbb{I}(\mathbb{V}(I)) \subseteq \sqrt{I}$$

for every ideal $I \subseteq \mathbb{K}[x_1, \dots, x_n]$. This can fail over non-algebraically closed fields, as the example $I = \langle x^2 + 1 \rangle \subset \mathbb{R}[x]$ shows.

The strategy of this worksheet is to isolate that difficulty: first review the geometry–algebra dictionary, and then prove the point Hilbert’s Nullstellensatz I for uncountable fields using field theory and a clever cardinality argument. We will then deduce what some call the weak Nullstellensatz:

Theorem 0.15 (Hilbert’s Weak Nullstellensatz). *If $\mathbb{K}$ is algebraically closed and $I \subset \mathbb{K}[x_1, \dots, x_n]$ is an ideal then $V(I) = \emptyset$ if and only if $I = \langle 1 \rangle$.*

Note the difficult direction of the weak Nullstellensatz is the forward direction, the reverse is immediate. Intuitively this is saying that a system of polynomial equations has no solution over $\mathbb{K}$ if and only if the “obvious” obstruction – namely you can write 1 in terms of your polynomials – occurs. Finally we will use Rabinowitsch’s trick to obtain the full statement. Along the way we will also pause to recall some field theory language, since the key obstruction is the difference between algebraic and transcendental field extensions.

Problem 1. Warm-Up: Recalling the Correspondence.

Recall the two order-reversing maps

$$I \longmapsto V(I) \quad \text{and} \quad V \longmapsto I(V).$$

Before we move to proving the main theorems let us warm up with a few examples.

- (a) Give the definitions of $V(I)$ and $I(V)$.
- (b) If $V = \{p\} \subset \mathbb{K}^n$ for $p = (a_1, \dots, a_n)$, describe $I(V)$ both intrinsically and by giving generators.
- (c) The ideals $\langle 0 \rangle$ and $\langle 1 \rangle$ are the extreme cases. Describe $V(\langle 0 \rangle)$ and $V(\langle 1 \rangle)$ as subsets of $\mathbb{K}^n$.
- (d) Over $\mathbb{R}$, describe $V(\langle x, y \rangle)$ and $V(\langle x^2 + y^2 \rangle) \subseteq \mathbb{R}^2$.
- (e) Over $\mathbb{F}_p$, describe $V(\langle 0 \rangle)$ and $V(\langle x^p - x, xy^p - x^p y \rangle) \subseteq \mathbb{F}_p^2$.

Problem 2. A Parameterized Example.

Consider the map

$$\phi : \overline{\mathbb{F}}_{17}^2 \longrightarrow \overline{\mathbb{F}}_{17}^3, \quad (u, v) \longmapsto (u^2, uv, v^2).$$

- (a) Show that every point in the image of ϕ satisfies the equation $xz - y^2 = 0$.
- (b) Let $(a, b, c) \in \overline{\mathbb{F}}_{17}^3$ satisfy $ac = b^2$. Show that if $a \neq 0$, then (a, b, c) lies in the image of ϕ . Handle the case $a = 0$ separately.

- (c) Deduce that the image of ϕ is equal to $\mathbb{V}(\mathbf{x}z - \mathbf{y}^2) \subseteq \overline{\mathbb{F}}_{17}^3$.
- (d) Show that $\mathbf{x}z - \mathbf{y}^2$ is irreducible in $\overline{\mathbb{F}}_{17}[\mathbf{x}, \mathbf{y}, \mathbf{z}]$ and conclude that the corresponding radical ideal is $\langle \mathbf{x}z - \mathbf{y}^2 \rangle$.

Problem 3. Reduction to the Core Difficulty.

Recall what you proved last time about the interaction of $\mathbb{V}$ and $\mathbb{I}$ over an arbitrary field.

- (a) Show that if $V \subseteq \mathbb{K}^n$ is algebraic, then $\mathbb{V}(\mathbb{I}(V)) = V$.
- (b) Show that for every ideal $I \subseteq \mathbb{K}[\mathbf{x}_1, \dots, \mathbf{x}_n]$ one has $I \subseteq \sqrt{I} \subseteq \mathbb{I}(\mathbb{V}(I))$.
- (c) Show via an example – other than $\langle \mathbf{x}^2 + 1 \rangle \subset \mathbb{R}[\mathbf{x}]$ – that the inclusion in 2b can be proper when $\mathbb{K}$ is not algebraically closed
- (d) Explain why proving Hilbert’s Nullstellensatz II reduces to showing that when $\mathbb{K}$ is algebraically closed, $\mathbb{I}(\mathbb{V}(I)) \subset \sqrt{I}$ for every ideal $I \subset \mathbb{K}[\mathbf{x}_1, \dots, \mathbf{x}_n]$.

Before the next question, recall some field theory language. If $\mathbb{K}$ is a field and $\mathbb{E}$ is a field containing $\mathbb{K}$, we call $\mathbb{E}/\mathbb{K}$ a *field extension*. In this situation $\mathbb{E}$ is naturally a vector space over $\mathbb{K}$, and we write $[\mathbb{E} : \mathbb{K}] := \dim_{\mathbb{K}} \mathbb{E}$ for its dimension, called the *degree* of the extension. If $[\mathbb{E} : \mathbb{K}]$ is finite we say the extension is *finite*.

Fixing a field extension $\mathbb{E}/\mathbb{K}$, we say an element $\zeta \in \mathbb{E}$ is *algebraic* over $\mathbb{K}$ if there exists a nonzero polynomial $f \in \mathbb{K}[\mathbf{x}]$ with $f(\zeta) = 0$; otherwise ζ is *transcendental* over $\mathbb{K}$. The extension $\mathbb{E}/\mathbb{K}$ is called *algebraic* if every element of $\mathbb{E}$ is algebraic over $\mathbb{K}$. Every finite extension is algebraic: if $[\mathbb{E} : \mathbb{K}] = d$, then for any $\zeta \in \mathbb{E}$ the $d + 1$ elements $\zeta^0, \zeta, \dots, \zeta^d$ are $\mathbb{K}$ -linearly dependent, giving a polynomial relation for ζ over $\mathbb{K}$.

More generally, elements $\zeta_1, \dots, \zeta_r \in \mathbb{E}$ are *algebraically independent* over $\mathbb{K}$ if no nonzero polynomial $f \in \mathbb{K}[\mathbf{x}_1, \dots, \mathbf{x}_r]$ satisfies $f(\zeta_1, \dots, \zeta_r) = 0$. In particular, a single element ζ is transcendental over $\mathbb{K}$ exactly when $\{\zeta\}$ is algebraically independent over $\mathbb{K}$. One helpful characterization of transcendental elements can be phrased in terms of our trusty friend evaluation maps.

Lemma 0.16. *Let $\mathbb{E}/\mathbb{K}$ be a field extension. An element $\zeta \in \mathbb{E}$ is transcendental over $\mathbb{K}$ if and only if the evaluation-at- ζ homomorphism below is injective:*

$$\begin{array}{ccc} \mathbb{K}[\mathbf{x}] & \xrightarrow{\text{ev}_{\zeta}} & \mathbb{E} \\ f & \longmapsto & f(\zeta) \end{array} .$$

A useful consequence of this lemma is that if $\zeta \in \mathbb{E}$ is transcendental over $\mathbb{K}$, then the map ev_ζ identifies $\mathbb{K}[x]$ with the subring $\mathbb{K}[\zeta] \subseteq \mathbb{E}$. Passing to fields of fractions then identifies the rational function field $\mathbb{K}(x)$ with the subfield $\mathbb{K}(\zeta) \subseteq \mathbb{E}$. You will prove the converse in the exercises as well.

A field $\mathbb{K}$ is *algebraically closed* if every nonconstant polynomial in $\mathbb{K}[x]$ has a root in $\mathbb{K}$. An equivalent formulation is that $\mathbb{K}$ has no nontrivial algebraic extensions. That is, if $\mathbb{E}/\mathbb{K}$ is algebraic, then $\mathbb{E} = \mathbb{K}$. The field $\mathbb{C}$ of complex numbers is the prototypical example.

A $\mathbb{K}$ -*algebra* is a ring R equipped with a ring homomorphism $\alpha : \mathbb{K} \rightarrow R$. Given such a map, R becomes a $\mathbb{K}$ -vector space by defining $\lambda \cdot a := \alpha(\lambda)a$ for $\lambda \in \mathbb{K}$ and $a \in R$. This scalar multiplication is compatible with the ring multiplication in the sense that

$$\lambda \cdot (ab) = (\lambda \cdot a)b = a(\lambda \cdot b)$$

for all $a, b \in R$ and $\lambda \in \mathbb{K}$. Conversely, any ring R that is simultaneously a $\mathbb{K}$ -vector space satisfying the above compatibility condition is a $\mathbb{K}$ -algebra; the structure map is $\alpha(\lambda) = \lambda \cdot 1_R$. A *homomorphism of $\mathbb{K}$ -algebras* $\phi : R \rightarrow S$ is a ring homomorphism that also respects the $\mathbb{K}$ -action, i.e. $\phi(\lambda \cdot_R a) = \lambda \cdot_S \phi(a)$ for all $\lambda \in \mathbb{K}$ and $a \in R$. Equivalently, the diagram

$$\begin{array}{ccc} & \mathbb{K} & \\ \alpha_R \swarrow & & \searrow \alpha_S \\ R & \xrightarrow{\phi} & S \end{array}$$

commutes, where α_S and α_R are the respective structure maps. In other words, ϕ is a ring homomorphism that is also $\mathbb{K}$ -linear. For example, $R = \mathbb{K}[x_1, \dots, x_n]$ is a $\mathbb{K}$ -algebra via the inclusion of constants, and any quotient R/I inherits a $\mathbb{K}$ -algebra structure from the surjection $\pi : R \rightarrow R/I$. If $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ is an ideal, the quotient $\mathbb{K}[x_1, \dots, x_n]/I$ is a $\mathbb{K}$ -algebra. It is a field if and only if I is maximal. In that case, the quotient map π gives a field extension:

$$\mathbb{K} \longrightarrow \mathbb{K}[x_1, \dots, x_n] \xrightarrow{\pi} \mathbb{K}[x_1, \dots, x_n]/I$$

gives a field extension of $\mathbb{K}$. Every element of this quotient is a $\mathbb{K}$ -linear combination of the images of monomials $x_1^{a_1} \cdots x_n^{a_n}$, so properties of this extension (such as its dimension over $\mathbb{K}$) are controlled by the combinatorics of monomials together with the relations imposed by I .

The problem below uses all of these ideas to prove a beautiful dictionary between algebra and geometry: when $\mathbb{K}$ is both uncountable and algebraically closed (e.g. $\mathbb{K} = \mathbb{C}$), the maximal ideals of $\mathbb{K}[x_1, \dots, x_n]$ are in natural bijection with the points of $\mathbb{K}^n$.

Problem 4. Hilbert's Nullstellensatz I ($\mathbb{K}$ uncountable).

Let $R := \mathbb{K}[x_1, \dots, x_n]$ and $\mathfrak{m} \subset R$ be a maximal ideal. Write $L := R/\mathfrak{m}$ for the residue field, which contains $\mathbb{K}$ via the natural quotient map. The goal is to show that if $\mathbb{K}$ is

uncountable and algebraically closed, then in fact $L \cong \mathbb{K}$.

- (a) Show that the monomials in R form a countable $\mathbb{K}$ -basis of R .
- (b) Deduce that L has at most countable dimension as a vector space over $\mathbb{K}$.
- (c) Let $\mathbb{E}/\mathbb{K}$ be a field extension and let $\zeta \in \mathbb{E}$. Explain why the following are equivalent:
 - (i) ζ is transcendental over $\mathbb{K}$;
 - (ii) $\{\zeta\}$ is algebraically independent over $\mathbb{K}$;
 - (iii) the evaluation map $\mathbb{K}[x] \rightarrow \mathbb{E}$ sending $f(x)$ to $f(\zeta)$ is injective.

In particular, if a field extension of $\mathbb{K}$ is not algebraic, then it contains a transcendental element.

- (d) Let $\mathbb{K}$ be any infinite field, and let $\zeta \in \mathbb{E}$ be transcendental over $\mathbb{K}$ for some field extension $\mathbb{E}/\mathbb{K}$. Prove that the set of elements in $\mathbb{E}$:

$$\left\{ \frac{1}{\zeta - \lambda} \mid \lambda \in \mathbb{K} \right\} \subseteq \mathbb{E}$$

is linearly independent over $\mathbb{K}$. (Hint: Suppose $\sum_{i=1}^m \frac{c_i}{\zeta - \lambda_i} = 0$ for distinct $\lambda_1, \dots, \lambda_m \in \mathbb{K}$ and coefficients $c_1, \dots, c_m \in \mathbb{K}$. Multiply both sides by $\prod_{j=1}^m (\zeta - \lambda_j)$ to obtain $\sum_{i=1}^m c_i \prod_{j \neq i} (\zeta - \lambda_j) = 0$. Now define $P(x) = \sum_{i=1}^m c_i \prod_{j \neq i} (x - \lambda_j) \in \mathbb{K}[x]$. Then $P(\zeta) = 0$. Since ζ is transcendental over $\mathbb{K}$, conclude that $P(x)$ must be the zero polynomial. Evaluate $P(x)$ at $x = \lambda_k$ for each k to show that $c_k = 0$.)

- (e) Now assume that $\mathbb{K}$ is uncountable. Using part 4d, show that if a field extension $\mathbb{E}/\mathbb{K}$ contains a transcendental element, then $[\mathbb{E} : \mathbb{K}]$ is uncountable.
- (f) Assume $\mathbb{K}$ is uncountable. Show if $\mathfrak{m} \subseteq R$ is a maximal ideal, then the extension $\mathbb{K} \hookrightarrow R/\mathfrak{m} := L$ is algebraic.
- (g) Now assume that $\mathbb{K}$ is both uncountable and algebraically closed. Show that every algebraic extension of $\mathbb{K}$ is trivial, so $L \cong \mathbb{K}$ as $\mathbb{K}$ -algebras.
- (h) With the notation from part 7g let $\psi : R \rightarrow \mathbb{K}$ be the ring map given by the composition

$$R \xrightarrow{\pi} R/\mathfrak{m} = L \xrightarrow{\sim} \mathbb{K}$$

where π is the quotient map and the second map is the identification of L with $\mathbb{K}$ from part 7g. Note $\ker(\psi) = \mathfrak{m}$. Define $\mathbf{a}_i := \psi(x_i) \in \mathbb{K}$ for $i = 1, \dots, n$. Prove that $\mathfrak{m} = \langle x_1 - \mathbf{a}_1, \dots, x_n - \mathbf{a}_n \rangle$.

- (i) Deduce that when $\mathbb{K}$ is uncountable and algebraically closed, the assignment $\mathfrak{p} \mapsto \mathfrak{m}_{\mathfrak{p}}$ gives a bijection between points of $\mathbb{K}^n$ and maximal ideals of $\mathbb{K}[x_1, \dots, x_n]$.

We have now proven Hilbert's Nullstellensatz I under the additional assumption that $\mathbb{K}$ is uncountable, using virtually only field theory! Note this applies to $\mathbb{C}$ and $\mathbb{C}_p$ as these are both algebraically closed and uncountable, but does not apply to $\overline{\mathbb{Q}}$ and $\overline{\mathbb{F}_p}$ which while algebraically closed are countable.

In what follows we will *only* assume $\mathbb{K}$ is algebraically closed, and will use Hilbert's Nullstellensatz I to deduce the weak Nullstellensatz. After that we will use what has come to be known as *Rabinowitsch's trick* to upgrade the weak Nullstellensatz to the full statement of Hilbert's Nullstellensatz II.

Problem 5. Weak Nullstellensatz. Let $\mathbb{K}$ be algebraically closed and let $I \subseteq \mathbb{K}[x_1, \dots, x_n]$ be an ideal.

- (a) Show that if I is proper, then I is contained in some maximal ideal $\mathfrak{m}$. (Hint: Use Zorn's Lemma.)
- (b) Assuming Hilbert's Nullstellensatz I, show that if I is proper then $\mathbb{V}(I) \neq \emptyset$. (Hint: $\mathbb{V}(\mathfrak{m}) \subset \mathbb{V}(I)$.)
- (c) Conclude that $\mathbb{V}(I) = \emptyset$ if and only if $I = \langle 1 \rangle$.

Problem 6. Hilbert's Nullstellensatz II. (Rabinowitsch's Trick)

Let $\mathbb{K}$ be algebraically closed and $R := \mathbb{K}[x_1, \dots, x_n]$. Let $I = \langle f_1, \dots, f_t \rangle \subset R$ be an ideal. Choose any element $g \in \mathbb{I}(\mathbb{V}(I))$, we will show that some power of g lies in I . Note if $g = 0$ there is nothing to prove as $0 \in I$. Thus, we assume g is non-zero.

- (a) In the larger polynomial ring $R[z]$, define $J := \langle f_1, \dots, f_t, 1 - gz \rangle$. Show that $\mathbb{V}(J) = \emptyset$ without appealing to the weak Nullstellensatz. (Hint: If $(a_1, \dots, a_{n+1}) \in \mathbb{V}(J)$ then what do we know about $g(a_1, \dots, a_n)$ since $g \in \mathbb{I}(\mathbb{V}(I))$.)
- (b) Use Question ?? to conclude that $J = \langle 1 \rangle \subset R[z]$. Deduce there exist polynomials $h_1, \dots, h_t, h_{t+1} \in R[z]$ such that the following relation holds.

$$1 = h_1 f_1 + \dots + h_t f_t + h_{t+1}(1 - gz) \in R[z].$$

- (c) Consider the ring $R[1/g]$ consisting of expressions of the form p/g^k where $p \in R$ and $k \geq 0$, in other words, we enlarge R by formally allowing division by powers

of g . Consider the map of $\mathbb{K}$ -algebras:

$$\begin{aligned} \mathbb{R}[z] &\xrightarrow{\psi} \mathbb{R}\left[\frac{1}{g}\right] \\ z &\longmapsto \frac{1}{g} \end{aligned}$$

Explain why ψ is a well-defined $\mathbb{K}$ -algebra homomorphism. (Hint: What does it take to define a $\mathbb{K}$ -algebra map out of a polynomial ring $\mathbb{R}[z]$?)

- (d) Prove that $\psi(1 - gz) = 0$ and deduce that $1 = \psi(h_1)f_1 + \cdots + \psi(h_t)f_t$ in $\mathbb{R}[1/g]$.
- (e) Explain why there exists $k \in \mathbb{N}$ such that $g^k\psi(h_i) \in \mathbb{R}$ for every $i = 1, \dots, t$. Multiply the equation in part 4d by g^k to conclude that $g^k \in I$. (Hint: Each $\psi(h_i)$ is an element of $\mathbb{R}[1/g]$, so it can be written as p_i/g^{k_i} for some $p_i \in \mathbb{R}$ and $k_i \geq 0$.)
- (f) Conclude that $g \in \sqrt{I}$, and hence $\mathbb{I}(\mathbb{V}(I)) \subseteq \sqrt{I}$.
- (g) Combine this with Question ?? to finish the proof of Hilbert's Nullstellensatz II assuming Hilbert's Nullstellensatz I. (The latter we have proven over every uncountable algebraically closed field, in particular over $\mathbb{C}$.)

You have now proved Hilbert's Nullstellensatz for every *uncountable* algebraically closed field, including $\mathbb{C}$. The only remaining gap in the proof over an arbitrary algebraically closed field is the point case: one still needs to show that every maximal ideal is of the form $\mathfrak{m}_p$ without assuming uncountability. We will return to that later, after developing more tools.

For the remainder of the worksheet, assume Hilbert's Nullstellensatz I and II over all algebraically closed fields and use them freely. The point now is to unpack what the theorem says about coordinate rings, closed subsets, irreducibility, and functoriality. In particular, we would like a version of the Nullstellensatz that describes algebraic subsets not of just $\mathbb{K}^n$, but of any algebraic set $V \subset \mathbb{K}^n$. Towards this we will define and study relative versions of the $\mathbb{I}$ and $\mathbb{V}$ operators. In particular, for an algebraic subset $W \subset V$ and an ideal $J \subset \mathbb{K}[V]$ define

$$\mathbb{I}_V(W) := \{ \phi \in \mathbb{K}[V] \mid \phi(p) = 0 \text{ for all } p \in W \}$$

and

$$\mathbb{V}_V(J) := \{ p \in V \mid \phi(p) = 0 \text{ for all } \phi \in J \}.$$

Using these we how to prove the following relative version of the Nullstellensatz connecting algebraic subsets of V and radical ideals in $\mathbb{K}[V]$.

Theorem 0.17 (Hilbert's Nullstellensatz III). *Let $\mathbb{K}$ be an algebraically closed field. If $V \subset \mathbb{K}^n$ is an algebraic subset then the maps $\mathbb{I}_V$ and $\mathbb{V}_V$ induce inverse, inclusion-reversing*

bijections:

$$\left\{ \begin{array}{c} \text{algebraic subsets} \\ \text{in } V \end{array} \right\} \begin{array}{c} \xrightarrow{\mathbb{I}_V} \\ \xleftarrow{\mathbb{V}_V} \end{array} \left\{ \begin{array}{c} \text{radical ideals} \\ \text{in } \mathbb{K}[V] \end{array} \right\}$$

Note that since the coordinate ring of $\mathbb{K}^n$ is canonically isomorphic to $\mathbb{K}[x_1, \dots, x_n]$ we could view Hilbert's Nullstellensatz II as a special case of this third version. However, the real difficulty in proving the third version boils down to applying the second. As you will see many of the following exercises boil down to the same ideas and arguments we used for $\mathbb{I}$ and $\mathbb{V}$, at times almost verbatim.

Problem 7. Properties of $\mathbb{I}_V$.

Assume $\mathbb{K}$ is algebraically closed, let $V \subseteq \mathbb{K}^n$ be an algebraic set with coordinate ring $\mathbb{K}[V]$. Let $W \subseteq V$ be an algebraic subset and $\rho_W : \mathbb{K}[V] \rightarrow \mathbb{K}[W]$ be the restriction homomorphism taking a function $\phi : V \rightarrow \mathbb{K}$ and restricting it to W to get a function $\phi|_W : W \rightarrow \mathbb{K}$. (If we wanted to feel fancy we could say $\phi|_W$ is the composition of $\phi : V \rightarrow \mathbb{K}$ with the inclusion $\iota : W \rightarrow V$, but in this setting that is needlessly complex.)

- (a) Show that $\mathbb{I}_V(W)$ is an ideal of $\mathbb{K}[V]$, and if $Z \subset W \subset V$ then $\mathbb{I}_V(W) \subset \mathbb{I}_V(Z)$.
- (b) Since V is an algebraic subset of itself, as a reality check what is $\mathbb{I}_V(V)$?
- (c) Prove that ρ_W is a surjective $\mathbb{K}$ -algebra homomorphism with kernel $\mathbb{I}_V(W)$.
- (d) Prove that ρ_W induces an $\mathbb{K}$ -algebra isomorphism $\mathbb{K}[V]/\mathbb{I}_V(W) \cong \mathbb{K}[W]$.
- (e) Show that if $W \subset V$ is an algebraic subset then the following diagram commutes:

$$\begin{array}{ccc} \mathbb{K}[x_1, \dots, x_n] & \xrightarrow{\rho} & \mathbb{K}[V] \\ & \searrow \rho' & \downarrow \rho_W \\ & & \mathbb{K}[W] \end{array}$$

where $\rho' : \mathbb{K}[x_1, \dots, x_n] \rightarrow \mathbb{K}[W]$ is the restriction of functions map defining $\mathbb{K}[W]$. Deduce that $\mathbb{I}_V(W) = \mathbb{I}(W)/\mathbb{I}(V)$.

- (f) Conclude that if $W \subset V$ is an algebraic subset then $\mathbb{I}_V(W)$ is a radical ideal of $\mathbb{K}[V]$.

Problem 8. Properties of $\mathbb{V}_V$.

Let $\mathbb{K}$ be an algebraically closed field, and let $V \subseteq \mathbb{K}^n$ be an algebraic set with coordinate ring $\mathbb{K}[V]$.

- (a) Let $J \subset \mathbb{K}[V]$ be an ideal. Show that $\mathbb{V}_V(J)$ is an algebraic subset of V by showing that $\mathbb{V}_V(J) = \mathbb{V}(\rho^{-1}(J)) \subseteq \mathbb{K}^n$. (Hint: Since $\ker(\rho) = \mathbb{I}(V) \subseteq \rho^{-1}(J)$, every point of $\mathbb{V}(\rho^{-1}(J))$ already lies in V .)
- (b) Again exploring the extremes what algebraic subsets of V are $\mathbb{V}_V(\langle 1 \rangle)$ and $\mathbb{V}_V(\langle 0 \rangle)$?
- (c) Show that if $I \subset J$ are ideals in $\mathbb{K}[V]$ then $\mathbb{V}_V(J) \subset \mathbb{V}_V(I)$.

Problem 9. Proving Hilbert's Nullstellensatz III.

Assume $\mathbb{K}$ is algebraically closed, let $V \subseteq \mathbb{K}^n$ be an algebraic set with coordinate ring $\mathbb{K}[V]$.

- (a) Use Hilbert's Nullstellensatz in $\mathbb{K}[x_1, \dots, x_n]$ to prove that $\mathbb{I}_V(\mathbb{V}_V(J)) = \sqrt{J}$ (Hint: First identify $\mathbb{I}_V(\mathbb{V}_V(J))$ with $\mathbb{I}(\mathbb{V}(\rho^{-1}(J)))/\mathbb{I}(V)$, then use $\mathbb{I}(\mathbb{V}(-)) = \sqrt{-}$.)
- (b) Show that if $W \subseteq V$ is an algebraic subset, then $\mathbb{V}_V(\mathbb{I}_V(W)) = W$.
- (c) Conclude that the assignments $W \mapsto \mathbb{I}_V(W)$ and $J \mapsto \mathbb{V}_V(J)$ induce inverse, inclusion-reversing bijections between algebraic subsets of V and radical ideals in $\mathbb{K}[V]$.
- (d) Prove that under the bijections in the previous parts the maximal ideals of $\mathbb{K}[V]$ correspond to the points of V , thought of as algebraic subsets. Explain why we should think of this as a relative version of Hilbert's Nullstellensatz I.

Problem 10. Irreducibility and Prime Ideals.

An algebraic set $V \subset \mathbb{K}^n$ is *irreducible* if there do not exist two proper algebraic subsets $W_1, W_2 \subsetneq V$ such that $V = W_1 \cup W_2$. Equivalently, V is not the union of two proper algebraic subsets. An algebraic set that is not irreducible is said to be *reducible*.

- (a) Which of the following algebraic sets in $\mathbb{K}^n$ (with $\mathbb{K}$ algebraically closed) do you think are irreducible. Draw pictures when possible.
 - (i) $V = \mathbb{V}(x^2 - y^2) \subset \mathbb{C}^2$.
 - (ii) $V = \mathbb{V}(y - x^3) \subset \mathbb{C}^2$.
 - (iii) $V = \mathbb{V}(xz, yz) \subset \mathbb{C}^3$.
 - (iv) $V = \mathbb{V}(x^2 + y^2 - 1) \subset \mathbb{C}^2$.
 - (v) $V = \mathbb{V}(xy - z^2) \subset \mathbb{C}^3$.
 - (vi) $V = \mathbb{V}(3x^2 - 2xy) \subset \mathbb{C}^2$.

- (b) Assume $\mathbb{I}(V)$ is a prime ideal. Show that V is irreducible. (Hint: Suppose for contradiction that $V = W_1 \cup W_2$ with $W_1, W_2 \subsetneq V$. Since $W_i \subsetneq V$, there exist $f_i \in \mathbb{I}(W_i) \setminus \mathbb{I}(V)$. What can you say about the product $f_1 f_2$?)
- (c) Show that if $fg \in \mathbb{I}(V)$, then $V \subseteq \mathbb{V}(g) \cup \mathbb{V}(f)$. (Hint: Use that $\mathbb{V}$ is inclusion reversing, together with some facts you proved when checking the Zariski topology is a topology.)
- (d) Prove that if V is irreducible then $\mathbb{I}(V)$ is prime. (Hint: Given $fg \in \mathbb{I}(V)$ we must show that $f \in \mathbb{I}(V)$ or $g \in \mathbb{I}(V)$. By 3c if $fg \in \mathbb{I}(V)$ then $V \subset \mathbb{V}(f) \cup \mathbb{V}(g)$ what does V being irreducible imply here?)
- (e) Conclude that an algebraic set V is irreducible if and only if $\mathbb{I}(V)$ is prime. State a bijection between irreducible algebraic sets and prime ideals. Why do we not need to say radical?

Problem 11. Algebraic Sets and Spec.

Continue to assume that $\mathbb{K}$ is an algebraically closed field and that $V \subset \mathbb{K}^n$ is an algebraic subset with coordinate ring $\mathbb{K}[V]$. In this question we will prove an inclusion-reversing bijection between $\text{Spec}(\mathbb{K}[V])$ and the irreducible algebraic subsets of V . Keep the notation from the previous problem, and use the Hilbert's Nullstellensatz III from Questions ??-??.

- (a) If $W_1, W_2 \subseteq V$ are algebraic, show that $\mathbb{I}_V(W_1 \cup W_2) = \mathbb{I}_V(W_1) \cap \mathbb{I}_V(W_2)$. (Note: You proved the same holds of $\mathbb{I}$ when you checked the Zariski topology on algebraic sets is well-defined.)
- (b) Let $W \subseteq V$ be an algebraic subset. Show that if $\mathbb{I}_V(W)$ is prime, then W is irreducible.
- (c) Let $W \subseteq V$ be an algebraic subset. Prove that given $f, g \in \mathbb{K}[V]$, if $fg \in \mathbb{I}_V(W)$, then $W \subseteq \mathbb{V}_V(f) \cup \mathbb{V}_V(g)$. (This should feel very familiar to Question ??, except that now we are working with $\mathbb{V}_V$ instead of $\mathbb{V}$.)
- (d) Prove that if W is irreducible, then $\mathbb{I}_V(W)$ is a prime ideal of $\mathbb{K}[V]$.
- (e) Deduce that the assignment $[P] \mapsto \mathbb{V}_V(P)$ induces an inclusion reversing bijection between $\text{Spec}(\mathbb{K}[V])$ and the set of irreducible algebraic subsets of V .
- (f) Show that the correspondence in the previous part induces a homeomorphism between $\text{mSpec}(\mathbb{K}[V])$, with the subspace Zariski topology inherited from $\text{Spec}(\mathbb{K}[V])$, and V with its Zariski topology.

Problem 12. Functor-of-Points Viewpoint. Let $\mathbb{K}$ be an arbitrary field, and let $I \subset \mathbb{K}[x_1, \dots, x_n]$ be an ideal. For a field extension $\mathbb{E}/\mathbb{K}$ define:

$$\mathcal{V}_I(\mathbb{E}) := \{(\mathbf{a}_1, \dots, \mathbf{a}_n) \in \mathbb{E}^n \mid f(\mathbf{a}_1, \dots, \mathbf{a}_n) = 0 \text{ for all } f \in I\} \subset \mathbb{E}^n$$

where we are viewing the elements of I as elements of $\mathbb{E}[x_1, \dots, x_n]$ via the natural inclusion, $\mathbb{K}[x_1, \dots, x_n] \subset \mathbb{E}[x_1, \dots, x_n]$. Extend $\mathcal{V}_I$ to a function of sets by defining

$$\left\{ \begin{array}{c} \text{field extensions} \\ \text{of } \mathbb{K} \end{array} \right\} \xrightarrow{\mathcal{V}_I} \mathbf{Set}$$

$$\mathbb{E} \longmapsto \mathcal{V}_I(\mathbb{E}).$$

- (a) For the ideal $I = \langle x^2 + y^2 \rangle \subseteq \mathbb{Q}[x, y]$, describe the sets $\mathcal{V}_I(\mathbb{Q})$, $\mathcal{V}_I(\mathbb{R})$, and $\mathcal{V}_I(\mathbb{C})$.
- (b) For the ideal $J = \langle x^2 + 1 \rangle \subseteq \mathbb{Q}[x]$, describe the sets $\mathcal{V}_J(\mathbb{Q})$, $\mathcal{V}_J(\mathbb{R})$, and $\mathcal{V}_J(\mathbb{C})$.
- (c) Let $\iota : \mathbb{E} \hookrightarrow \mathbb{F}$ be a $\mathbb{K}$ -embedding of field extensions, i.e., a field homomorphism that fixes $\mathbb{K}$ pointwise. Show that ι induces a well-defined map of sets

$$\iota_* : \mathcal{V}_I(\mathbb{E}) \longrightarrow \mathcal{V}_I(\mathbb{F}), \quad (\mathbf{a}_1, \dots, \mathbf{a}_n) \longmapsto (\iota(\mathbf{a}_1), \dots, \iota(\mathbf{a}_n)).$$

(Hint: If $f \in \mathbb{K}[x_1, \dots, x_n]$ and $f(\mathbf{a}_1, \dots, \mathbf{a}_n) = 0$, why is $f(\iota(\mathbf{a}_1), \dots, \iota(\mathbf{a}_n)) = 0$? Use the fact that ι fixes the coefficients of f .)

- (d) Show that $\mathcal{V}_I$ is a *functor* from the category of field extensions of $\mathbb{K}$ (with $\mathbb{K}$ -embeddings as morphisms) to the category of sets.
- (e) Revisit part 1a; the inclusion $\mathbb{R} \hookrightarrow \mathbb{C}$ gives a map $\mathcal{V}_I(\mathbb{R}) \rightarrow \mathcal{V}_I(\mathbb{C})$. Describe this map explicitly. Is it injective? Is it surjective?
- (f) Generalize the map $\mathcal{V}_I$ by enlarging the source from the category of field extensions to the category of $\mathbb{K}$ -algebras, $\mathcal{V}_I : \mathbb{K}\text{-alg} \rightarrow \mathbf{Set}$. Prove that your generalization remains a functor.

The ideas presented in Question 12 is one standard description of the affine scheme of finite type cut out by I . If these words do not mean anything to you, imagine a geometric space similar to algebraic set $\mathbb{V}(I)$, but somehow remembering the ideal I itself not just $\sqrt{I}$. This viewpoint is often called the *functor of points* because we think of I as remembering not only its $\mathbb{K}$ -valued solutions, but also its solutions in every test $\mathbb{K}$ -algebra. That functorial viewpoint is what survives even when $\mathbb{K}$ is not algebraically closed, and it is one of the main bridges from classical algebraic sets to affine schemes. However, much of this jump from

algebraic sets to schemes is beyond the scope of this course.

Worksheet 4.1: Modules Pt. I

Throughout this course, “ring” means *commutative* ring with unity. In this worksheet $\mathbb{K}$ will denote a field. A $\mathbf{R}$ -module is an abelian group $(\mathbf{M}, +)$ together with a scalar multiplication $\mathbf{R} \times \mathbf{M} \rightarrow \mathbf{M}$, $(r, \mathbf{m}) \mapsto r\mathbf{m}$, satisfying the following:

- (i) Associativity: $(rs)\mathbf{m} = r(s\mathbf{m})$ for all $r, s \in \mathbf{R}$ and $\mathbf{m} \in \mathbf{M}$,
- (ii) Distribution over $\mathbf{R}$: $(r + s)\mathbf{m} = r\mathbf{m} + s\mathbf{m}$ for all $r, s \in \mathbf{R}$ and $\mathbf{m} \in \mathbf{M}$,
- (iii) Distribution over $\mathbf{M}$: $r(\mathbf{m} + \mathbf{n}) = r\mathbf{m} + r\mathbf{n}$ for all $r \in \mathbf{R}$ and $\mathbf{m}, \mathbf{n} \in \mathbf{M}$, and
- (iv) Identity: $1_{\mathbf{R}}\mathbf{m} = \mathbf{m}$ for all $\mathbf{m} \in \mathbf{M}$.

Since $\mathbf{M}$ is an abelian group, it has a distinguished additive identity, which we normally denote by 0 or by $0_{\mathbf{M}}$ if we need to distinguish it from $0_{\mathbf{R}}$. When the base ring is clear from context, we might simply say that $\mathbf{M}$ is a module, or we might say that $\mathbf{M}$ is a module over $\mathbf{R}$. As we will see, modules are the analogue of vector spaces in which the scalars come from a ring instead of a field. They include ideals, quotient rings, abelian groups, vector spaces, and many other naturally occurring algebraic objects.

If $\mathbf{M}$ is an $\mathbf{R}$ -module, a subset $\mathbf{N} \subseteq \mathbf{M}$ is an $\mathbf{R}$ -submodule of $\mathbf{M}$ if and only if $r\mathbf{m} + s\mathbf{n} \in \mathbf{N}$ for all $r, s \in \mathbf{R}$ and $\mathbf{m}, \mathbf{n} \in \mathbf{N}$. That is, a submodule is a subgroup of $\mathbf{M}$ such that the scalar multiplication of $\mathbf{R}$ on $\mathbf{M}$ restricts to a scalar multiplication on $\mathbf{N}$.

Problem 1. Basic Identities. Let $\mathbf{R}$ be a ring and $\mathbf{M}$ an $\mathbf{R}$ -module. Prove the following identities.

1. $r0_{\mathbf{M}} = 0_{\mathbf{M}}$ for all $r \in \mathbf{R}$.
2. $0_{\mathbf{R}}\mathbf{m} = 0_{\mathbf{M}}$ for all $\mathbf{m} \in \mathbf{M}$.
3. $(-r)\mathbf{m} = r(-\mathbf{m})$ for all $r \in \mathbf{R}$ and all $\mathbf{m} \in \mathbf{M}$.

Solution.

1. Let $\mathbf{m} \in \mathbf{M}$.

$$r0_{\mathbf{M}} = r(\mathbf{m} - \mathbf{m}) = r\mathbf{m} - r\mathbf{m} = 0_{\mathbf{M}}$$

2. Let $r \in \mathbf{R}$.

$$0_{\mathbf{R}}\mathbf{m} = (r - r)\mathbf{m} = r\mathbf{m} - r\mathbf{m} = 0_{\mathbf{M}}$$

3.

$$(-r)m = (0_R - r)m = 0_R m - rm$$

From (ii) we have that $0_R m = 0_M$,

$$\begin{aligned} (-r)m &= 0_R m - rm = 0_M - rm \\ &= 0_M r - rm = r(0_M - m) \\ &= r(-m) \end{aligned}$$

Problem 2. Basic Examples of Modules.

1. If $\mathbb{K}$ is a field, explain why a $\mathbb{K}$ -module is the same as a $\mathbb{K}$ -vector space.
2. Explain why R is itself an R -module in a natural way.
3. Show that if $I \subseteq R$ is an ideal, then the multiplication on R naturally makes I into an R -module.
4. Prove that if $I \subseteq R$ is an ideal, then R/I is an R -module.
5. Prove that a $\mathbb{Z}$ -module is the same thing as an abelian group.
6. If $I \subseteq R$ is an ideal and M is an R -module, show that

$$IM := \left\{ \sum_{i=1}^n r_i m_i \mid n \geq 0, r_i \in I, m_i \in M \right\}$$

is a submodule of M .

Solution. **TODO:**

1.

Problem 3. Submodules Generated by Elements.

Let M be an R -module. Given elements $m_1, \dots, m_t \in M$, the submodule of M generated by $m_1, \dots, m_t$ is the set

$$Rm_1 + Rm_2 + \cdots + Rm_t := \{r_1 m_1 + \cdots + r_t m_t \mid r_1, \dots, r_t \in R\} \subseteq M.$$

We say that $m_1, \dots, m_t$ are *generators* for M if $Rm_1 + Rm_2 + \cdots + Rm_t = M$. A submodule $N \subseteq M$ is a *cyclic submodule* if $N = Rm$ for some $m \in M$.

1. Prove that $Rm_1 + Rm_2 + \cdots + Rm_t$ is an R -submodule of M .
2. Prove that $Rm_1 + Rm_2 + \cdots + Rm_t$ is the smallest R -submodule of M containing $m_1, \dots, m_t$.
3. Prove that a ring R is itself a cyclic R -module by finding a generator.
4. In the $\mathbb{Z}$ -module $\mathbb{Z}/12\mathbb{Z}$, compute the submodules generated by $\bar{8}$ and $\bar{6}$.
5. Let $R = \mathbb{K}[x]$ and $M = R/\langle x^3 \rangle$. Describe the submodules generated by $\bar{1}$, $\bar{x}$, and $\bar{x}^2$.

Solution.

Let $N = Rm_1 + Rm_2 + \cdots + Rm_t$.

1. Let $r, s \in R$ and $n, m \in N$. Then, by definition

$$m = r_1m_1 + \cdots + r_tm_t$$

and

$$n = s_1m_1 + \cdots + s_tm_t$$

with $r_1, \dots, r_t, s_1, \dots, s_t \in R$. Then by distributivity and associativity of R , and the abelian nature of M we have,

$$\begin{aligned} pm + qn &= p(r_1m_1 + \cdots + r_tm_t) + q(s_1m_1 + \cdots + s_tm_t) \\ &= ((pr_1)m_1 + \cdots + (pr_t)m_t) + ((qs_1)m_1 + \cdots + (qs_t)m_t) \\ &= ((pr_1 + qs_1)m_1 + \cdots + (pr_t + qs_t)m_t) \end{aligned}$$

but since R is closed under addition and multiplication $pr_1 + qs_1, \dots, pr_t + qs_t \in R$. Hence, $pm + qn \in N$ and N is a submodule.

2. Let L be the smallest R -submodule of M containing $m_1, \dots, m_t$. Then by minimality of L we have $L \subseteq N$. To show the reverse containment consider $m \in N$. By definition,

$$m = r_1m_1 + \cdots + r_tm_t$$

with $r_1, \dots, r_t \in R$. But since $m_1, \dots, m_t \in L$, all R -linear combinations of them is in L (by definition of a submodule), hence $m \in L$. Since m was an arbitrary element in N this implies that $N = L$, hence N is the smallest R -submodule of M containing $m_1, \dots, m_t$.

3. The ring R is a cyclic R -module with the generator 1_R .

By definition,

$$R \cdot 1_R = \{r \cdot 1_R \mid r \in R\}$$

But $r \cdot 1_R = r$ for all $r \in R$, i.e., $R \cdot 1_R = R$.

4. Consider $\mathbb{Z}/12\mathbb{Z} \cdot \bar{8}$,

$$\mathbb{Z}/12\mathbb{Z} \cdot \bar{8} = \{r \cdot \bar{8} \mid r \in \mathbb{Z}/12\mathbb{Z}\} = \{\bar{0}, \bar{8}, \bar{4}\}$$

Consider $\mathbb{Z}/12\mathbb{Z} \cdot \bar{6}$,

$$\mathbb{Z}/12\mathbb{Z} \cdot \bar{6} = \{r \cdot \bar{6} \mid r \in \mathbb{Z}/12\mathbb{Z}\} = \{\bar{0}, \bar{6}\}$$

5. $R \cdot \bar{1} = \{r \cdot \bar{1} = \bar{r} \mid r \in R\} = M$

$$R \cdot \bar{x} = \{r \cdot \bar{x} = \overline{rx} \mid r \in R\} = \{\overline{r_1x} + \overline{r_2x^2} \mid r_1, r_2 \in R\} = \mathbb{K} \cdot \bar{x}$$

$$R \cdot \bar{x}^2 = \{r \cdot \bar{x}^2 = \overline{rx^2} \mid r \in R\} = \mathbb{K} \cdot \bar{x}^2$$

A *homomorphism* of $\mathbf{R}$ -modules is a group homomorphism $\phi : M \rightarrow N$ such that $\phi(r\mathbf{m}) = r\phi(\mathbf{m})$ for all $r \in \mathbf{R}$ and $\mathbf{m} \in M$. Two $\mathbf{R}$ -modules M and N are *isomorphic* if and only if there exist $\mathbf{R}$ -module homomorphisms $\phi : M \rightarrow N$ and $\psi : N \rightarrow M$ such that $\phi \circ \psi = \text{Id}_N$ and $\psi \circ \phi = \text{Id}_M$. Equivalently, there is a bijective homomorphism of $\mathbf{R}$ -modules $\phi : M \rightarrow N$. We call such maps *isomorphisms* of $\mathbf{R}$ -modules. If $\phi : M \rightarrow N$ is a homomorphism of $\mathbf{R}$ -modules, there are two natural submodules associated to it: its *kernel* and its *image*:

$$\ker(\phi) := \{\mathbf{m} \in M \mid \phi(\mathbf{m}) = 0_N\} \subseteq M \quad \text{and} \quad \text{img}(\phi) := \{\phi(\mathbf{m}) \mid \mathbf{m} \in M\} \subseteq N.$$

One should readily check that both the kernel and image of a module homomorphism are submodules. As with linear maps, group homomorphisms, and ring homomorphisms, the kernel measures the failure of injectivity and the image measures the failure of surjectivity.

Given a submodule $N \subseteq M$, the quotient abelian group M/N inherits an $\mathbf{R}$ -module structure via

$$r(\mathbf{m} + N) := r\mathbf{m} + N.$$

Well-definedness uses that N is a submodule, not merely a subgroup. The projection $\pi : M \rightarrow M/N$ is a homomorphism with $\ker(\pi) = N$. The quotient M/N is the module obtained by forcing $N = 0$. Precisely: if $f : M \rightarrow P$ is a homomorphism with $N \subseteq \ker(f)$, there is a unique $\bar{f} : M/N \rightarrow P$ with $f = \bar{f} \circ \pi$:

$$\begin{array}{ccc} M & \xrightarrow{f} & P \\ \pi \downarrow & \nearrow \bar{f} & \\ M/N & & \end{array}$$

Phrased differently, there is a natural isomorphism

$$\text{Hom}_{\mathbf{R}}(M/N, P) \cong \{f \in \text{Hom}_{\mathbf{R}}(M, P) \mid N \subseteq \ker(f)\};$$

That is, homomorphisms out of M/N are exactly homomorphisms out of M that kill N . This is usually summarized as the First Isomorphism Theorem.

Theorem 0.18 (First Isomorphism Theorem). *Every homomorphism $\phi : M \rightarrow N$ of $\mathbf{R}$ -modules induces an isomorphism*

$$M/\ker(\phi) \xrightarrow{\sim} \text{img}(\phi), \quad \mathbf{m} + \ker(\phi) \mapsto \phi(\mathbf{m}).$$

In other words, quotienting by the kernel accounts for all the redundancy: every surjection is a quotient map.

Theorem 0.19 (Third Isomorphism Theorem). *Let $N \subseteq M$ be a submodule and $\pi: M \twoheadrightarrow M/N$ the projection. There is an inclusion-preserving bijection*

$$\{\text{submodules of } M \text{ containing } N\} \xrightarrow{1:1} \{\text{submodules of } M/N\}$$

given by $L \mapsto L/N = \pi(L)$ with inverse $\bar{L} \mapsto \pi^{-1}(\bar{L})$. Moreover, for $N \subseteq L \subseteq M$:

$$M/L \cong (M/N)/(L/N).$$

The correspondence says that the submodule lattice of M/N is the interval $[N, M]$ in the submodule lattice of M . The supplementary isomorphism $(M/N)/(L/N) \cong M/L$ is sometimes called the *Third Isomorphism Theorem*: quotienting in two stages is the same as quotienting all at once.

Problem 4. Basic Examples of Module Homomorphisms.

1. If $\mathbb{K}$ is a field, explain why a homomorphism of $\mathbb{K}$ -modules is the same as a $\mathbb{K}$ -linear map.
2. Explain why a homomorphism of $\mathbb{Z}$ -modules is the same as a homomorphism of abelian groups.

Solution.

1. If $\mathbb{K}$ is a field, we know that a $\mathbb{K}$ module is a $\mathbb{K}$ -vector space. Moreover, by definition, a homomorphism $\varphi: N \rightarrow M$ where N, M are $\mathbb{K}$ -modules ($\mathbb{K}$ -vector spaces) has the following property: for any $r, s \in \mathbb{K}$ and $a, b \in N$ we have,

$$\varphi(ra + sb) = \varphi(ra) + \varphi(sb) = r\varphi(a) + s\varphi(b)$$

which is the condition defining a $\mathbb{K}$ -linear map.

2. We proved in Problem 2 that $\mathbb{Z}$ -modules are the same as abelian groups, then any $\mathbb{Z}$ -module homomorphism must preserve the group operation and hence is the same as an abelian group homomorphism.

Problem 5. Cyclic Modules & Annihilators. Let M be an R -module. The *annihilator* of an element $m \in M$ is $\text{Ann}_R(m) := \{r \in R \mid rm = 0\}$.

1. Prove that $\text{Ann}_R(m)$ is an ideal of R .
2. What is $\text{Ann}_R(0_M)$?
3. Consider the multiplication by m function:

$$\phi_m : R \rightarrow M \quad \text{given by} \quad r \mapsto rm.$$

Prove that ϕ_m is a homomorphism of R -modules.

4. Show that $\ker(\phi_m) = \text{Ann}_R(m)$ and $\text{img}(\phi_m) = Rm$.
5. Prove that a submodule $N \subseteq M$ is cyclic if and only if $N \cong R/I$ for some ideal $I \subseteq R$.

Solution.

1. To show that $\text{Ann}_R(m)$ is an ideal of R we need to check that it is a subgroup under addition, and is closed under multiplication from elements in R .

(a) **Identity:** Note that $0 \in \text{Ann}_R(m)$ since $0 \cdot m = 0$.

(b) **Closure under Addition and Inverses:** If $a, b \in \text{Ann}_R(m)$, then

$$(a - b)m = am - bm = 0 - 0 = 0$$

so $a - b \in \text{Ann}_R(m)$.

Hence $\text{Ann}_R(m)$ is a subgroup of $(R, +)$.

(c) Let $r \in R$ and $a \in \text{Ann}_R(m)$. Then

$$(ra)m = r(am) = r \cdot 0 = 0$$

so $ra \in \text{Ann}_R(m)$.

Hence $\text{Ann}_R(m)$ is an ideal of R .

2. $\text{Ann}_R(0_M) = \{r \in R \mid r0_M = 0\} = R$. That is, $\text{Ann}_R(0_M)$ is the whole ring.
3. Let $a, b \in R$. Consider

$$\phi_m(a + b) = (a + b)m = am + bm = \phi_m(a) + \phi_m(b)$$

Hence ϕ_m is a group homomorphism. And, for any $r, a \in R$ we have,

$$\phi_m(ra) = (ra)m = r(am) = r\phi_m(a)$$

Hence, ϕ_m is a homomorphism of R -modules.

4. $\ker(\phi_{\mathfrak{m}}) = \{r \in R \mid r\mathfrak{m} = 0\}$ which is precisely the definition of $\text{Ann}_R(\mathfrak{m})$.

And $\text{im}(\phi_{\mathfrak{m}}) = \{r\mathfrak{m} \mid r \in R\} = R\mathfrak{m}$.

5. ($\implies$)

Assume submodule $N \subseteq M$ is cyclic. Then $N = R \cdot \mathfrak{m}$ for some $\mathfrak{m} \in M$. From part (4) we know that $\ker(\phi_{\mathfrak{m}}) = \text{Ann}_R(\mathfrak{m})$ and $\text{im}(\phi_{\mathfrak{m}}) = R\mathfrak{m}$, that is by the first isomorphism theorem,

$$N = R\mathfrak{m} = \text{im}(\phi_{\mathfrak{m}}) \cong R / \ker(\phi_{\mathfrak{m}}) = R / \text{Ann}_R(\mathfrak{m})$$

where we know that $\text{Ann}_R(\mathfrak{m})$ is an ideal in R (part 1)

($\impliedby$)

Assume $N \cong R/I$ for some ideal $I \subseteq R$. Then there exists some isomorphism of R -modules $\varphi : R/I \rightarrow N$. Note that every element in R/I can be written as, $r(1 + I)$ for some $r \in R$ and therefore every element of N is of the form

$$\varphi(r(1 + I)) = r\varphi(1 + I)$$

Let $\varphi(1 + I) = \mathfrak{n} \in N$. That is every element is of the form $r\mathfrak{n}$ for some $r \in R$, then $N \subseteq R\mathfrak{n}$. Moreover since $\mathfrak{n} \in N$ and N is closed under scalar multiplication we have $R\mathfrak{n} \subseteq N$. Hence, N is cyclic.

Let $\{M_i\}_{i \in I}$ be a family of R -modules indexed by a set I . The *direct product* is

$$\prod_{i \in I} M_i = \{(m_i)_{i \in I} \mid m_i \in M_i\},$$

with componentwise addition and scalar multiplication. The *direct sum* is the submodule

$$\bigoplus_{i \in I} M_i = \left\{ (m_i)_{i \in I} \in \prod_{i \in I} M_i \mid m_i = 0 \text{ for all but finitely many } i \right\}.$$

For each $j \in I$ there is a projection map π_j and an inclusion map ι_j :

$$\pi_j : \prod_{i \in I} M_i \rightarrow M_j, \quad (m_i)_{i \in I} \mapsto m_j,$$

and

$$\iota_j : M_j \rightarrow \bigoplus_{i \in I} M_i, \quad m \mapsto (0, \dots, 0, m, 0, \dots).$$

The product is characterized by maps to the factors, while the direct sum is characterized by maps from the summands. These are the universal properties that make products and sums so useful.

Problem 6. Direct Sums and Products of Modules.

Let $\{M_i\}_{i \in I}$ be a family of R -modules indexed by a set I .

1. Prove that both $\prod_{i \in I} M_i$ and $\bigoplus_{i \in I} M_i$ are R -modules.
2. Explain why, if I is finite, $\prod_{i \in I} M_i = \bigoplus_{i \in I} M_i$.
3. Consider the $\mathbb{Z}$ -modules $M_n = \mathbb{Z}/n\mathbb{Z}$ for $n \geq 1$. Describe explicitly an element of $\prod_{n=1}^{\infty} M_n$ that is not an element of $\bigoplus_{n=1}^{\infty} M_n$.
4. Let $M_i = R$ for all $i \in \mathbb{N}$. Show that the submodule of $\prod_{i \in \mathbb{N}} R$ generated by the standard basis vectors $e_1, e_2, e_3, \dots$ is $\bigoplus_{i \in \mathbb{N}} R$, not the whole product, provided $R \neq 0$.
5. For two modules M and N , write down the inclusion maps ι_M, ι_N and projection maps π_M, π_N for $M \oplus N$. Verify the following identities

$$\pi_M \circ \iota_M = \text{Id}_M, \pi_N \circ \iota_N = \text{Id}_N, \pi_M \circ \iota_N = 0, \pi_N \circ \iota_M = 0, \text{ and } \iota_M \circ \pi_M + \iota_N \circ \pi_N = \text{Id}_{M \oplus N}.$$

Solution.

1.

Problem 7. Universal Properties of Products and Sums. Let $\{M_i\}_{i \in I}$ be a family of R -modules indexed by I .

1. Let L be an R -module and suppose that for every $i \in I$ we are given a homomorphism $\phi_i : L \rightarrow M_i$. Prove that there exists a unique homomorphism $\phi : L \rightarrow \prod_{i \in I} M_i$ such that the following diagram commutes for all $i \in I$:

$$\begin{array}{ccc} L & \xrightarrow{\phi} & \prod_{i \in I} M_i \\ & \searrow \phi_i & \downarrow \pi_i \\ & & M_i \end{array} \quad .$$

2. Let L be an R -module and suppose that for every $i \in I$ we are given a homomorphism $\psi_i : M_i \rightarrow L$. Prove that there exists a unique homomorphism $\psi : \bigoplus_{i \in I} M_i \rightarrow L$ such that the following diagram commutes for all $i \in I$:

$$\begin{array}{ccc} \bigoplus_{i \in I} M_i & \xrightarrow{\psi} & L \\ \uparrow \iota_i & \nearrow \psi_i & \\ M_i & & \end{array} \quad .$$

3. Deduce natural isomorphisms of R -modules

$$\text{Hom}_R \left(L, \prod_{i \in I} M_i \right) \cong \prod_{i \in I} \text{Hom}_R(L, M_i)$$

and

$$\text{Hom}_R \left(\bigoplus_{i \in I} M_i, L \right) \cong \prod_{i \in I} \text{Hom}_R(M_i, L).$$

4. Explain why a family of maps $\psi_i : M_i \rightarrow L$ does not generally define a map $\prod_{i \in I} M_i \rightarrow L$. Use the case $M_i = L = \mathbb{Z}$ and $\psi_i = \text{Id}_{\mathbb{Z}}$ to illustrate the problem.
5. When I is finite, explain why the two universal properties describe the same module. This is why finite direct sums are also finite direct products.

Over a field, every module (i.e. vector space) has a basis, and a linear map is determined by its values on a basis. Over a general ring, modules need not have bases at all—but those that do are of fundamental importance. The goal of this section is to set up the theory of free modules and to understand precisely how they differ from vector spaces.

Let M be an R -module. A subset $B \subseteq M$ *generates* (or *spans*) M if every element of M can be written as a finite R -linear combination of elements of B , i.e. if

$$M = \left\{ \sum_{i=1}^n r_i m_i \mid n \geq 0, r_i \in R, m_i \in B \right\}.$$

We say B is *linearly independent* if, whenever $m_1, \dots, m_n \in B$ are distinct and $\sum_{i=1}^n r_i m_i = 0$, we have $r_i = 0$ for all i . A *basis* for M is a subset that is both generating and linearly independent, or equivalently, a subset $\{b_s\}_{s \in S}$ such that every element of M can be written *uniquely* as a finite R -linear combination $\sum_s r_s b_s$.

Over a field, these notions interact exactly as in undergraduate linear algebra: every spanning set contains a basis, every linearly independent set extends to one, and all bases have the same cardinality. Over a general ring, the first two statements can fail; in particular, many modules do not admit bases at all. A module that does admit a basis is called *free*. Given a set S , the *free R -module on S* is

$$R^{(S)} := \bigoplus_{s \in S} R.$$

Concretely, $R^{(S)}$ consists of all functions $S \rightarrow R$ with finite support. It has a distinguished element e_s for each $s \in S$, where e_s has a 1 in the s -coordinate and 0 elsewhere. Every element of $R^{(S)}$ can then be written uniquely as a finite sum

$$\sum_{s \in S} r_s e_s,$$

so $\{e_s \mid s \in S\}$ is a basis, called the *standard basis*. When $S = \{1, \dots, n\}$ we write R^n instead of $R^{(S)}$.

The key feature of a free module is that a homomorphism out of it is completely determined by where the basis goes. Precisely: for every function of sets $\alpha: S \rightarrow M$ (where M is any R -module), there is a unique R -module homomorphism $\tilde{\alpha}: R^{(S)} \rightarrow M$ such that $\tilde{\alpha}(e_s) = \alpha(s)$ for all $s \in S$. In categorical language, the functor $S \mapsto R^{(S)}$ is left adjoint to the forgetful functor from R -modules to sets. This universal property characterizes $R^{(S)}$ up to unique isomorphism and is the reason free modules play a role analogous to that of vector spaces—they are the modules we can map out of “for free.”

Problem 8. Free Modules and Their Universal Property.

1. Verify directly that every element of $\mathbf{R}^{(S)}$ can be written uniquely as a finite $\mathbf{R}$ -linear combination of the standard basis elements $\mathbf{e}_s$.
2. Prove the universal property of $\mathbf{R}^{(S)}$ stated above.
3. Deduce that there is a natural isomorphism

$$\mathrm{Hom}_{\mathbf{R}}(\mathbf{R}^{(S)}, \mathbf{M}) \cong \prod_{s \in S} \mathbf{M}.$$

Why is the right side a product rather than a direct sum?

4. Show that an $\mathbf{R}$ -module $\mathbf{M}$ is generated by $\mathbf{m}_1, \dots, \mathbf{m}_n$ if and only if there is a surjective homomorphism $\mathbf{R}^n \rightarrow \mathbf{M}$ sending $\mathbf{e}_i$ to $\mathbf{m}_i$.
5. Conclude that every finitely generated $\mathbf{R}$ -module is a quotient of a finite free module.
6. More generally, show that every $\mathbf{R}$ -module is a quotient of a free $\mathbf{R}$ -module.
7. For $\mathbf{M} = \mathbf{R}/\mathbf{I}$, write $\mathbf{M}$ as a quotient of a free module and identify the kernel.
8. For $\mathbf{R} = \mathbb{K}[x]$ and $\mathbf{M} = \mathbf{R}/\langle x^3 \rangle$, describe $\mathbf{M}$ as a module with one generator and one relation.

Problem 9. Free Modules Are Not Quite Vector Spaces.

1. Show that $\mathbb{Z}/n\mathbb{Z}$ is not a free $\mathbb{Z}$ -module for any $n > 1$.
2. More generally, if $\mathbf{R}$ is an integral domain and $0 \neq \mathbf{I} \subsetneq \mathbf{R}$ is an ideal, show that $\mathbf{R}/\mathbf{I}$ is not a free $\mathbf{R}$ -module.
3. In the $\mathbb{Z}$ -module $\mathbb{Z}$, show that $\{2, 3\}$ is a generating set but not a basis.
4. In the $\mathbb{Z}$ -module $\mathbb{Z}$, show that $\{2\}$ is linearly independent but does not generate $\mathbb{Z}$.
5. Let $\mathbf{R} = \mathbb{Z}/6\mathbb{Z}$. Show that the element $\bar{2} \in \mathbf{R}$ is nonzero but not part of a basis for the free $\mathbf{R}$ -module $\mathbf{R}$.
6. Explain why the previous examples show that bases over rings behave differently from bases over fields.

Problem 10. Matrices and Maps Between Finite Free Modules.

1. Let $A = (a_{ij})$ be an $m \times n$ matrix with entries in R . Define a homomorphism

$$\phi_A : R^n \rightarrow R^m, \quad e_j \mapsto \sum_{i=1}^m a_{ij} e_i.$$

Prove that every homomorphism $R^n \rightarrow R^m$ arises uniquely in this way.

2. Under this correspondence, prove that composition of homomorphisms corresponds to multiplication of matrices.
3. When $R = \mathbb{K}$ is a field, explain how this recovers the usual matrix description of linear maps between finite-dimensional vector spaces.
4. Let $A = \begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$ and view A as a homomorphism $\mathbb{Z}^2 \rightarrow \mathbb{Z}^2$. Describe $\text{coker}(A) = \mathbb{Z}^2 / \text{img}(A)$.
5. Let $R = \mathbb{K}[x]$ and consider multiplication by x as a map $R^2 \rightarrow R^2$. Write its matrix with respect to the standard basis, and compute its cokernel.

For two R -modules M and N , we write $\text{Hom}_R(M, N)$ for the set of R -module homomorphisms $M \rightarrow N$. Since R is commutative, $\text{Hom}_R(M, N)$ is itself an R -module: addition and scalar multiplication are defined pointwise by

$$(\phi + \psi)(m) = \phi(m) + \psi(m), \quad (r\phi)(m) = r\phi(m).$$

This construction is one of the most important ways of building new modules from old ones. There is already a great deal to understand: Hom out of a free module is easy, Hom out of a quotient records annihilators, and Hom interacts cleanly with direct sums and products through their universal properties.

Problem 11. $\text{Hom}_R(M, N)$ as a Module.

1. Prove that $\text{Hom}_R(M, N)$ is an abelian group under pointwise addition. What is the zero element?
2. Prove that the formula $(r\phi)(m) = r\phi(m)$ makes $\text{Hom}_R(M, N)$ into an R -module.
3. Show that evaluation at $1 \in R$ defines an isomorphism of R -modules

$$\text{Hom}_R(R, M) \cong M.$$

4. More generally, prove that

$$\text{Hom}_R(R/I, M) \cong \{m \in M \mid Im = 0\}.$$

The isomorphism should again be given by evaluation at $\bar{1} \in R/I$.

5. Deduce that $\text{Hom}_R(R^n, M) \cong M^n$.
6. Prove that $\text{End}_R(R) \cong R$ as rings, where the ring structure on $\text{End}_R(R)$ is given by addition and composition.
7. If $I \subseteq R$ is an ideal, identify $\text{Hom}_R(R/I, R)$ in terms of an annihilator ideal of R .

Problem 12. Hom Computations.

1. Describe $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}/n\mathbb{Z})$ explicitly for positive integers m, n . Show that it is isomorphic to $\mathbb{Z}/d\mathbb{Z}$, where $d = \gcd(m, n)$.
2. Prove that $\text{Hom}_{\mathbb{Z}}(\mathbb{Q}, \mathbb{Z}) = 0$.
3. Prove that $\text{Hom}_{\mathbb{Z}}(\mathbb{Z}, M) \cong M$ for every abelian group M .
4. Let $R = \mathbb{K}[x]$ and let $M = R/\langle f \rangle$, $N = R/\langle g \rangle$. Use the preceding computation of $\text{Hom}_R(R/I, -)$ to describe $\text{Hom}_R(M, N)$ as a submodule of N .
5. In the special case $f = x^a$ and $g = x^b$, compute $\text{Hom}_{\mathbb{K}[x]}(\mathbb{K}[x]/\langle x^a \rangle, \mathbb{K}[x]/\langle x^b \rangle)$ up to isomorphism.

Problem 13. Functoriality of Hom. Let M, M', N, N' be R -modules.

1. If $\alpha : M' \rightarrow M$ is a homomorphism, prove that precomposition defines an R -module homomorphism

$$\alpha^* : \text{Hom}_R(M, N) \rightarrow \text{Hom}_R(M', N), \quad \phi \longmapsto \phi \circ \alpha.$$

2. If $\beta : N \rightarrow N'$ is a homomorphism, prove that postcomposition defines an R -module homomorphism

$$\beta_* : \text{Hom}_R(M, N) \rightarrow \text{Hom}_R(M, N'), \quad \phi \longmapsto \beta \circ \phi.$$

3. Check that $(\text{Id}_M)^* = \text{Id}_{\text{Hom}_R(M, N)}$ and $(\text{Id}_N)_* = \text{Id}_{\text{Hom}_R(M, N)}$.
4. If $M'' \xrightarrow{\gamma} M' \xrightarrow{\alpha} M$, prove that $(\alpha \circ \gamma)^* = \gamma^* \circ \alpha^*$.

5. If $N \xrightarrow{\beta} N' \xrightarrow{\delta} N''$, prove that $(\delta \circ \beta)_* = \delta_* \circ \beta_*$.
6. Let $M^* := \text{Hom}_R(M, R)$ be the *dual module*. If $M = R^n$, describe the dual basis of M^* and prove that $M^* \cong R^n$.

Many natural constructions in algebra are *bilinear*: matrix multiplication, the dot product, and the evaluation pairing between a module and its dual. In each case, the map is additive in each variable separately, but not additive as a function of both variables together—so it is not a homomorphism from the product module. The tensor product is the device that resolves this tension. It provides a module $M \otimes_R N$ that converts bilinear maps *out of* $M \times N$ into honest linear maps out of $M \otimes_R N$.

Let R be a commutative ring with identity, and let M , N , and P be R -modules. An R -bilinear map from $M \times N$ to P is a function

$$B: M \times N \rightarrow P$$

satisfying the following three conditions for all $r \in R$, $m, m' \in M$, and $n, n' \in N$:

- (i) $B(m + m', n) = B(m, n) + B(m', n)$,
- (ii) $B(m, n + n') = B(m, n) + B(m, n')$,
- (iii) $B(rm, n) = rB(m, n) = B(m, rn)$.

In other words, B is R -linear in each argument when the other is held fixed. A *tensor product* of M and N is a pair $(M \otimes_R N, \tau)$ consisting of an R -module $M \otimes_R N$ and a bilinear map

$$\tau: M \times N \rightarrow M \otimes_R N, \quad (m, n) \mapsto m \otimes n,$$

that is *universal* in the following sense: for every R -module P and every R -bilinear map $B: M \times N \rightarrow P$, there exists a *unique* R -module homomorphism $\tilde{B}: M \otimes_R N \rightarrow P$ such that $B = \tilde{B} \circ \tau$.

$$\begin{array}{ccc} M \times N & \xrightarrow{\tau} & M \otimes_R N \\ & \searrow B & \swarrow \tilde{B} \\ & P & \end{array}$$

The content of this property is that τ is the “most general” bilinear map out of $M \times N$: every other one factors through it, and does so in exactly one way. One important consequence is that to define a linear map *out of* a tensor product, it suffices to specify its values on the generators $m \otimes n$ and verify that the assignment is bilinear—the universal property then guarantees a unique, well-defined homomorphism. This strategy is used repeatedly in the exercises below.

The universal property tells us what a tensor product must *do*, but not that one exists. Here is a direct construction. Begin with the free $\mathbf{R}$ -module F on the set $M \times N$; its basis consists of formal symbols $[m, n]$, one for each pair $(m, n) \in M \times N$. Let $K \subseteq F$ be the submodule generated by all elements of the forms

$$\begin{aligned} [m + m', n] - [m, n] - [m', n], \\ [m, n + n'] - [m, n] - [m, n'], \\ [rm, n] - r[m, n], \\ [m, rn] - r[m, n], \end{aligned}$$

for all $r \in \mathbf{R}$, $m, m' \in M$, and $n, n' \in N$. Set $M \otimes_{\mathbf{R}} N = F/K$, and write $m \otimes n$ for the image of $[m, n]$ in the quotient. By construction, the map $(m, n) \mapsto m \otimes n$ is bilinear, and every element of $M \otimes_{\mathbf{R}} N$ is a finite $\mathbf{R}$ -linear combination of such “simple tensors.”

The identities that $m \otimes n$ satisfies are therefore precisely the bilinearity relations:

$$\begin{aligned} (m + m') \otimes n &= m \otimes n + m' \otimes n, \\ m \otimes (n + n') &= m \otimes n + m \otimes n', \\ (rm) \otimes n &= r(m \otimes n) = m \otimes (rn). \end{aligned}$$

Not every element of $M \otimes_{\mathbf{R}} N$ is a simple tensor $m \otimes n$; a general element is a finite sum $\sum_i m_i \otimes n_i$. A common source of errors is to assume that every tensor can be written as a single $m \otimes n$.

Problem 14. Constructing Tensor Products.

- Using the generators-and-relations construction above, prove the identities

$$(m + m') \otimes n = m \otimes n + m' \otimes n, \quad m \otimes (n + n') = m \otimes n + m \otimes n',$$

and

$$(rm) \otimes n = r(m \otimes n) = m \otimes (rn).$$

- Prove that $0 \otimes n = 0$, $m \otimes 0 = 0$, and $(-m) \otimes n = -(m \otimes n) = m \otimes (-n)$.
- Prove that the construction described above satisfies the universal property of the tensor product.
- Prove that the tensor product is unique up to unique isomorphism: if T and T' both satisfy the universal property for M and N , then there is a unique isomorphism $T \cong T'$ compatible with the two bilinear maps from $M \times N$.
- Use the universal property to construct a natural isomorphism

$$M \otimes_{\mathbf{R}} N \cong N \otimes_{\mathbf{R}} M.$$

6. If $f : M \rightarrow M'$ and $g : N \rightarrow N'$ are homomorphisms, construct a homomorphism

$$f \otimes g : M \otimes_{\mathbb{R}} N \rightarrow M' \otimes_{\mathbb{R}} N'$$

satisfying $(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$.

Problem 15. Basic Tensor Product Computations.

1. Prove that the map

$$R \otimes_{\mathbb{R}} M \rightarrow M, \quad r \otimes m \mapsto rm$$

is an isomorphism of $\mathbb{R}$ -modules.

2. Prove that $R^n \otimes_{\mathbb{R}} M \cong M^n$.

3. More generally, prove that

$$\left(\bigoplus_{i \in I} M_i \right) \otimes_{\mathbb{R}} N \cong \bigoplus_{i \in I} (M_i \otimes_{\mathbb{R}} N).$$

4. Let $I \subseteq R$ be an ideal. Prove that

$$(R/I) \otimes_{\mathbb{R}} M \cong M/IM,$$

where IM is the submodule defined earlier. What is the image of $\bar{r} \otimes m$ under this isomorphism?

5. Deduce that for ideals $I, J \subseteq R$ one has $(R/I) \otimes_{\mathbb{R}} (R/J) \cong R/(I + J)$.

6. Compute $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$ for positive integers m, n .

7. Compute $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z}$.

8. If V and W are finite-dimensional vector spaces over a field $\mathbb{K}$, prove that

$$\dim_{\mathbb{K}}(V \otimes_{\mathbb{K}} W) = \dim_{\mathbb{K}}(V) \dim_{\mathbb{K}}(W).$$

Problem 16. Tensor Products as Linearized Bilinear Maps.

1. Let $e_1, \dots, e_m$ be the standard basis of $\mathbb{R}^m$ and $f_1, \dots, f_n$ the standard basis of

- R^n . Show that $R^m \otimes_R R^n$ is free with basis $\{e_i \otimes f_j \mid 1 \leq i \leq m, 1 \leq j \leq n\}$.
2. Let $B : R^m \times R^n \rightarrow P$ be an R -bilinear map. Prove that B is uniquely determined by the mn elements $B(e_i, f_j) \in P$.
 3. If M is generated by $m_1, \dots, m_a$ and N is generated by $n_1, \dots, n_b$, prove that $M \otimes_R N$ is generated by the elements $m_i \otimes n_j$.
 4. Let $R = \mathbb{K}[x]$. Compute $R/\langle x^a \rangle \otimes_R R/\langle x^b \rangle$ up to isomorphism.
 5. Let $I, J \subseteq R$ be ideals. The multiplication map $I \times J \rightarrow IJ$, $(a, b) \mapsto ab$, is R -bilinear. Use the universal property to construct a natural surjective homomorphism

$$I \otimes_R J \twoheadrightarrow IJ.$$
 6. Show that if $I = R$, then the map in the previous part is an isomorphism. Then let $R = \mathbb{K}[\varepsilon]/\langle \varepsilon^2 \rangle$ and let $I = J = \langle \bar{\varepsilon} \rangle$. Compute $I \otimes_R I$ and IJ , and conclude that the multiplication map $I \otimes_R J \rightarrow IJ$ need not be injective.